Full Custom IC Design Flow Tutorial Using Synopsys Custom Tools

By:
Hamid Mahmoodi
Mojan Norouzi
Michael Chan
Casey Hardy

Nano-Electronics & Computing Research Lab
School of Engineering
San Francisco State University
San Francisco, CA

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Introduction

This manual assumes you are able to do some basic things in a Linux environment such as create a folder, change directories....etc. If you want to learn how to use Linux here are many good tutorials available on the web, such as this one: http://tldp.org/LDP/gs/node5.html. Figure I.1 shows the design flow this tutorial will be implementing. In the first three parts of this manual you will design and simulate a CMOS inverter using Custom DesignerSE in conjunction with Hspice and WaveView to visually assemble the circuit schematic, simulate it, and view the output waveforms. For further help you are encouraged to go to "Help" in the menu bar of CosmosSE. You will use Custon DesignerSE to create a layout, and use Hercules to run a design rule check (DRC) on the layout based on the technology process. You will also use Hercules to make sure our inverter layout matches our schematic by running a Layout Versus Schematic (LVS) check. Finally, you will use the inverter we create in a gate level design of a buffer.

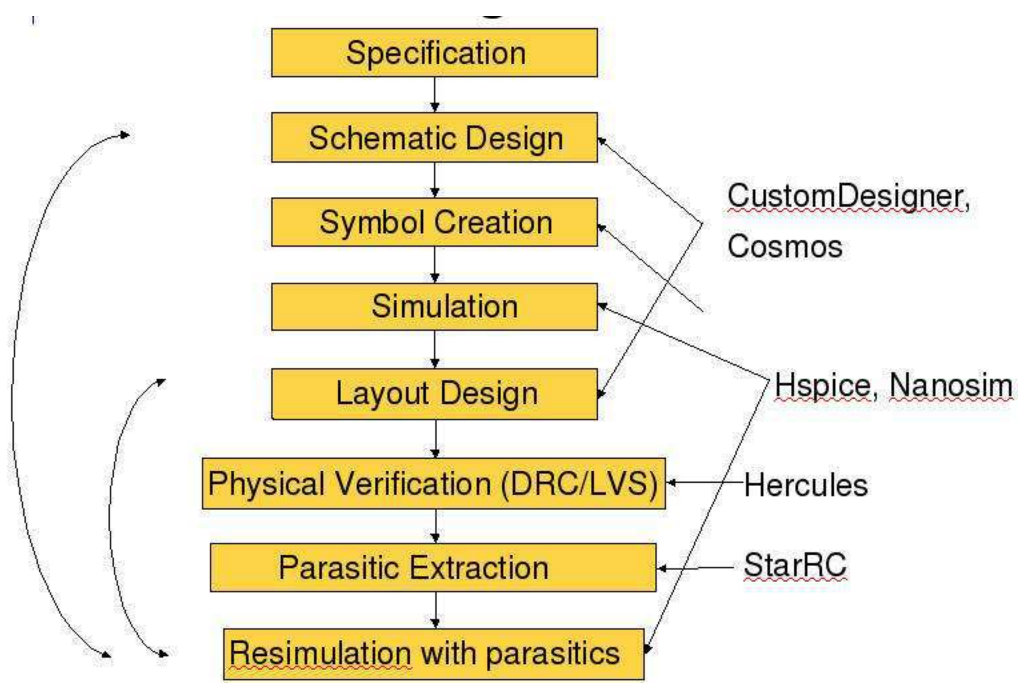


Figure I.1: Custom design flow

Part 1: Setting Up Your Workspace

The first step is to login. Please refer to the login tutorial if you are having trouble logging in or running the following commands. If you are using a Linux machine not connected to hafez.sfsu.edu try using the following command to ensure you can use x- server:

ssh -l username -X hafez.sfsu.edu

To setup the library we will use the following command in the shell window:

cp /packages/synopsys/setup/lib.defs ./

cp /packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/display.tcl ./

(Notice that there is a space between **cp** and **/packages** for the second copy command above, and also the space before **./**)

Notice that the above commands need to be run once after your first login. You do not need to rerun this command for future logins. Running the first copy command more than once can overwrite saved directories resulting in lost files later on.

To setup all the software we will use the following commands in the shell window. These commands must be run every time you use the Synopsys software:

csh

source /packages/synopsys/setup/full_custom.csh

Notice that the csh and source commands set the environment and need to be run every time you login. To run an instance of Custom Designer simply type "**cdesigner &**". Your command window should look like the one shown in Fig.1.

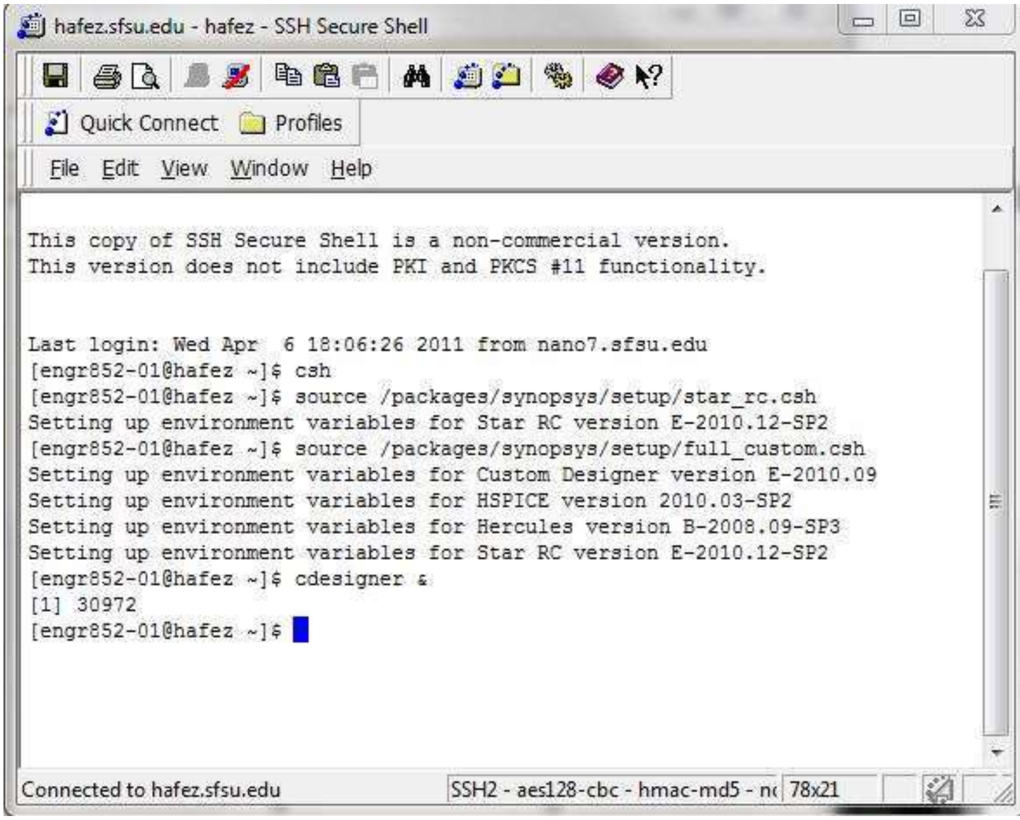


Figure 1: Shell commands

Custom Designer Console should open up (Fig.2).

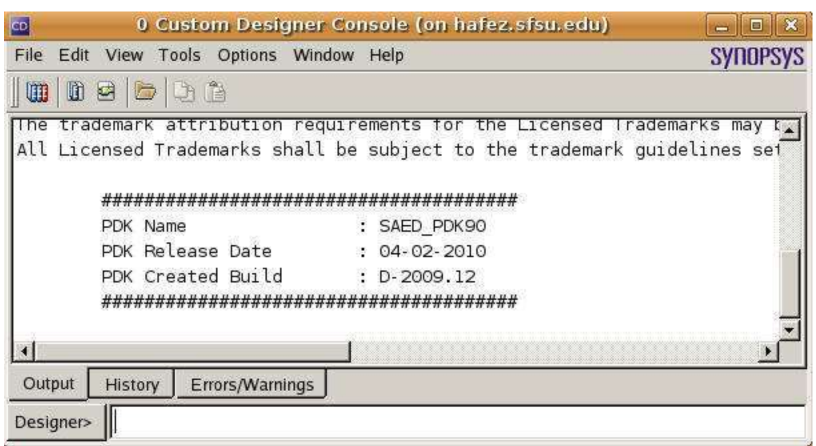


Figure 2: Custom Designer Console

Use **File → New→ Library** to create a new library. A window will appear. Several lines in the window are to be filled out to create a new library and several are not which are inactive and are filled by default. Fill out the name field with a library name and fill out the directory field for a place to save it. For the "Import File" field, click the dot so you can edit the text field and copy and paste this file directory in the field:

/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/techfiles/saed90nm_1p9m_cd.tf

After you have filled out the name, directory, and import file, click **OK**. See Fig.3.

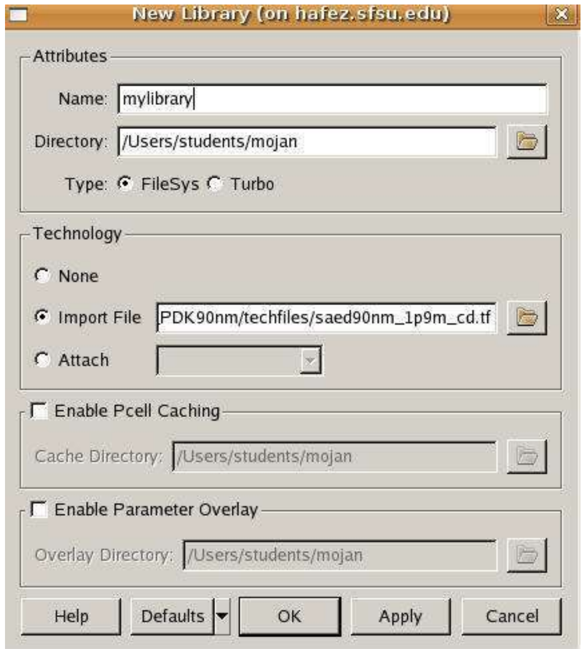


Figure 3: New Library

Part 2: Creating a Cell View

Use **File→ New→ CellView** to create a new Cell View under the library that you create in part 1 (In this case called mylibrary). Enter a name for "Cell Name" and choose schematic for "View Name". For the "Editor" choose SE-schematic (see Fig.4). Click **OK**.

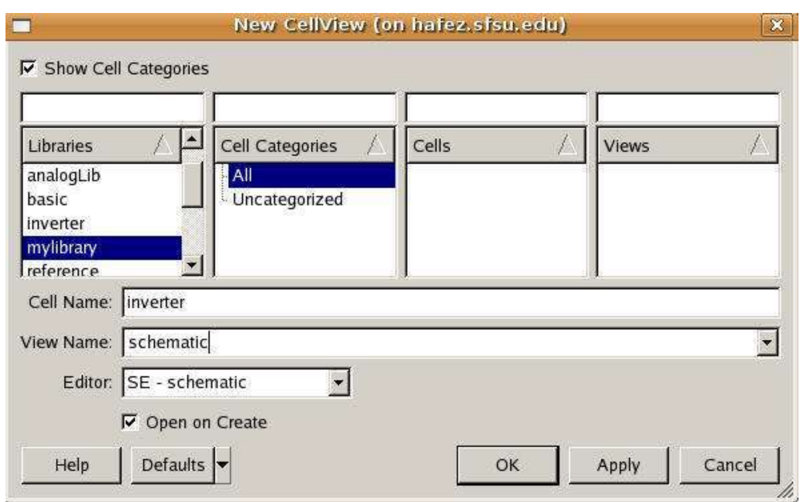


Figure 4: New Cell View

After you click Ok new window will open up called schematic editor (Fig.5)

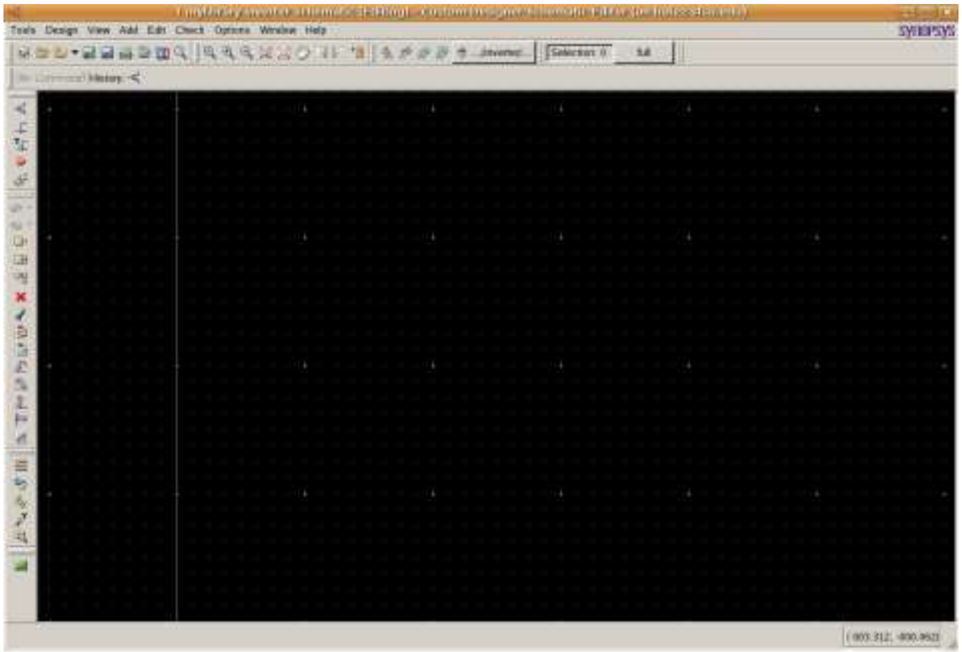


Figure 5: Schematic Editor Window

In here we need to make sure that we are in right library.
From your **console window** select **Tools →Technology Manager** and then make sure the attachment of your library (in this case "mylibrary") is SAED_PDK_90. If not, please click on it and change it (Fig.6).

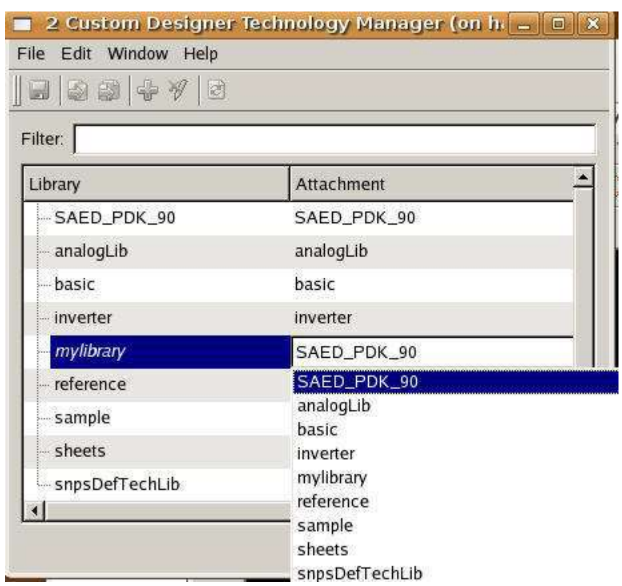


Figure 6: Technology Manager

Now go ahead and click on **Add → Instance** or simply click on this icon  in the schematic window. Select "SAED_PDK_90" for the library and select pmos4t and nmos4t for the cell while placing their respective parts on the schematic. For pmos4t width, assign 0.5um and for nmos4t width, assign 0.25um (Fig.7). You can also modify these properties later using the property editor by going to **Edit →Properties → Property Editor** and selecting the component you want to modify in schematic view.

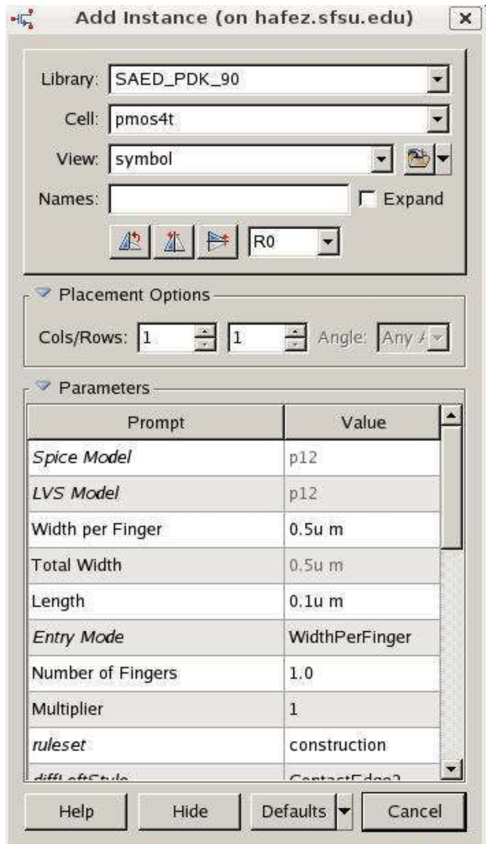


Figure 7: Add parts

After placing the pmos and nmos transistors, the schematic should look like figure 8 below.

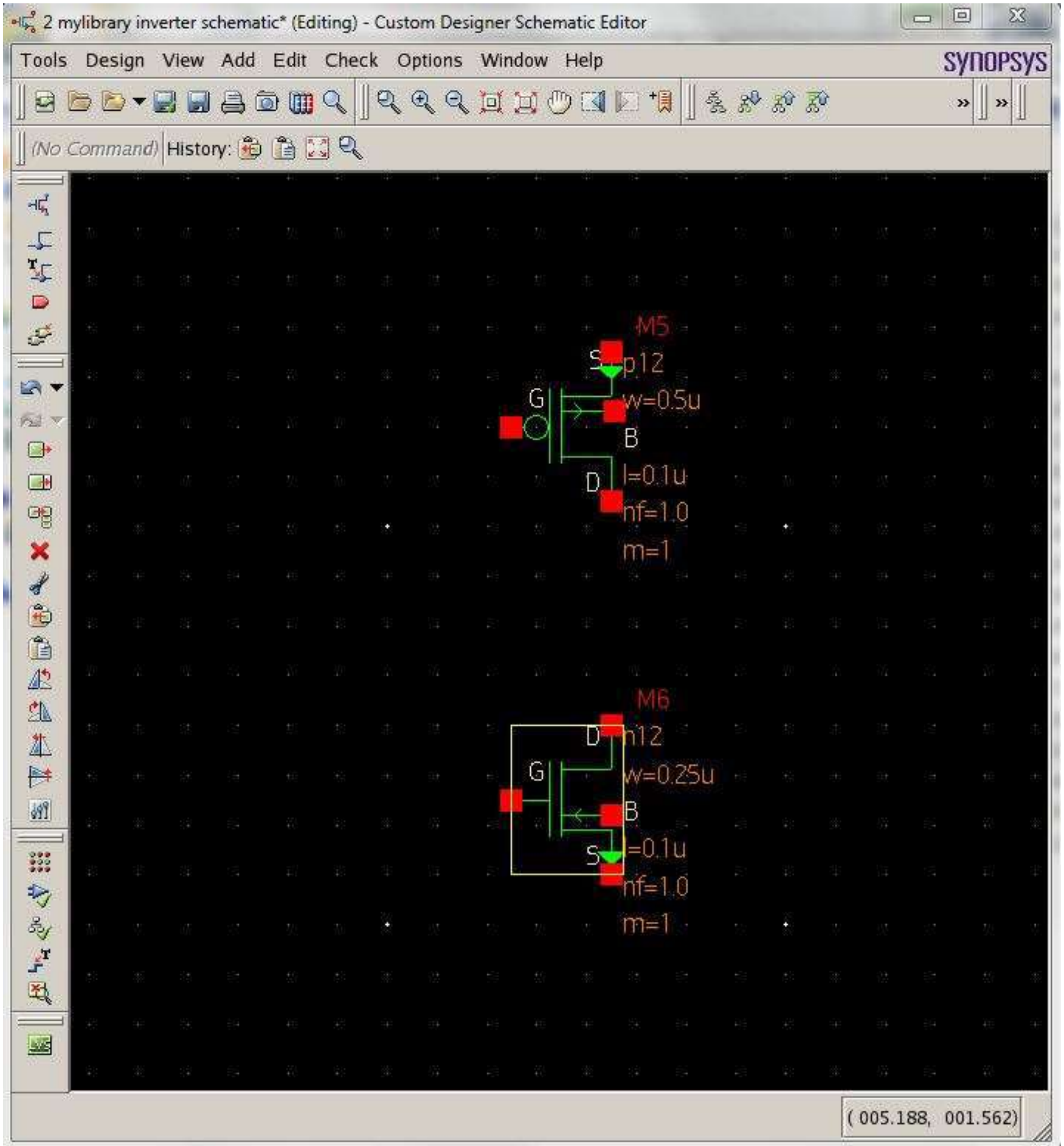


Figure 8: PMOS and NMOS

Next add wires to the schematic, click and draw wires to the circuit using the mouse pointer, see figure 9 for wire connections. To deselect wire adding, press ESC. Now you need to give names to your wires. Please click on . Then you will see a wire name space on top to enter wire names. See Fig.10 for wire labeling.

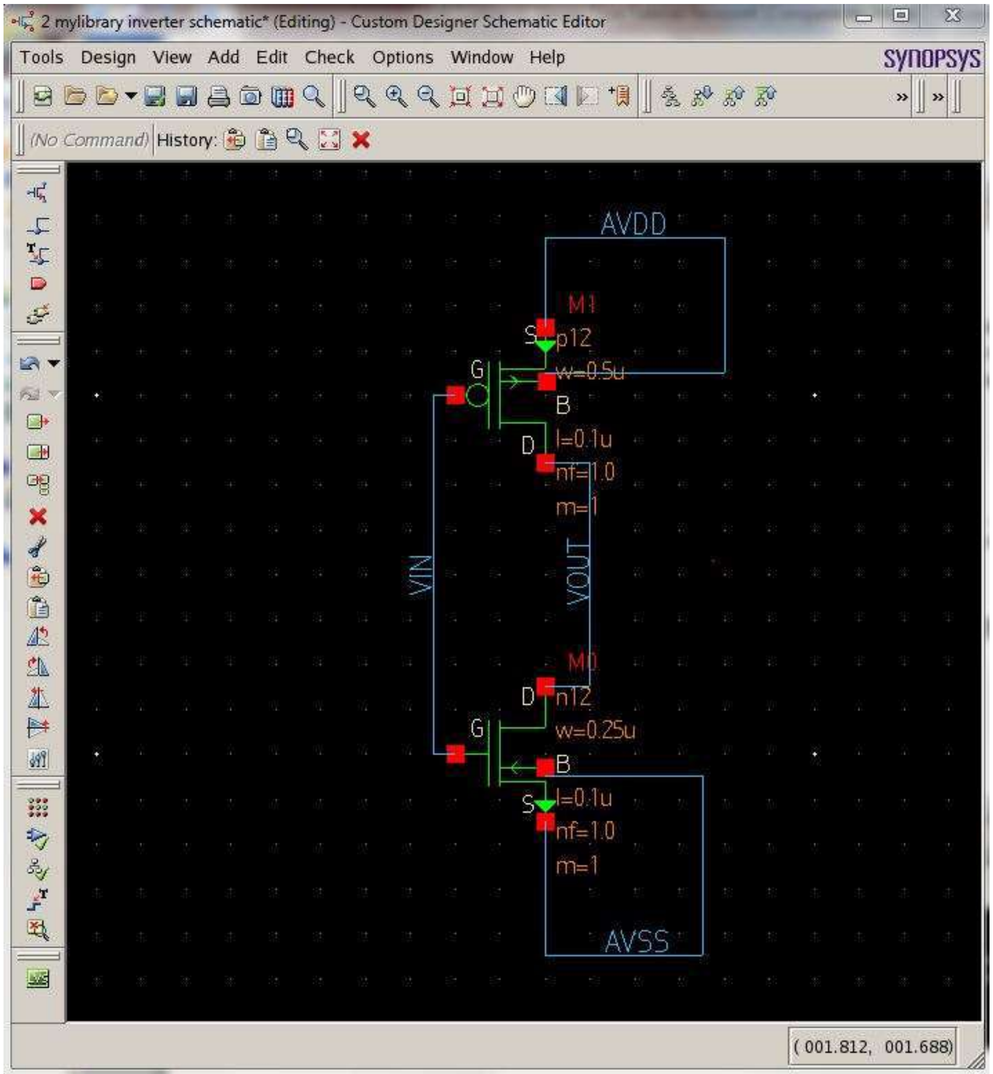


Figure 9: Wire Connections

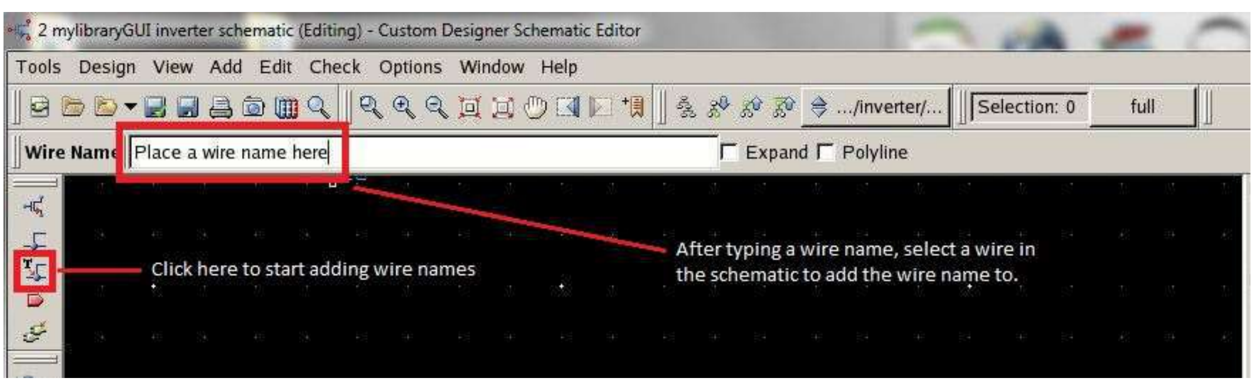


Figure 10: Wire Names

Adding Pins

To add pins to the schematic, go to **Add → Pin** to add pins for input (**VIN, AVDD, AVSS**) and output (**VOUT**) for your schematic. You can type in a name for the pin and select whether the pin is an input or output port. Then place the pin on a wire or if the wire in schematic view already has a name you can click on the wire and the pin will get the name of the wire. Note that the pin names in schematic view should match the label names in layout view (**AVDD, AVSS, VIN, VOUT**, etc) for future reference. See figure 11 and figure 12 for reference on how to add pins. Afterwards your circuit should look similar to figure 13 below after you add the pins.

As a general convention, use uppercase letters for naming pins instead of lowercase letters.

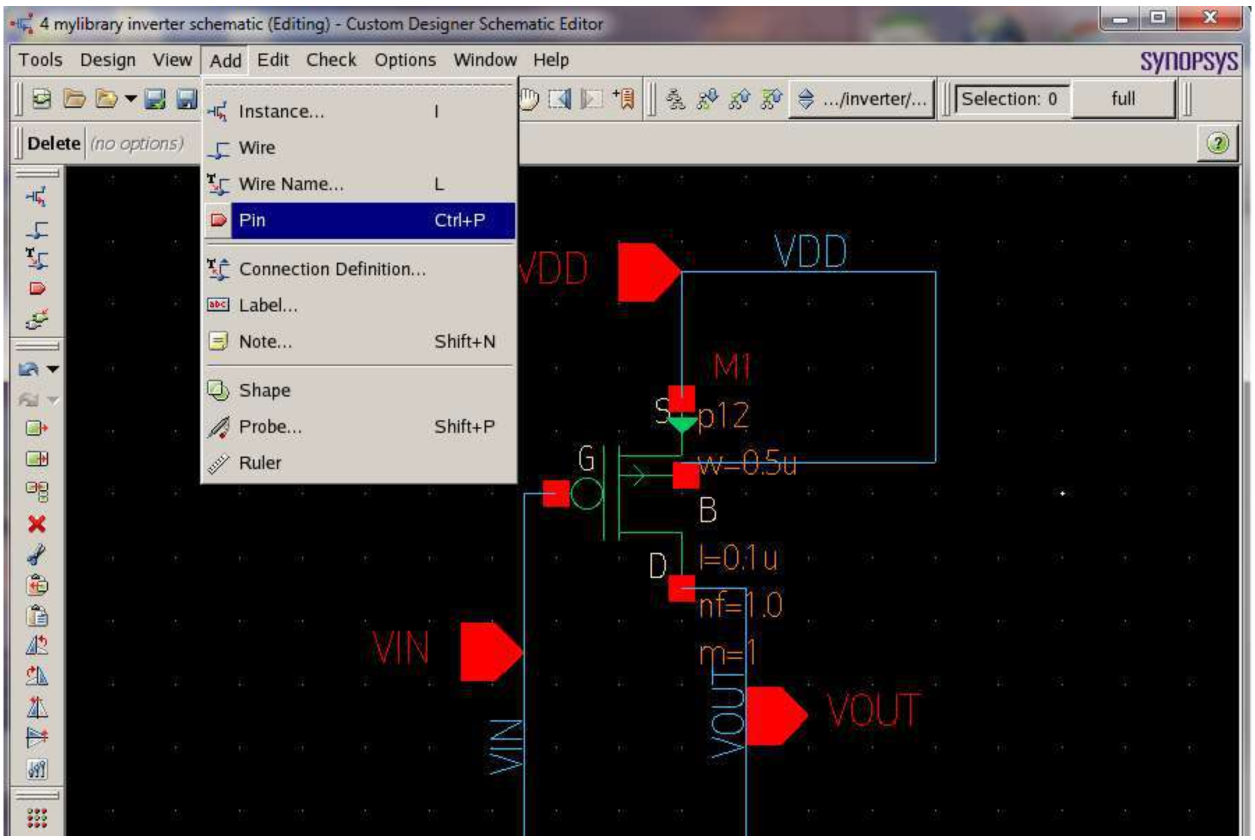


Figure 11: Adding Pins

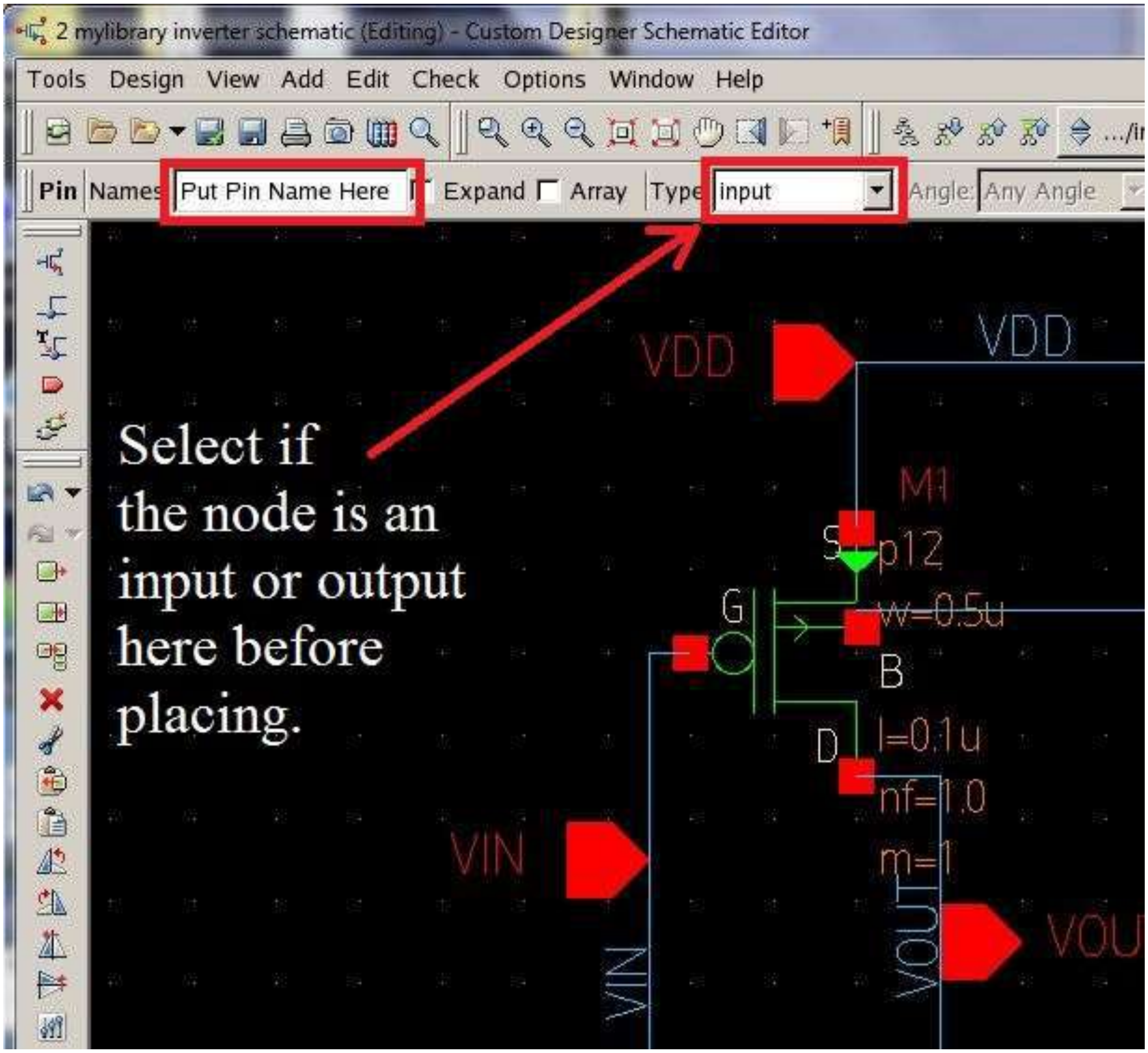


Figure 12: Adding Pins

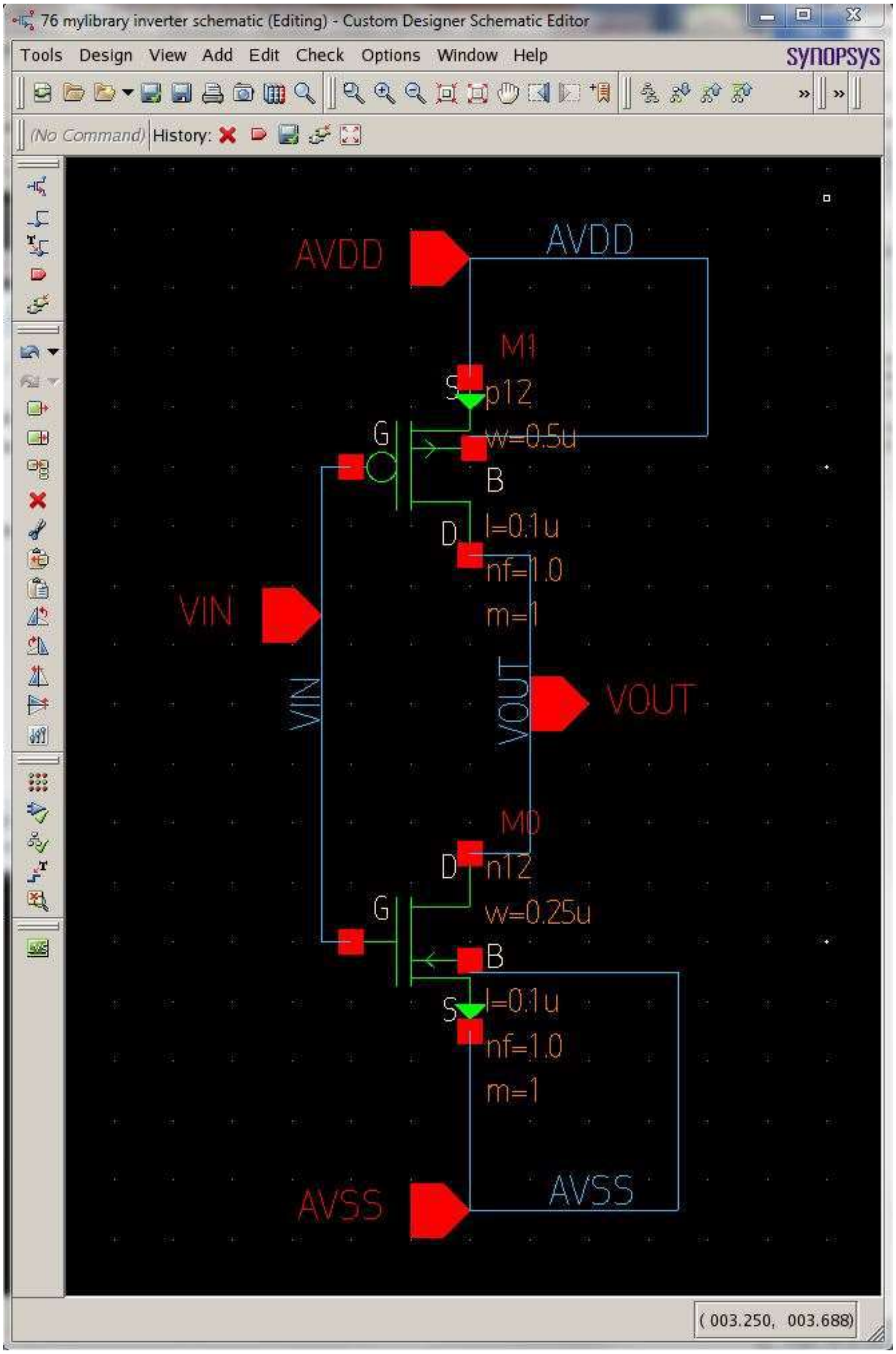


Figure 13: Schematic with Pins

Save your schematic by clicking  or go to **Design → Save**.

Now we want to create a symbol for the inverter schematic to use for future designs instead of redrawing it every time. To create a symbol for the inverter, go to **Design →New CellView → From CellView**. Make sure library and cell names match and click **OK**. See figure 14 below for reference.

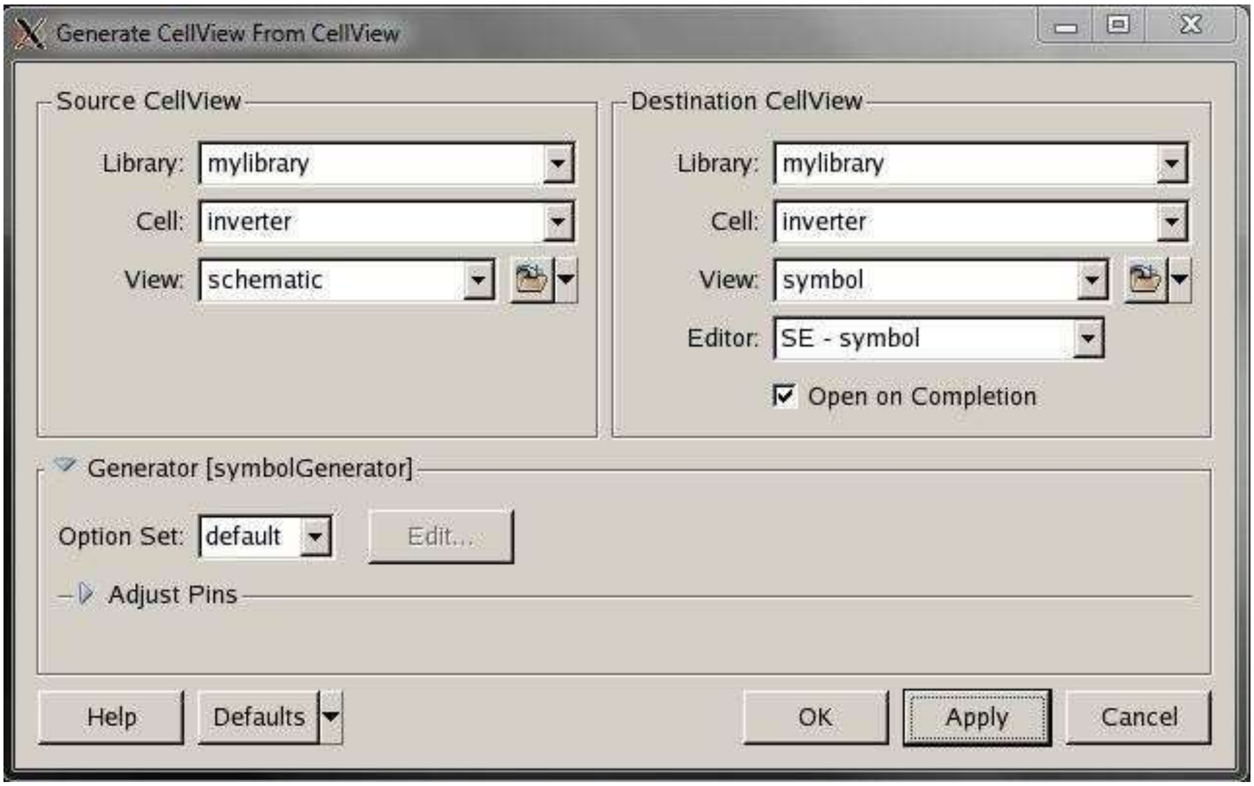


Figure 14: Generate CellView Window

Now we have a transistor level model of an inverter (symbol). See figure 15 for reference.

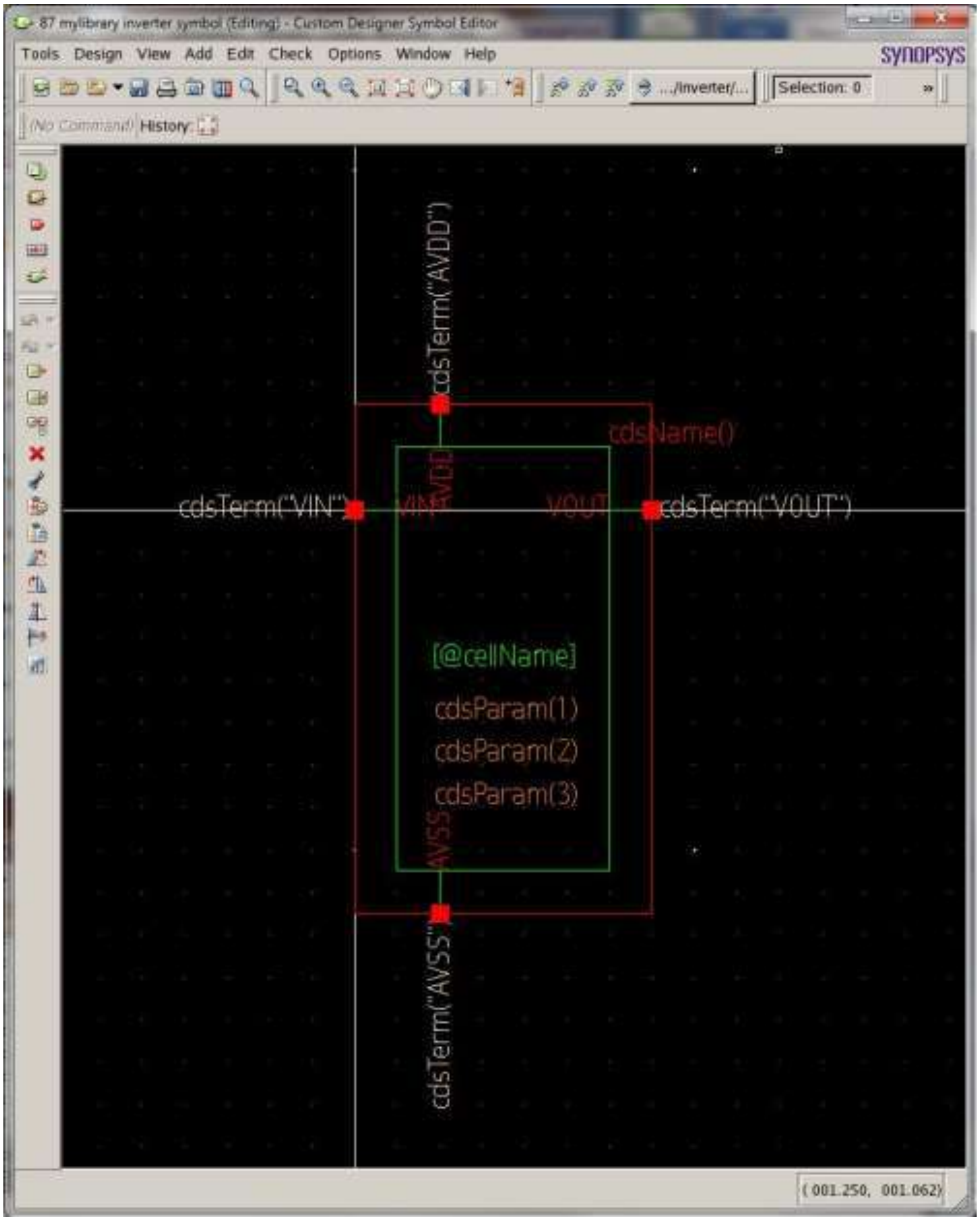


Figure 15: Inverter Symbol

You may also modify the appearance of the inverter symbol here by using the shape tools, **Add → Shape**. Now save and close the symbol window.

From the Custom Design Console, go to **File → New → CellView** and select mylibrary under the library column. Enter a new name for "Cell Name" in this case I used **inverter_testbench** and choose **schematic** for "View Name". For the editor choose SE-schematic. Click **OK**. See figure 16 for reference. Note that this schematic file will be used as a sandbox for circuits using the inverter symbol we created earlier.

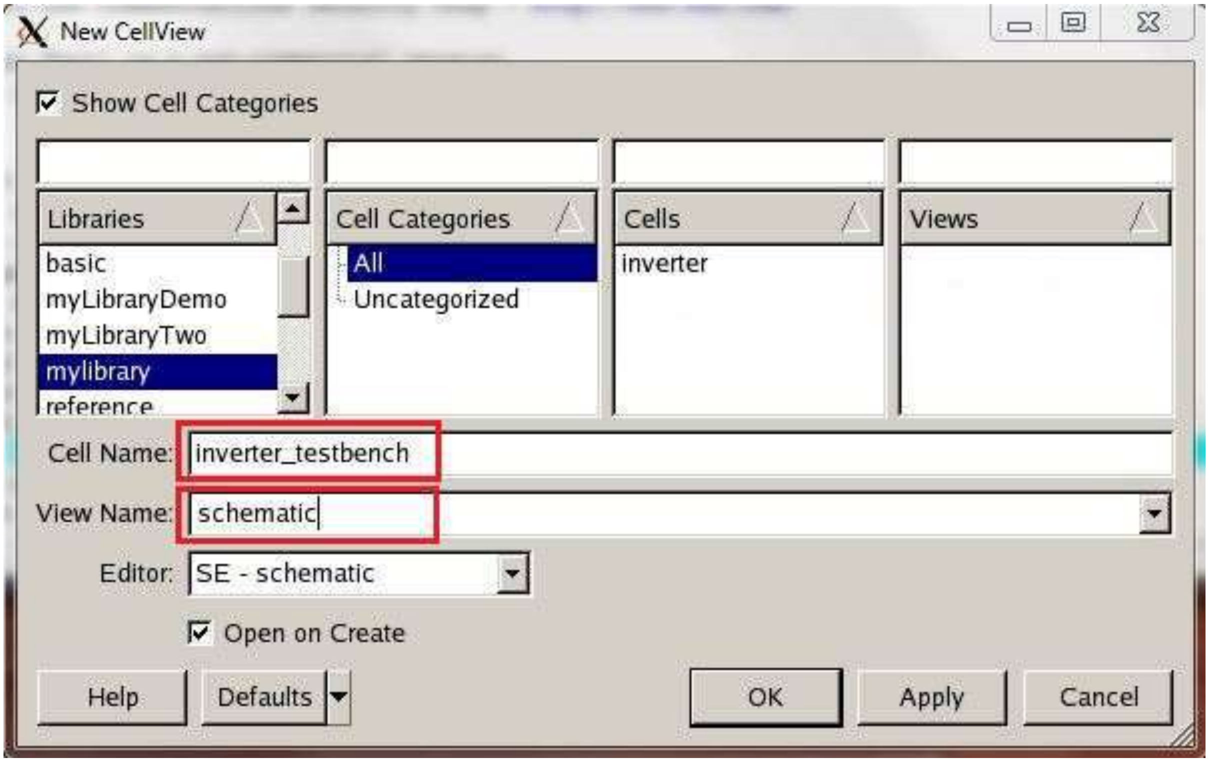


Figure 16: Creating a New Cell

Now that you have a new schematic window, go to **Add → Instance**. In the add instance window, select “mylibrary” as the library, “inverter” as the cell, and “symbol” for the view to select the inverter you just made and place it on the schematic. See figure 17 for reference.

Also in the add instance window, select “analogLib” for the library and choose: vsource, vpulse and gnd for the cell while placing the part for each selection on the schematic.

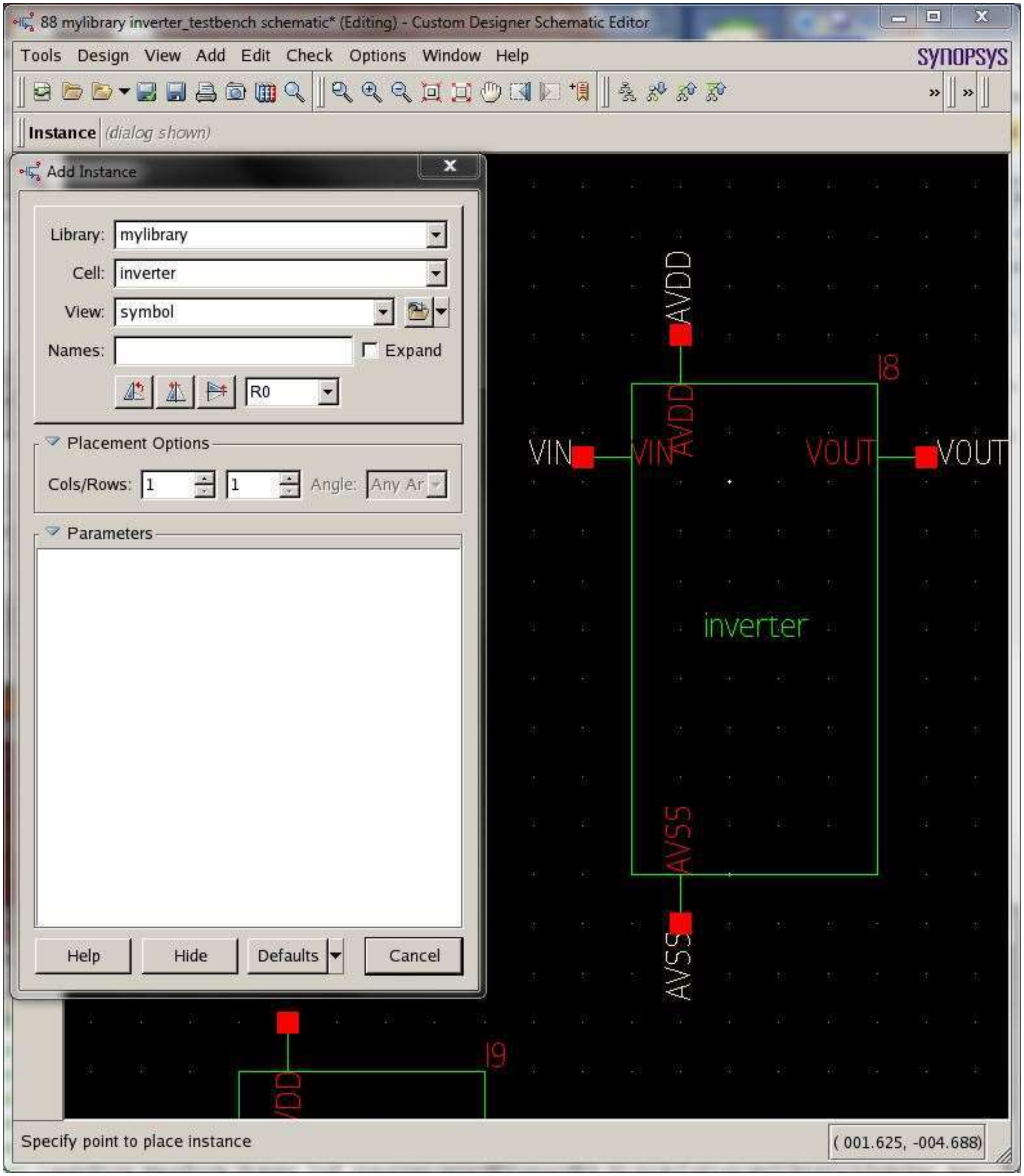


Figure 17: Drawing an Instance of an Inverter

Now add wires to the circuit using **Add → Wire** and use **Add →Pin** to add an output pin on the VOUT signal of the inverter so the schematic looks like figure 18 below. Don't worry if your values for vpulse and vsource don't match up with figure 18 since we will be modifying them next.

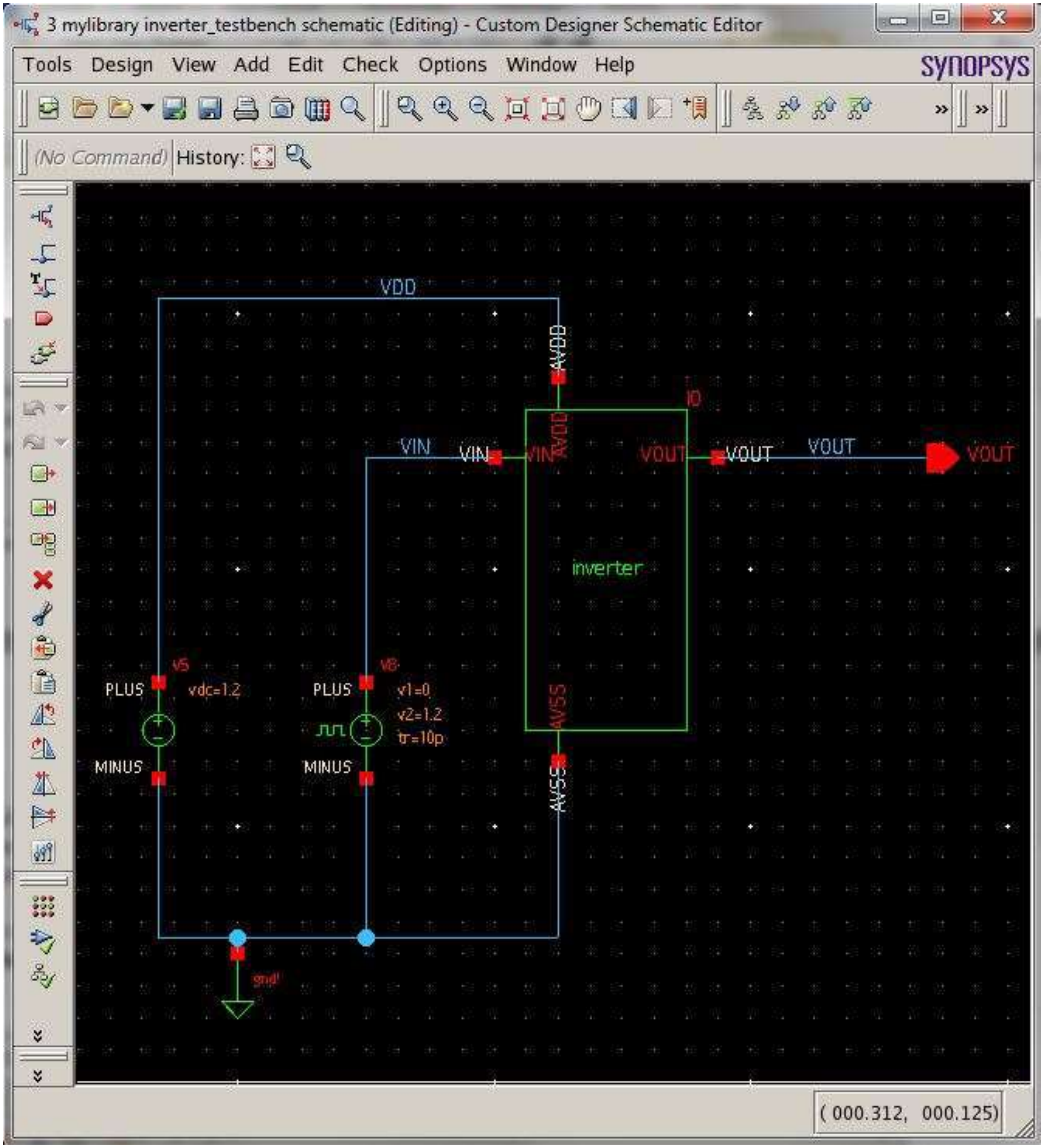


Figure 18: Test Circuit using Inverter Symbol

By clicking on  and selecting a component in schematic view, you can edit a component's property. You can also use **Edit →Properties → Property Editor** to edit the properties of the parts. Select vpulse and modify its properties as shown in Fig.19. Select vsource and modify its properties as shown in Fig.19.

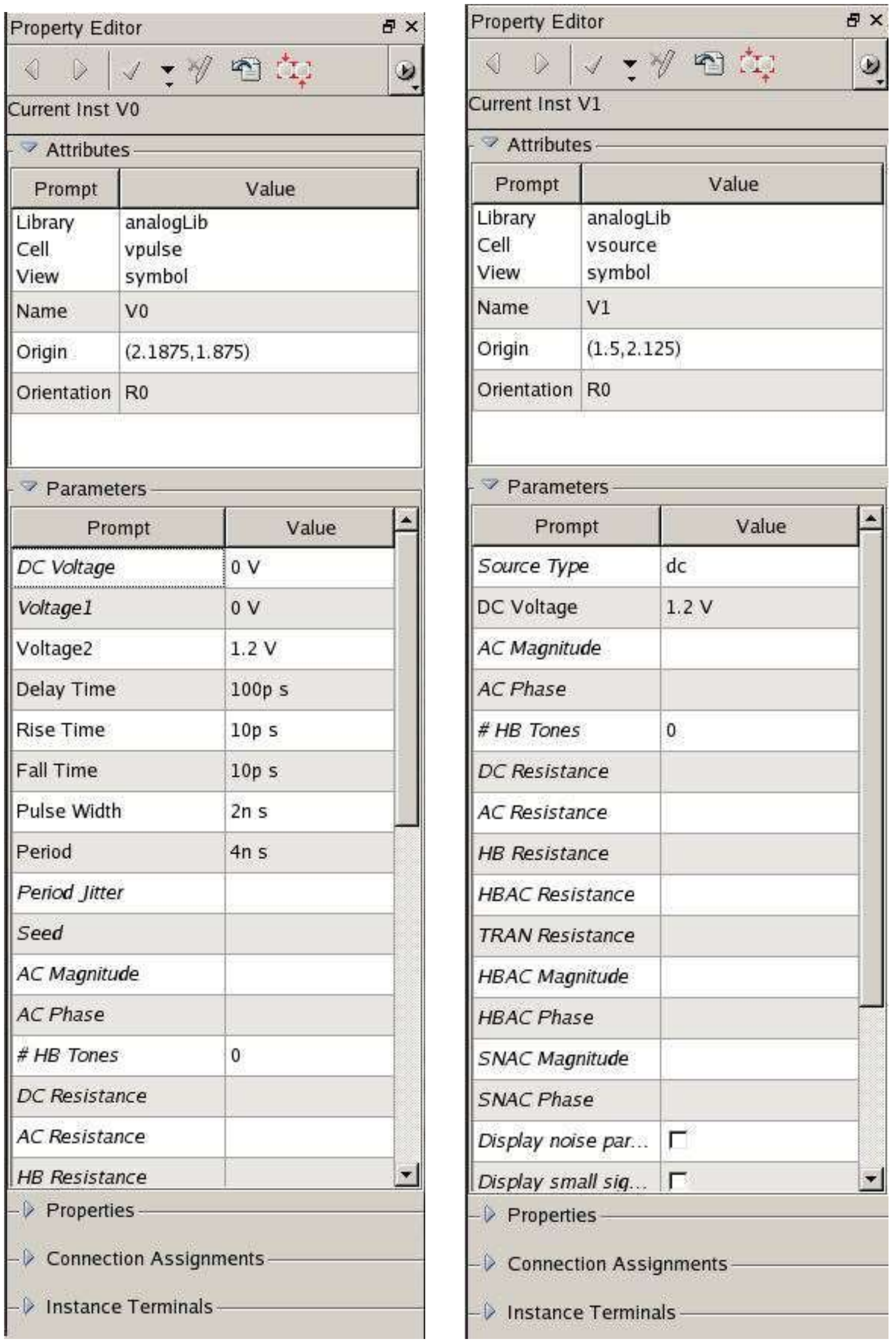


Figure 19: Edit property

Your final schematic should look like figure 18 above with the applied property edits for vpulse and vsource. The circuit is now ready for simulation. Also don't forget to save your schematic by clicking  or go to **Design → Save**.

Part 3: Simulation and Analysis Environment (SAE)

In SAE, we will run a DC sweep and a transient analysis.

To run SAE, go to **Tools menu →SAE** in the schematic window. This will launch the window in Fig. 20, which consists of three primary sections. Section one contains the design variables. Section Two contains the values being measures in the circuit. Section three displays the types of analysis being run. See figure 20 below for reference.

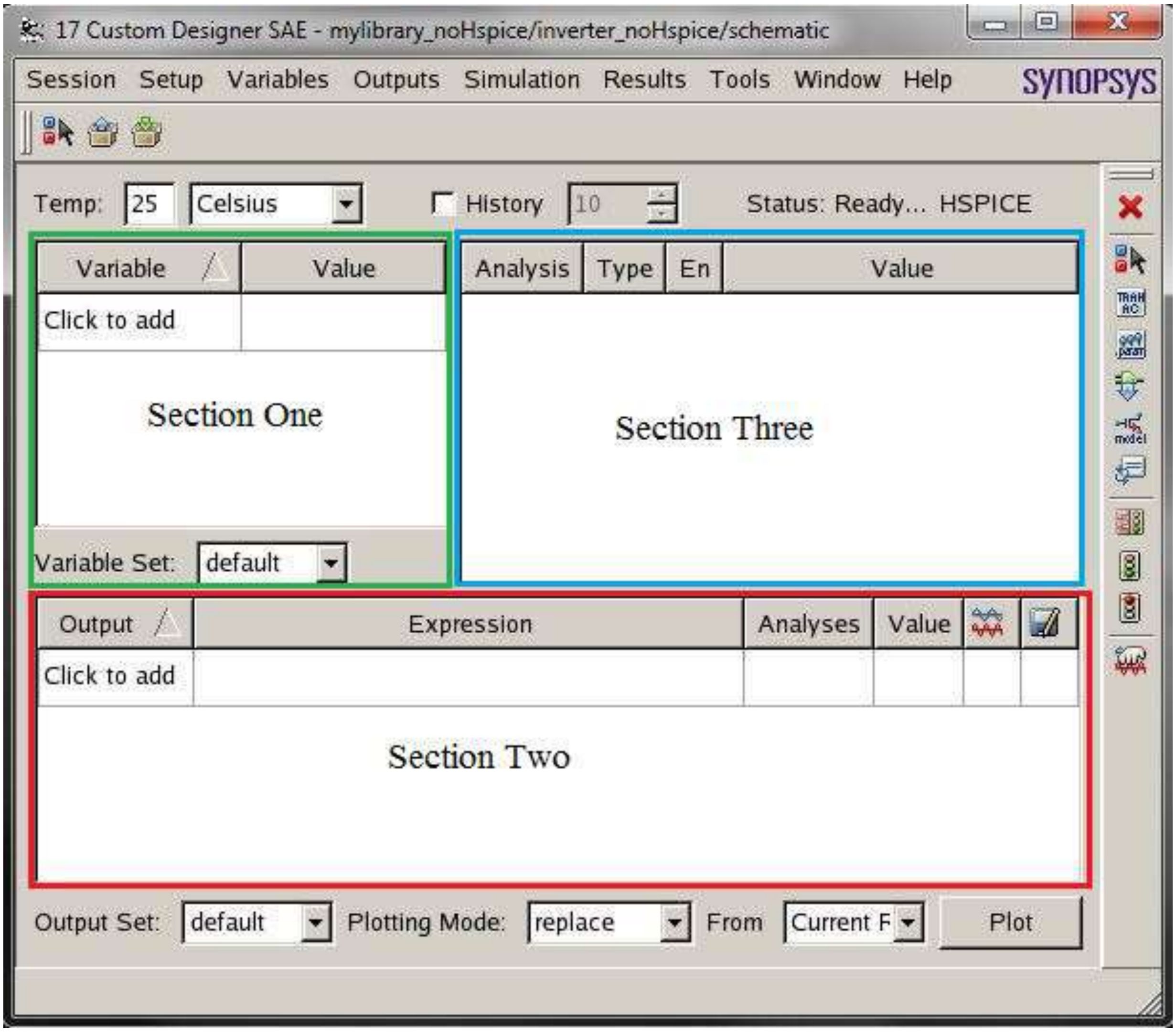


Figure: 20: SAE Main Window

To setup the model files, go to **Setup → Models Files**. A model files window will open as shown in figure 21 below. Next click on section one as shown in figure 21 and browse the directory and select the SAED90nm.lib file. You can find the file in the following directory: /packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/hspice/SAED90nm.lib

In section two of the model files window (see figure 21), select **TT_12** as your transistor type.

Click **OK** when done.

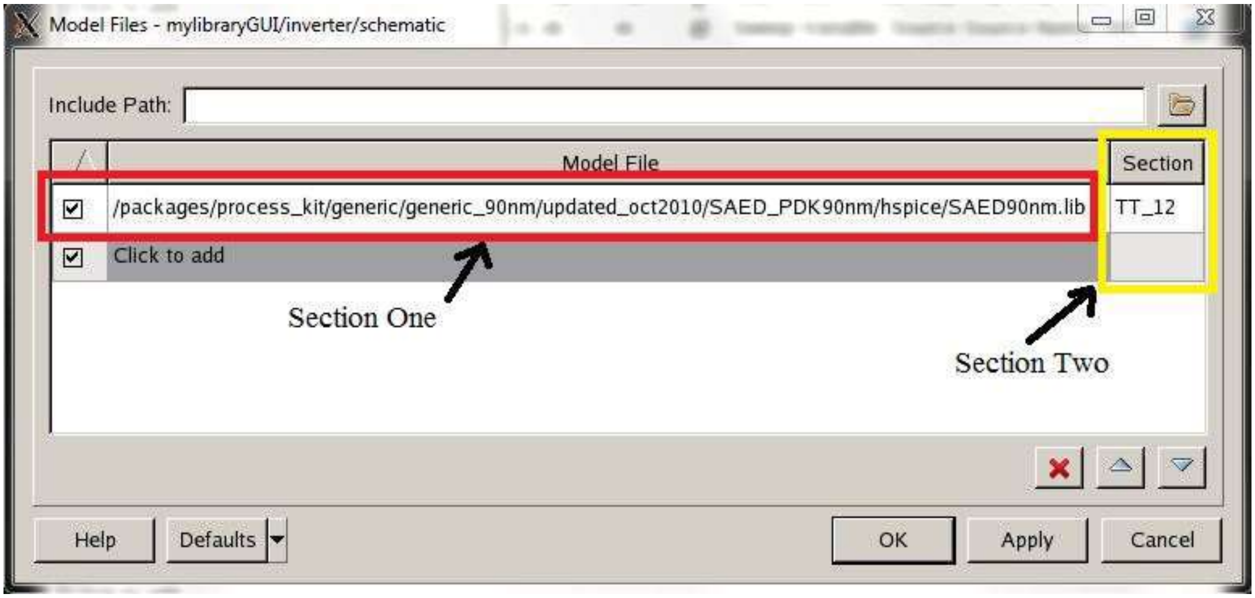


Figure 21: Models Files Window

Next to setup analysis type, go to **Setup → Analysis** and the window in figure 22 will appear. Stay on the general tab and select **tran** to setup the transient analysis type. Setup the remaining options as shown in figure 22 and click **Apply**.

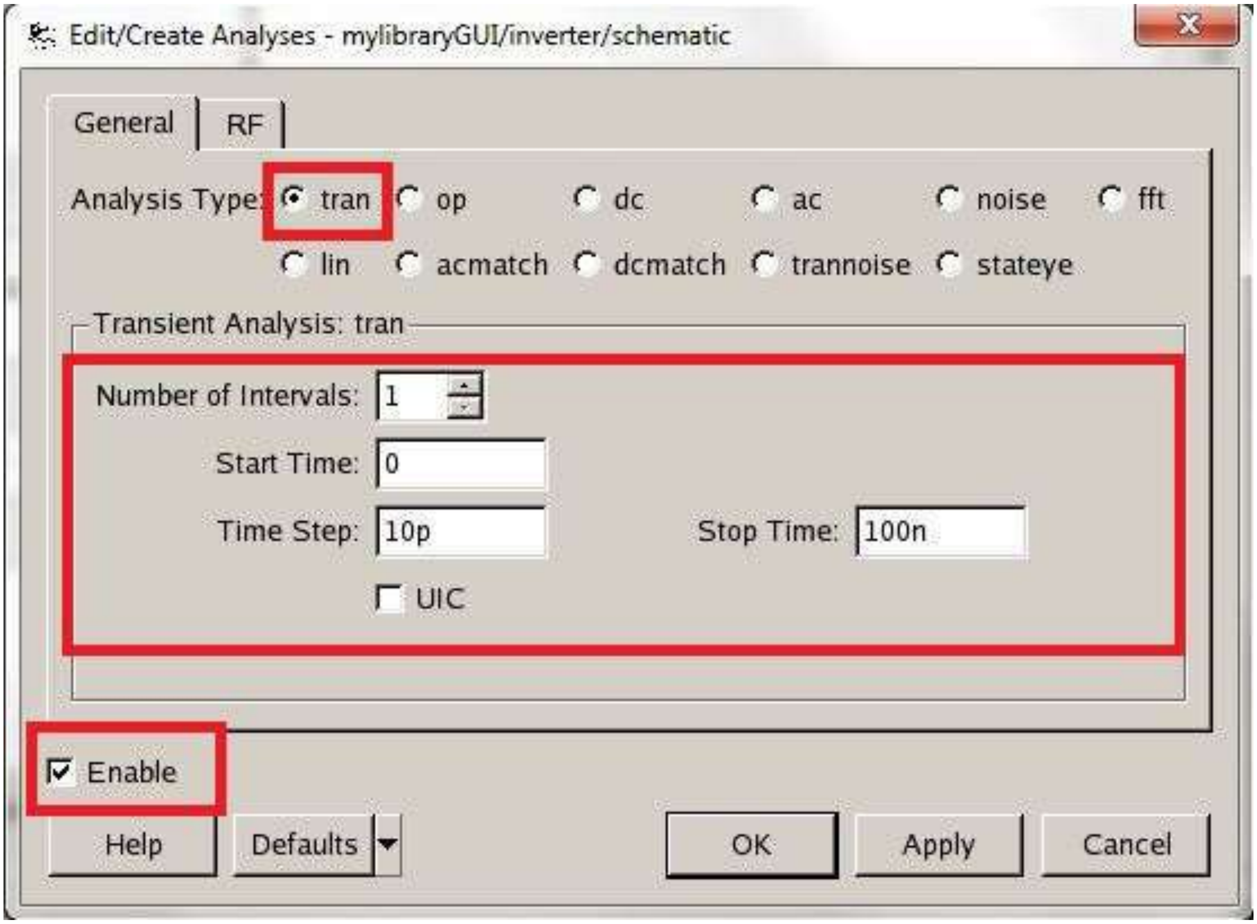


Figure 22: Transient Simulation Window

In the same window, select **dc** to setup a DC sweep. Fill out the options as shown in figure 23 below. Click **OK** when done.

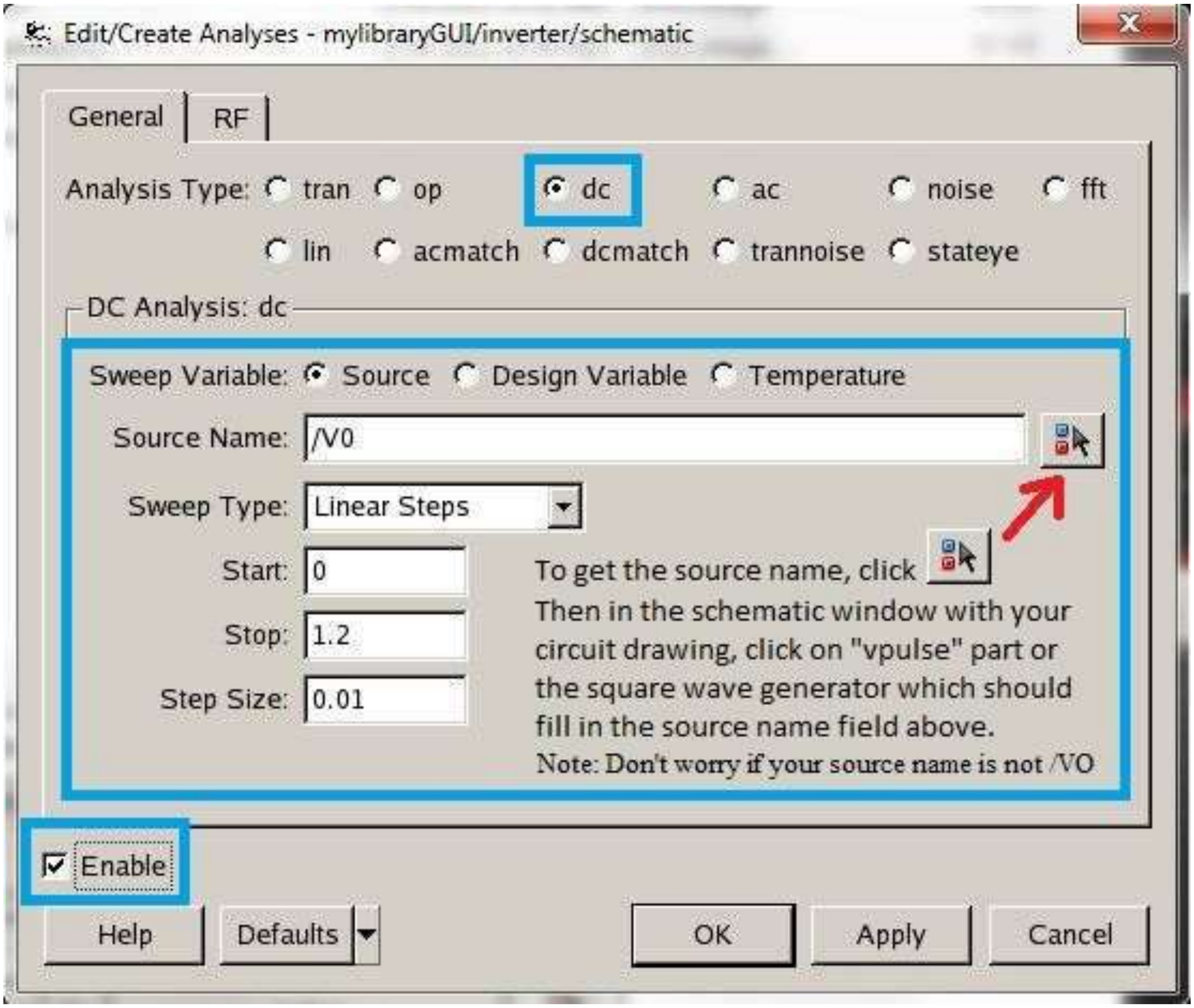


Figure 23: DC Simulation Window

The following step is to choose the desired simulations results and select the nodes in the circuit to measure. In section TWO of Fig: 20 do the following steps:

To setup the circuit output voltage:

1) Click under the output field, and write "Vout" or a name for an output variable of the inverter.

2) Click under the expression column and choose the output node from the schematic. In this case, click on  and select the wire labeled "VOUT" as shown below in figure 24 in the schematic window. You can also write an equation that uses the values of some nodes in that schematic.

3) Under analysis, select the type of analysis you want to run. In this case, select **dc** and **tran** to run the transient and DC analysis for this variable.

To setup the circuit input voltage:

1) Click under the output field in a new row, and write "Vin" or a name for an input variable of the inverter.
2) In the same row, click under the expression column and choose the input node from the schematic. In this case, click on  and select the wire labeled "VIN" as shown below in figure 24 in the schematic window.
3) Under analysis, select the type of analysis you want to run. In this case, select **dc** and **tran** to run the transient and DC analysis for this variable.

To setup the circuit source current:
1) Click under the output field in a new row, and write "isupply" or a name for a current variable.
2) In the same row, click under the expression column and choose the voltage source from the schematic. In this case, click on  and select the voltage source labeled "V5" as shown below in figure 24 in the schematic window. Notice that current is being measure rather than voltage.
3) Under analysis, select the type of analysis you want to run. In this case, select **dc** and **tran** to run the transient and DC analysis for this variable.

Afterwards, the SAE window should look something similar to figure 25 below. Note that the expression values in figure 25 may not match with your values which is fine since those are dependent on the names used in the schematic for the voltage sources and wires.

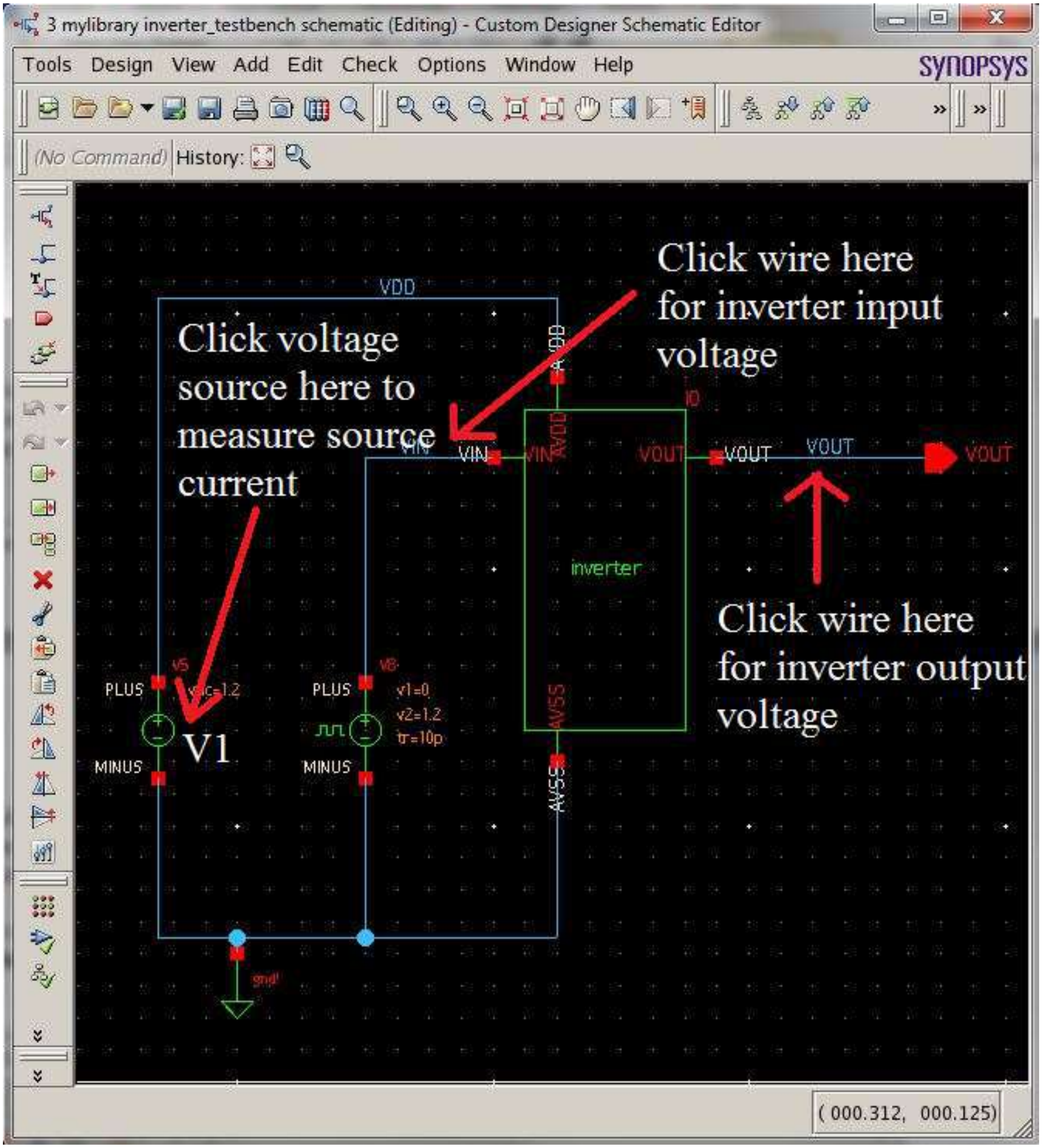


Figure 24: Clicking on the Schematic Window

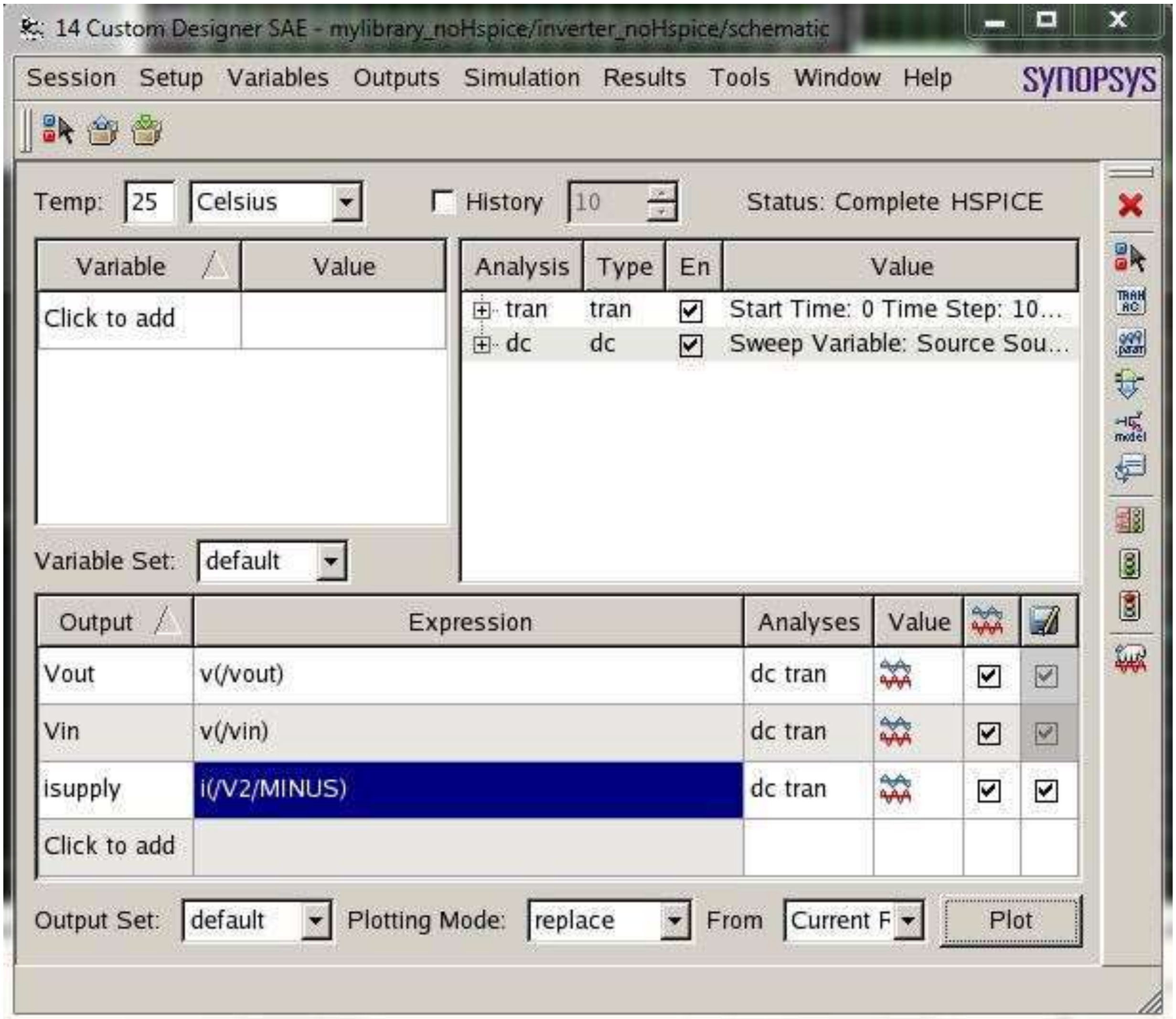


Figure 25: Updated SAE Window

Now save your simulation setups by going to **Session → Save State**. In the new window that opens, select **OpenAccess** from the three main categories at the top and give a name for the session in the name field. Click **OK** when done. See figure 26 below for reference.

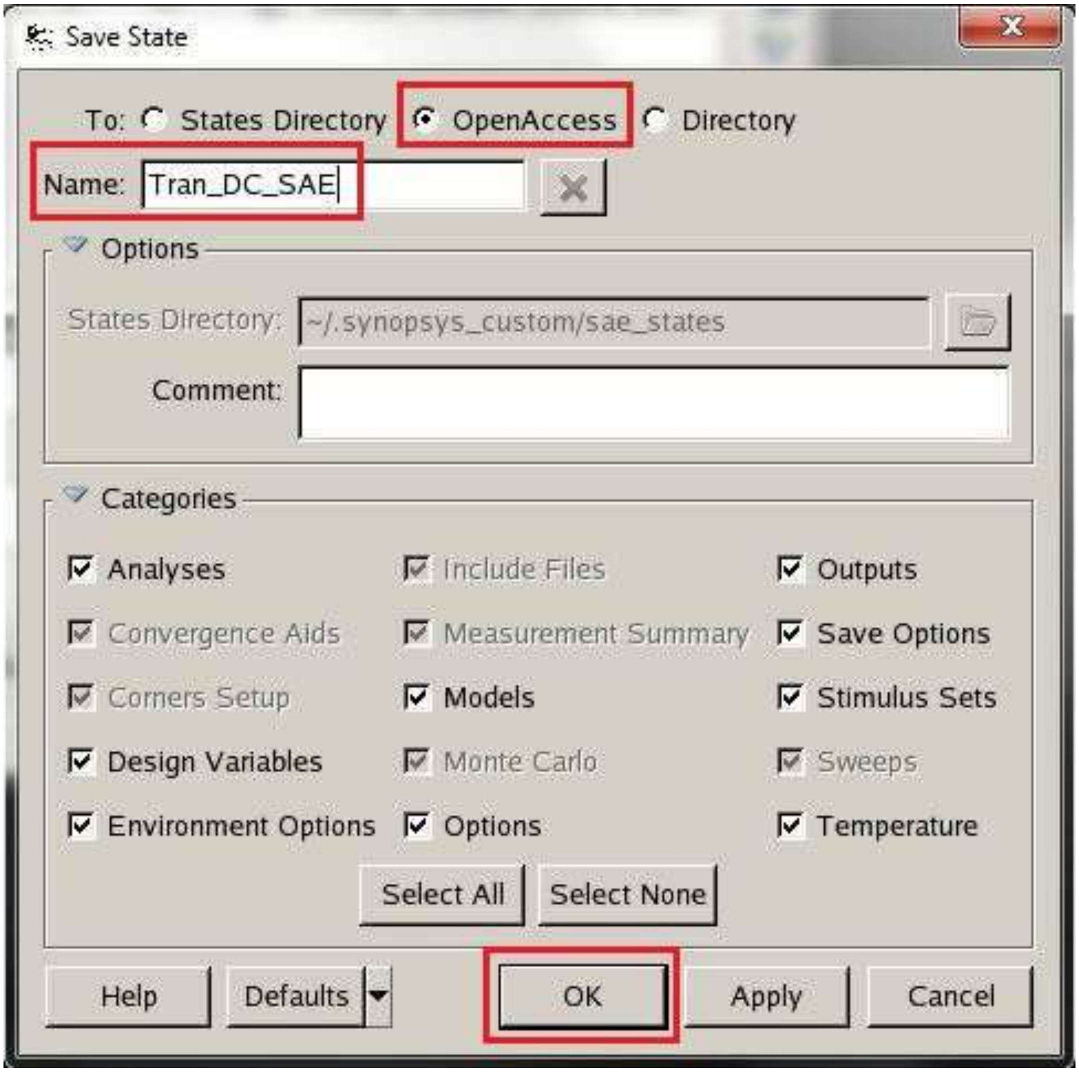


Figure 26: Save State Window

To run the simulation, go to **Simulation → Netlist and Run**. The status of simulation is reported in Custom Designer Console window. If you see "Simulation completed successfully" it means that your simulation is successfully done (Fig.27).

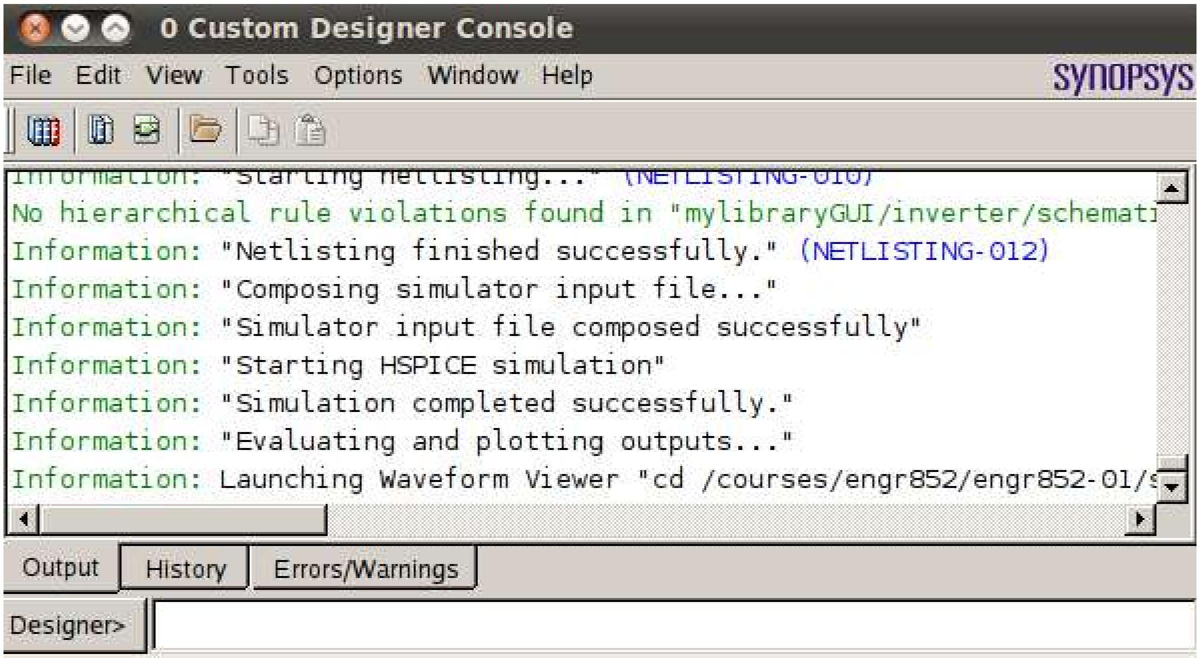


Figure 27: Custom Designer Console

After running simulation successfully, a WaveView window will open up. You can select the type of analysis you want to view by selecting the respective tab in the bottom left corner of the window.

For the transient analysis waveform, click the **tran** tab in the bottom left corner. See figure 28 for the transient analysis. For the DC sweep waveform, click the **dc** tab in the bottom left corner. See figure 29 for the dc sweep analysis

You now have run transient and dc analysis successfully.

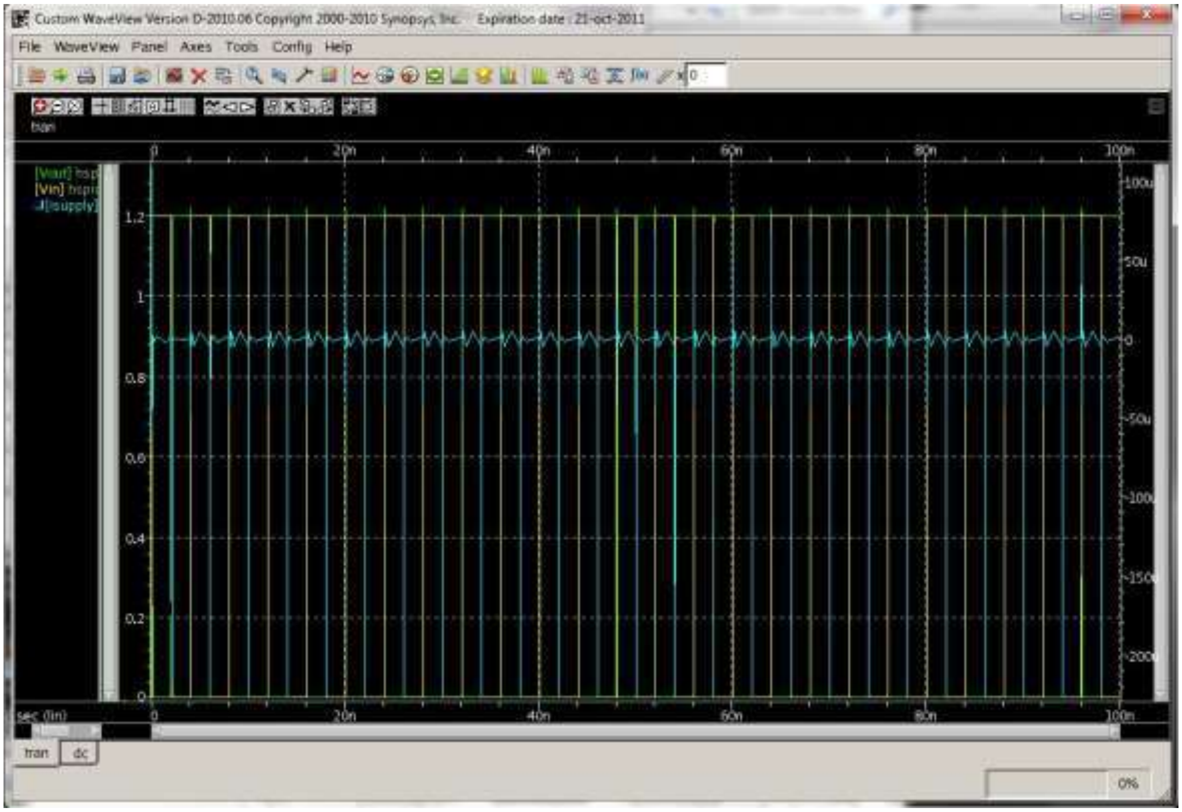


Figure 28: Transient Analysis Wave in WaveView

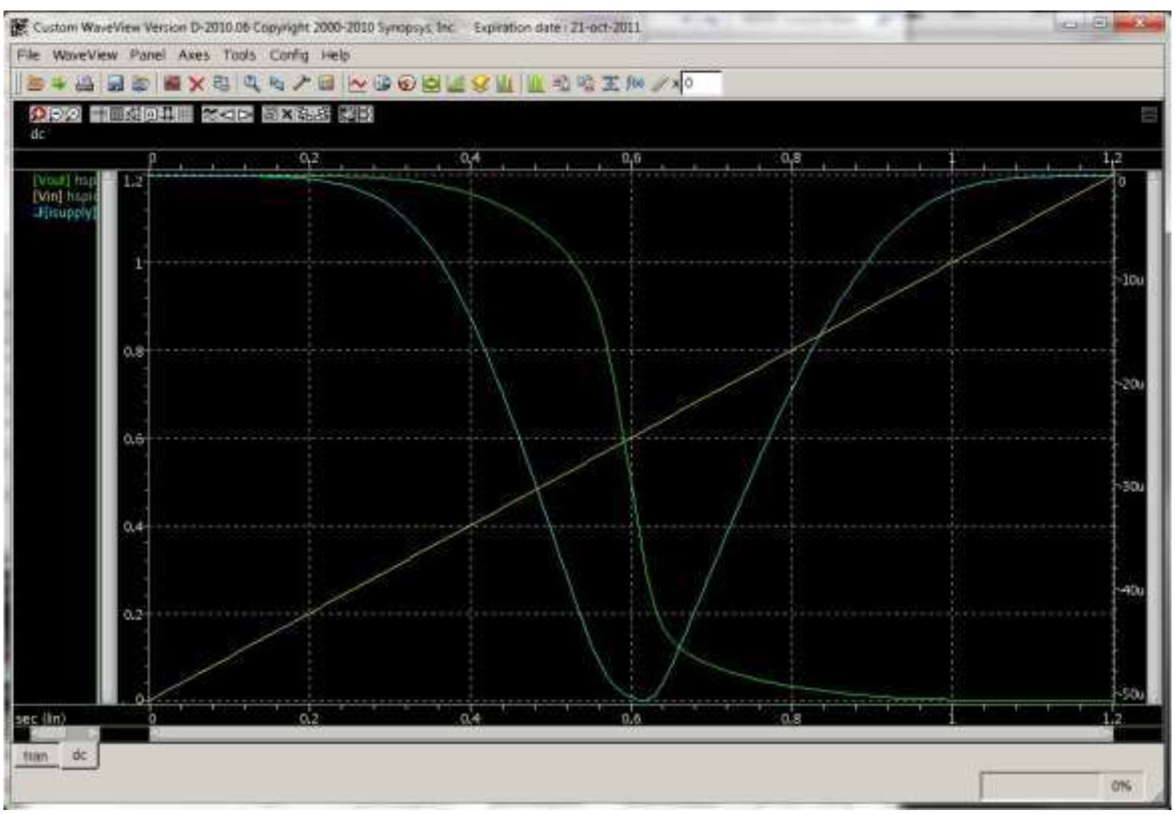


Figure 29: DC Sweep Analysis Wave in WaveView

Part 4: WaveView Measurement Tools and Features

The measurement tool in the WaveView window provides many methods for measuring the waveforms. Here is a list of several features and measurements you can do in WaveView:

Zooming In and Out:
To zoom in on your waveform, click to do a vertical or horizontal zoom by dragging a box over the waveform with the mouse. Click to unzoom. Also you can press **X** for a full unzoom of the waveform.

Grouping and Ungrouping Signals:
First off, it is handy to know how to group and ungroup waveforms in WaveView. See figure 30 for ungrouping waveforms and see figure 31 for grouping waveforms.

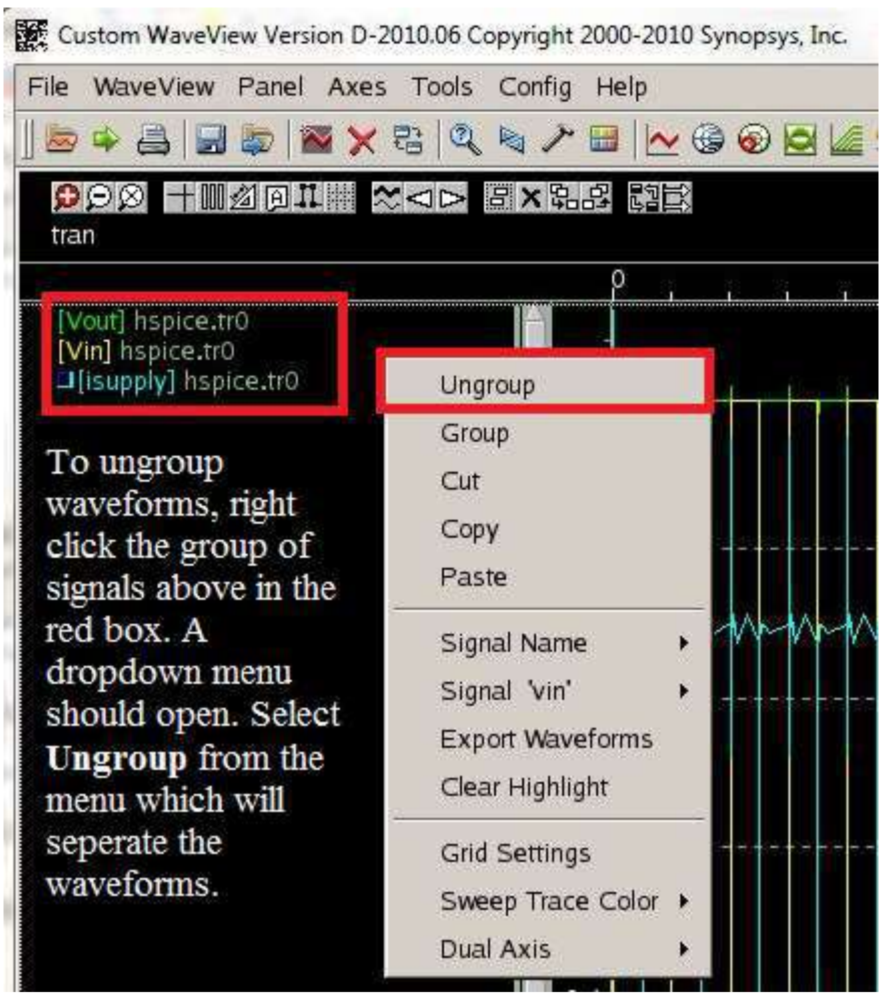


Figure 30: Ungrouping Waveforms

To group the waveforms in WaveView together, select the names of the signals to be grouped in the signals list and hold down **Ctrl** for each additional signal to add. Once you have all the signals you want to group selected, right click on the mouse and select **Group** from the dropdown menu to group the selected waveforms.

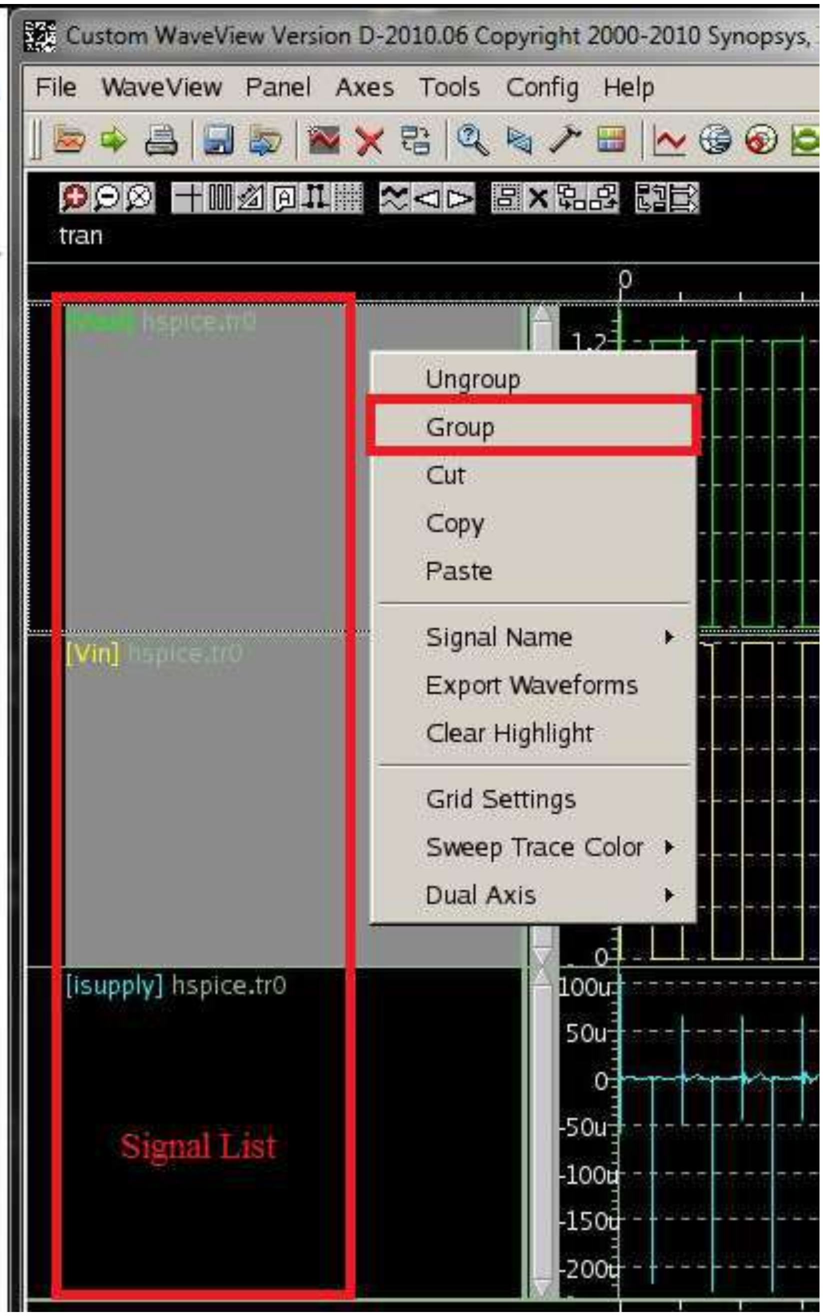
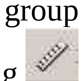


Figure 31: Grouping Waveforms

Delay Measurements of Vin and Vout at 50% to 50%:
To measure delay between the input and output signals of the inverter at 50% select the **tran** tab in the bottom left hand corner of the WaveView window. Group the Vout and Vin waveforms together so the two waves overlap each other (see figure 31 on how to group signals). Open the measurement tool by going to **Tools → Measurement…** or by clicking  in the WaveView window. Click the **All** tab in the measurement tool window and fill out the options as shown below in figure 32. Click **Ok** when done.

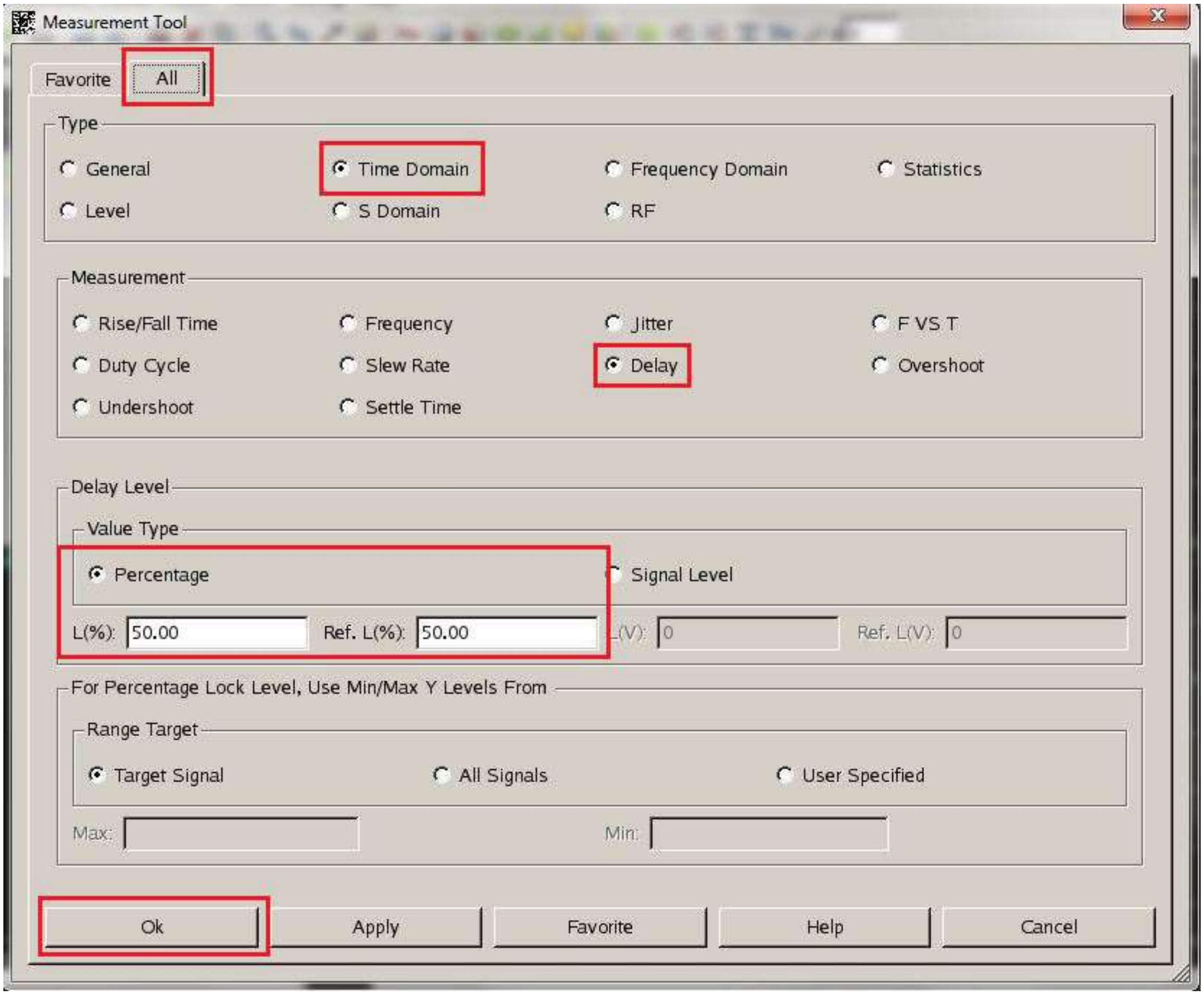


Figure 32: Delay Measurement Tool

After clicking **Ok**, a delay measurement box will appear on the waveform. Just drag the box to the waveform you want to measure (in this case the Vin and Vout overlapping waveforms) and the delay value will appear in the box. You can also drag the delay measurement box along different valid points of the waveform to get more delay values. See figure 33 below for the delay measurements box.

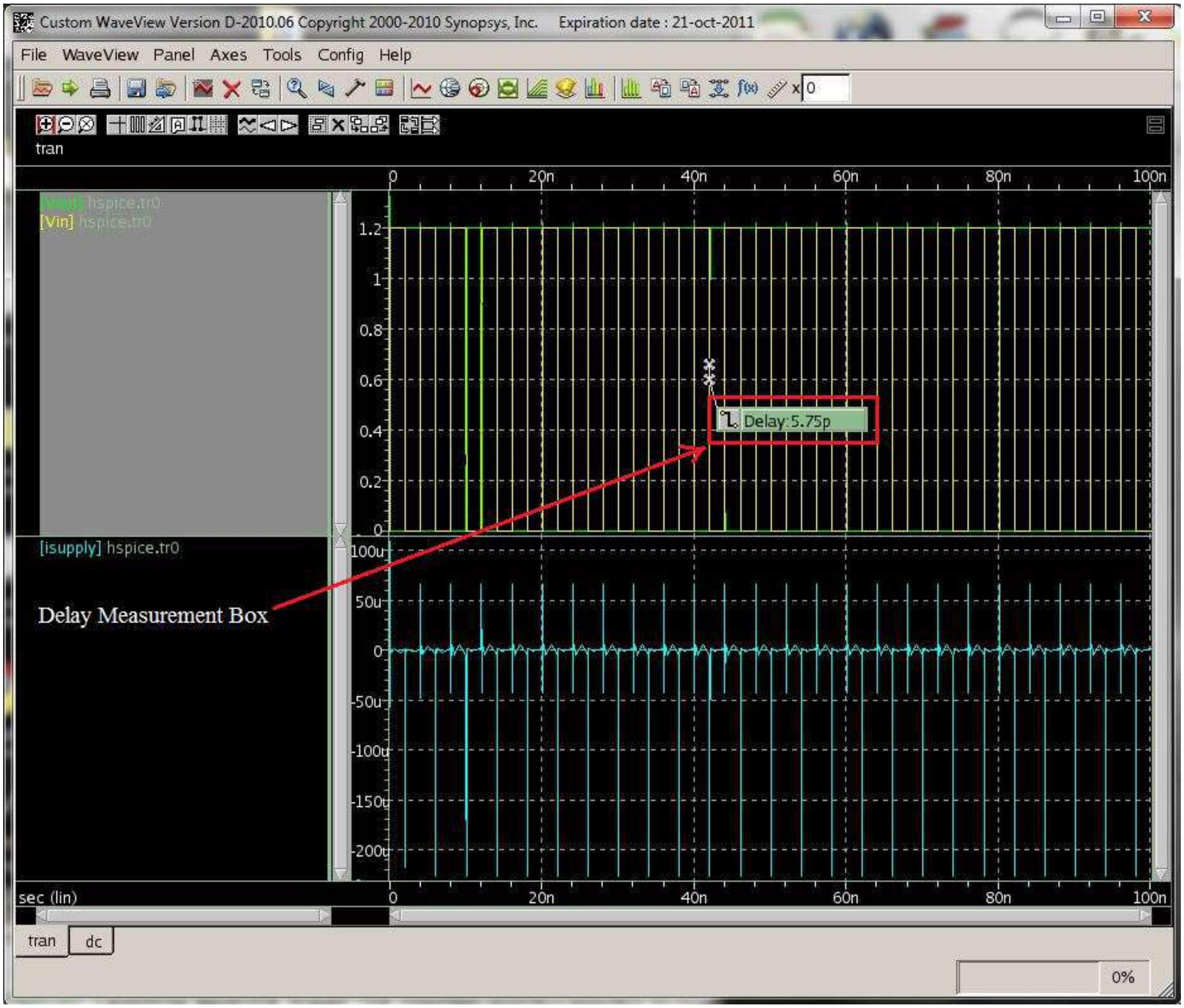


Figure 33: Delay Waveform Measurement

Rise/Fall Time Measurements at 90% and 10% for Vout:
To measure fall and rise time, select the **tran** tab in the bottom left hand corner of the WaveView window and ungroup all the waveforms as described in figure 30. Open the measurement tool by going to **Tools → Measurement…** or by clicking  in the WaveView window. Click the **All** tab in the measurement tool window and fill out the options as shown below in figure 34. Click **Ok** when done.

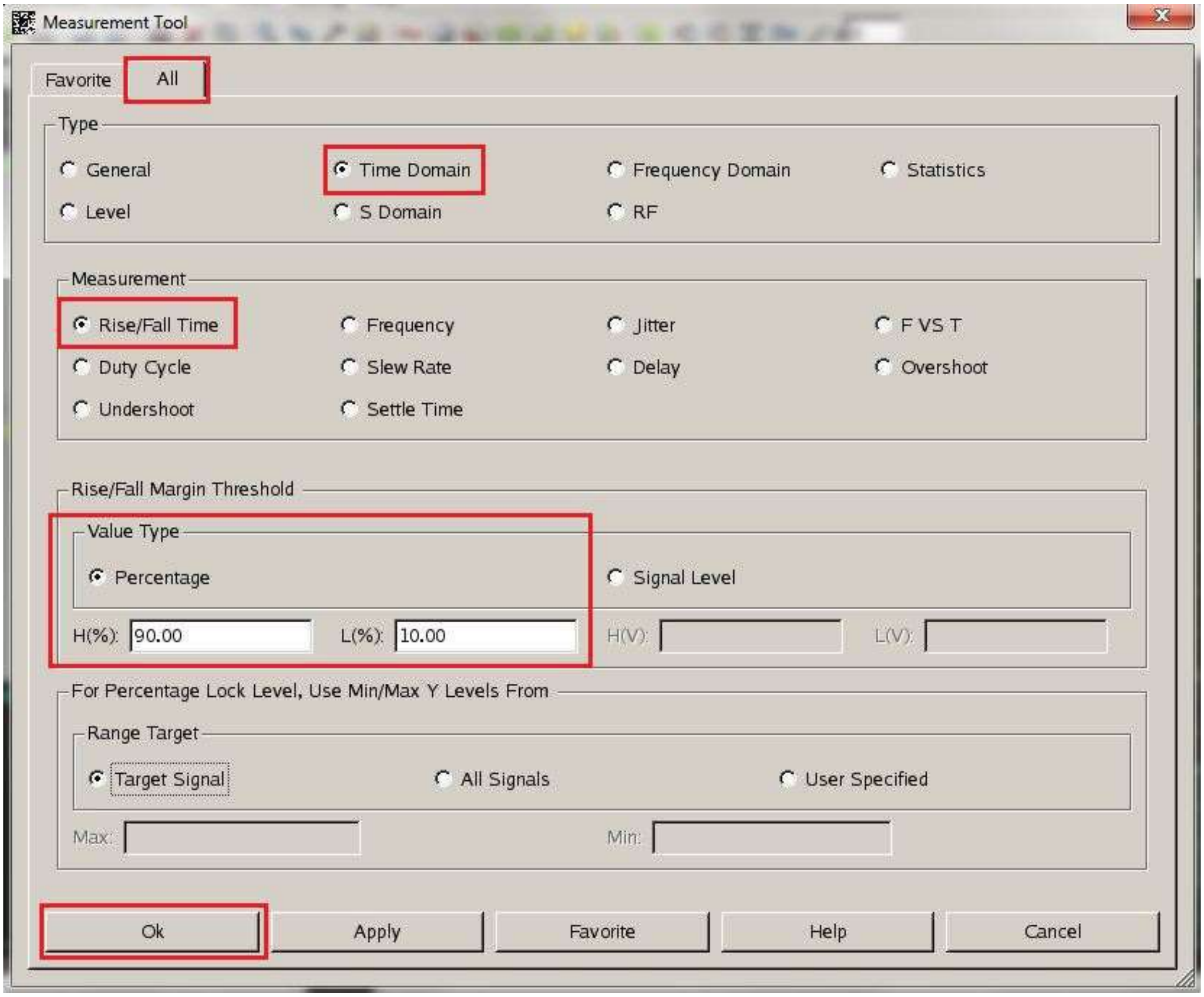


Figure 34: Rise/Fall Measurement Tool

After clicking **Ok**, a rise/fall measurement box will appear on the waveform. You can drag the rise/fall measurement box along the Vout waveform to get the rising and falling delay times. Notice that when the rise/fall measurement box shows a rising red curve the tool is measuring rising delay time from 10% of the signal to 90% of the signal. Also when the rise/fall measurement box shows a falling green curve the tool is measuring the falling delay time from 90% of the signal to 10% of the signal. See figure 35 for reference, it shows two measurement boxes and zooms in on Vout.

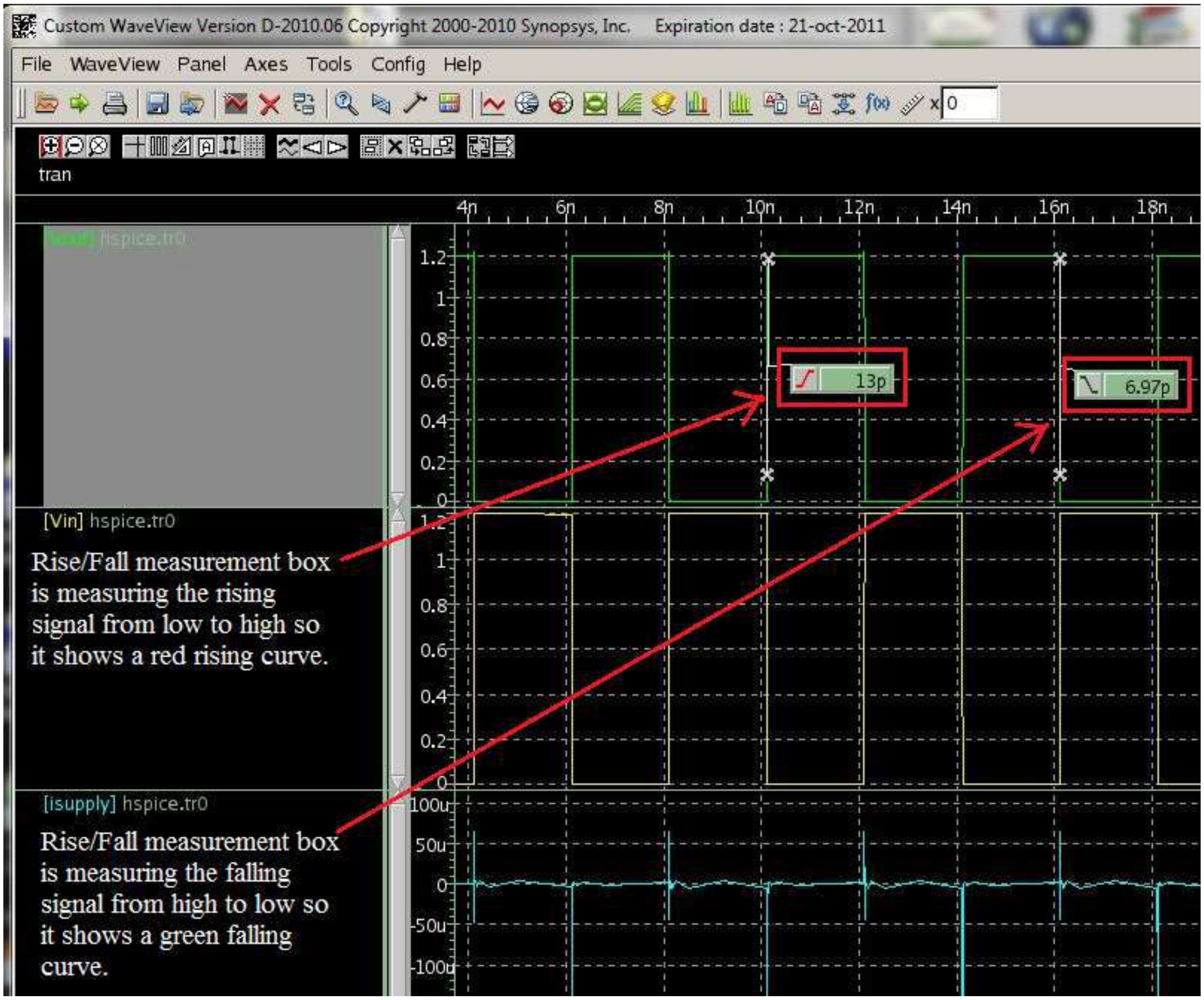


Figure 35: Rise/Fall Waveform Measurement

Average Current Measurement:
To measure average current, select the **tran** tab in the bottom left hand corner of the WaveView window and ungroup all the waveforms as described in figure 30. Delete the Vout and Vin waveforms so only the isupply waveform shows. You can delete a waveform by selecting its name in the signal list on the left side of the WaveView window and pressing **delete** on the keyboard and clicking ok. You can always recover these signals later by clicking **Plot** on the SAE window.

Open the measurement tool by going to **Tools → Measurement…** or by clicking  in the WaveView window. Scroll down in the menu window on the left and then click on the **Average** measurement within the **Level** submenu as shown in Figure 36. Click **Ok** when done.

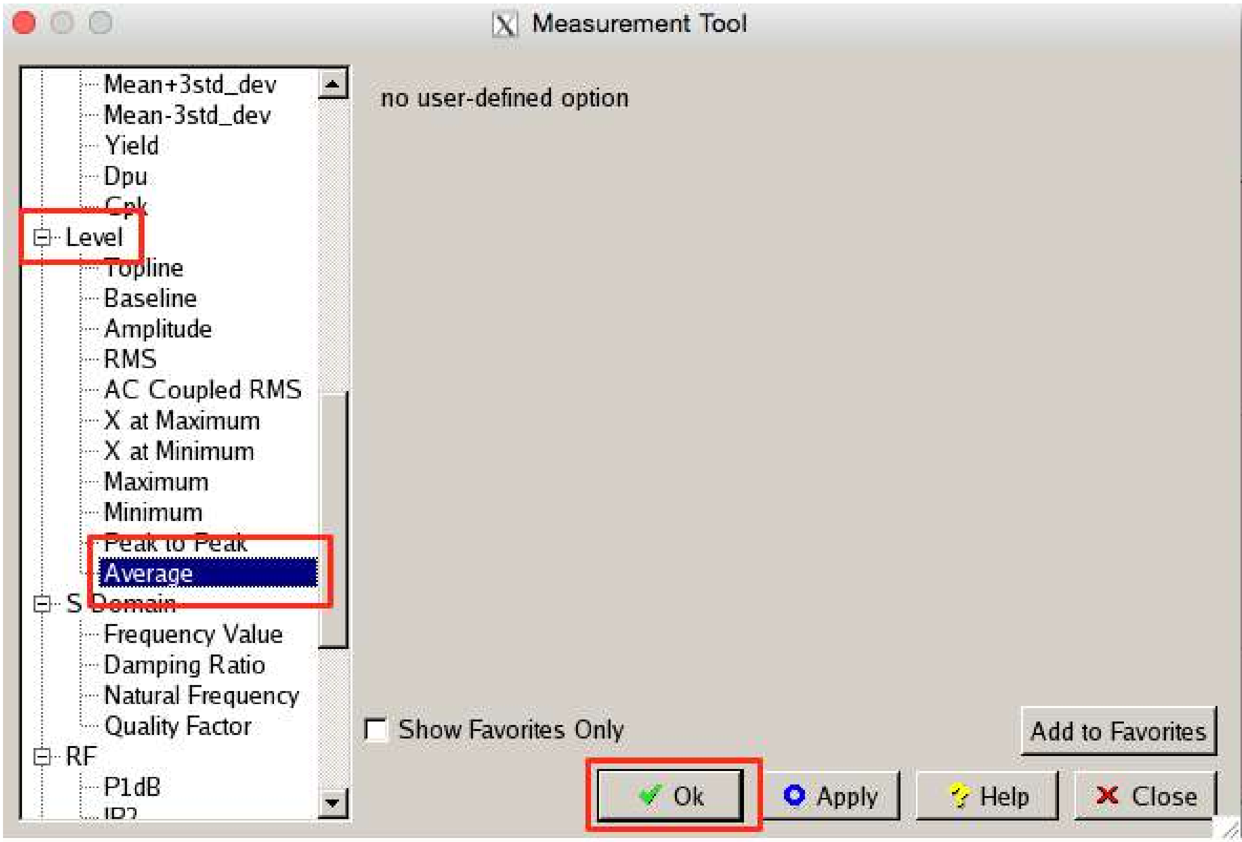


Figure 36: Average Measurement Tool

After clicking **Ok**, an average measurement box will appear on the waveform. Drag the box toward the waveform until it displays the average value. See figure 37 below for reference.

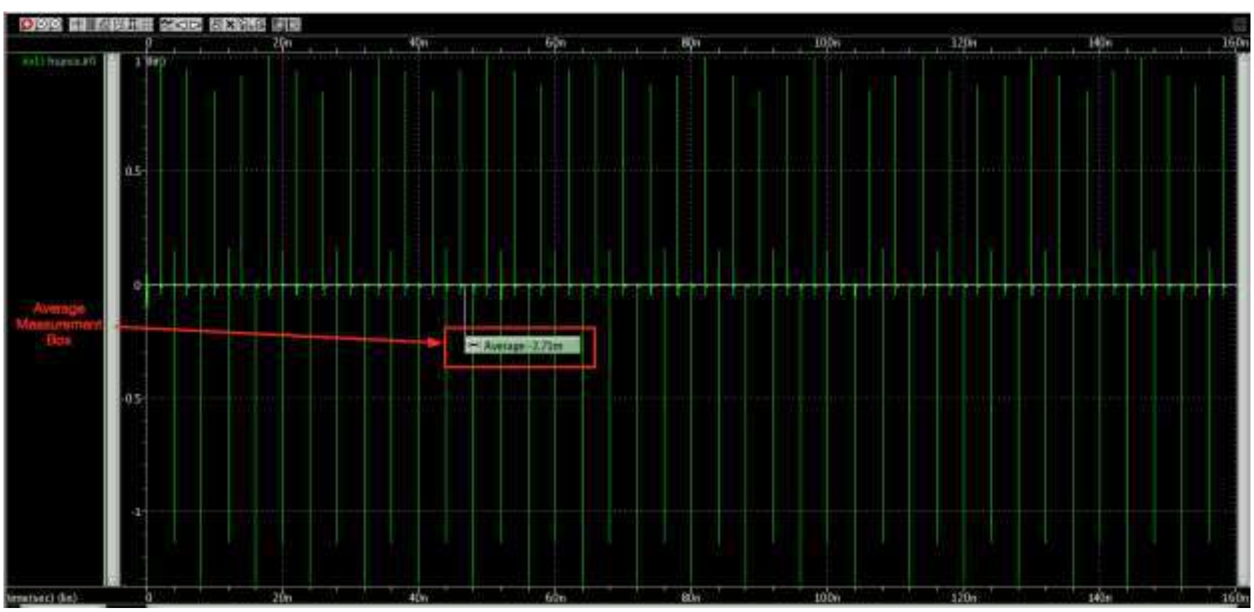


Figure 37: Average Waveform Measurement

Frequency Measurement:
To measure frequency, select the **tran** tab in the bottom left hand corner of the WaveView window and ungroup all the waveforms as described in figure 30.

Open the measurement tool by going to **Tools → Measurement…** or by clicking  in the WaveView window. Click the **All** tab in the measurement tool window and fill out the options as shown below in figure 38. Click **Ok** when done.

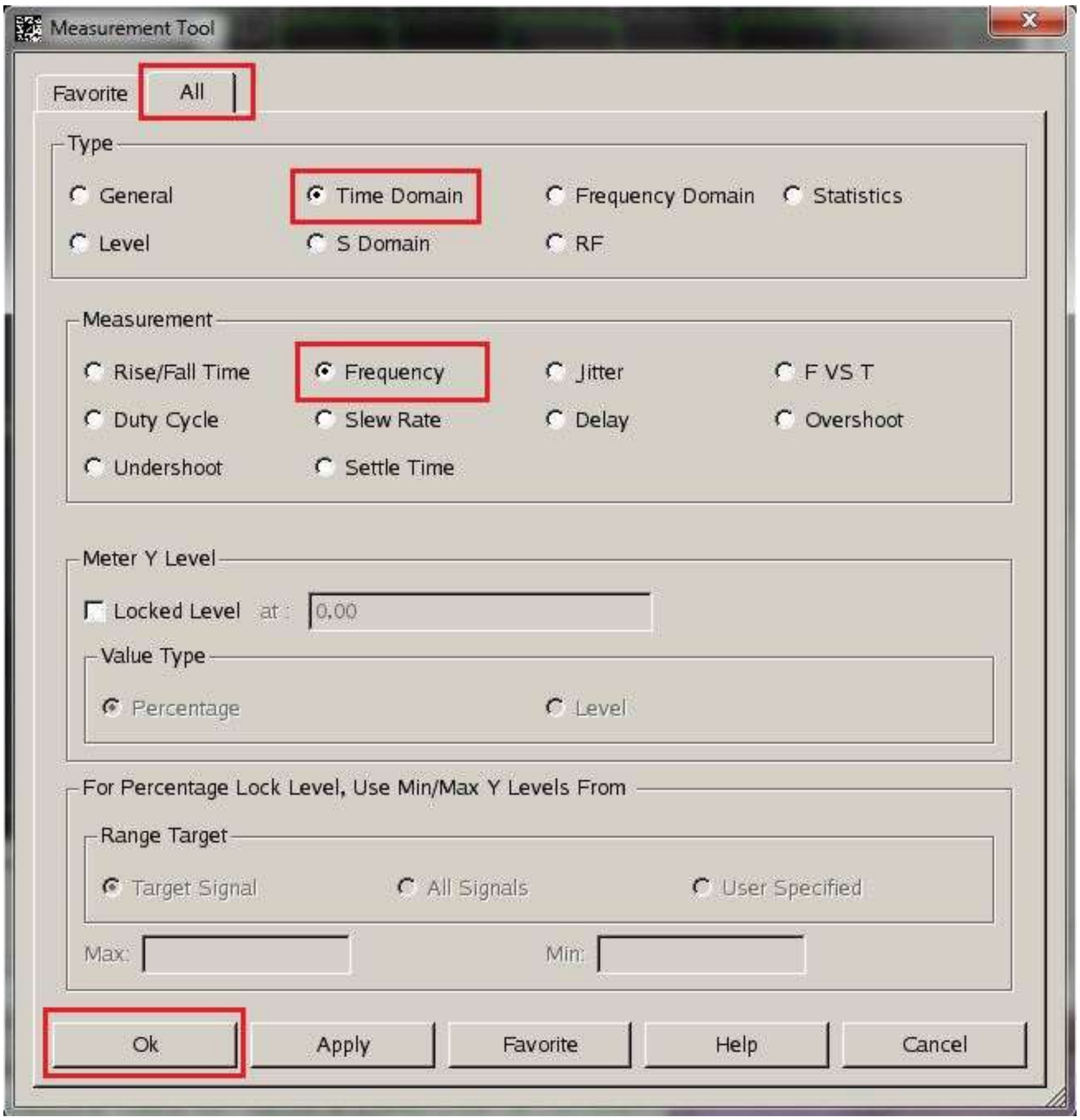


Figure 38: Frequency Measurement Tool

After clicking **Ok**, a frequency measurement box will appear on the waveform. Drag the box toward the waveform you want to measure until it displays the frequency value. See figure 39 below for reference.

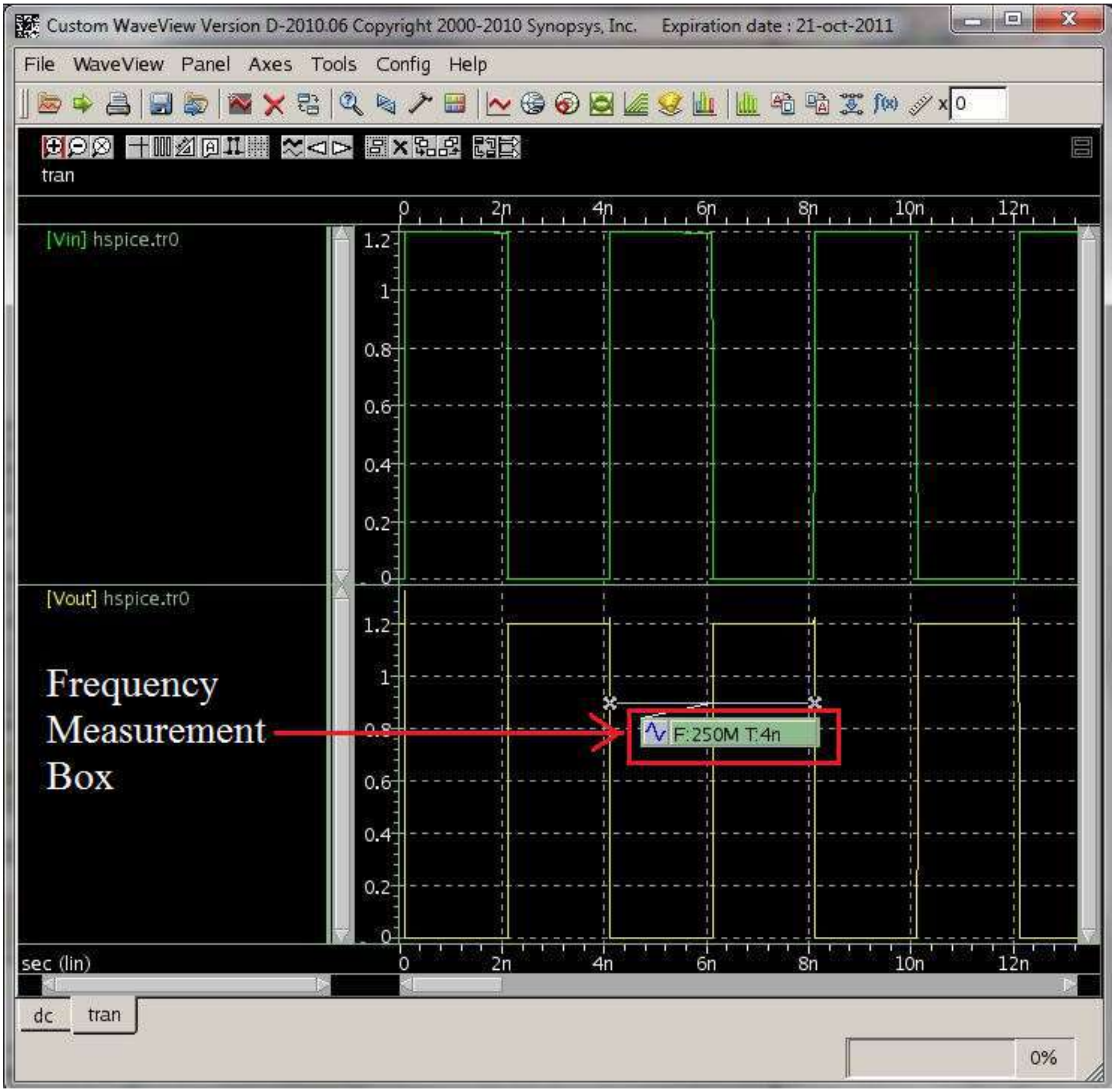


Figure 39: Frequency Waveform Measurement

Part 5: Creating Layout

To draw a layout, it is strongly recommended that you make yourself familiar with the Lambda Rules. This will help in reducing the layout design cycle time and debugging the errors identified by the Design Rules Check (DRC). You can find the guide with the rules for the 90nm technology we are using here:

/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/documentatio
n
File: DesignRules.pdf

It is also suggested that you run the DRC throughout the layout process, instead of waiting until you have completed your layout to run it. This may help in finding errors you are making early on, and correct your mistake before you repeat throughout the layout.
Important: If you have errors in your DRC report, look for the coordinates that are listed by each error. These will allow you to pinpoint where the error is on your layout.

First, open Custom DesignerLE. If your project library is already open in Custom DesignerSE, go to **File → New → CellView** on you Custom Designer Console. Make sure the name of the cell is the same as the schematic cell you are making the layout for. In this case, it is inverter (see Fig.40). Click **OK** and the new layout will open (see Fig.41).

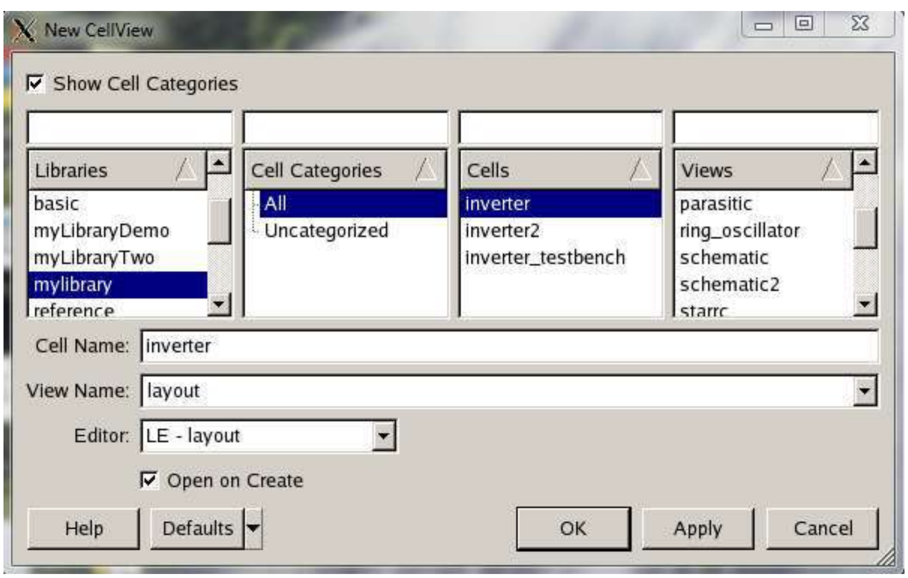


Figure 40: Creating a New Layout

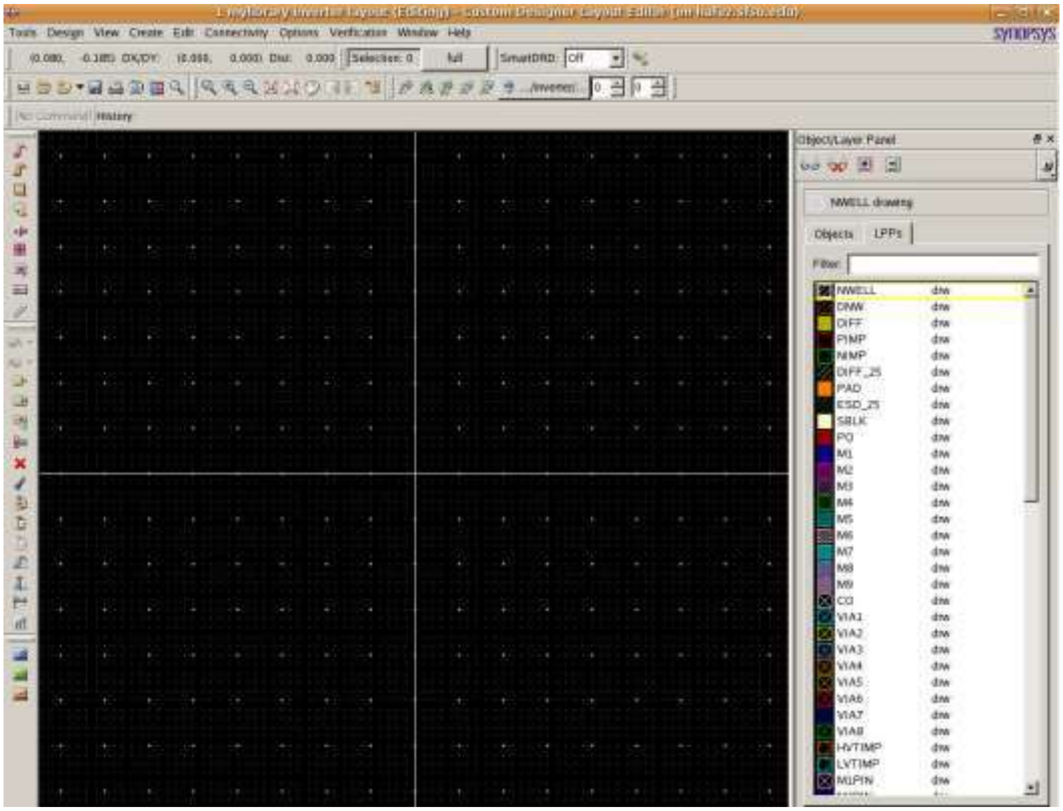


Figure 41: Layout Window

Important: It is recommended that you turn on the SmartDRD feature while laying out your cell by clicking on 

at the top of the layout window. When enabled, this feature annotates DRCs in real time during placement and helps minimize the number of times you need to refer to the Design Rules document when placing/routing objects. The SmartDRD feature includes two modes: Visual and Assist. The Visual mode only annotates the object spacing and DRC violations while the Assist mode attempts to create barriers between objects to help enforce DRC compliance.

Go to ruler by clicking on  . Click once to start drawing the ruler and again to end it. Draw two rulers about the lengths shown in figure 42 (about 3 micrometers by 3 micrometers is a good start). Select the NWELL layer in the layers panel on the right and select **Create → Rectangle** (Fig.42).

Draw a rectangle that approximately fits the dimensions of the rulers you set. You can always adjust the dimensions of this rectangle by using the tools (stretch) that existed on left side of layout window then clicking the side you want to stretch.

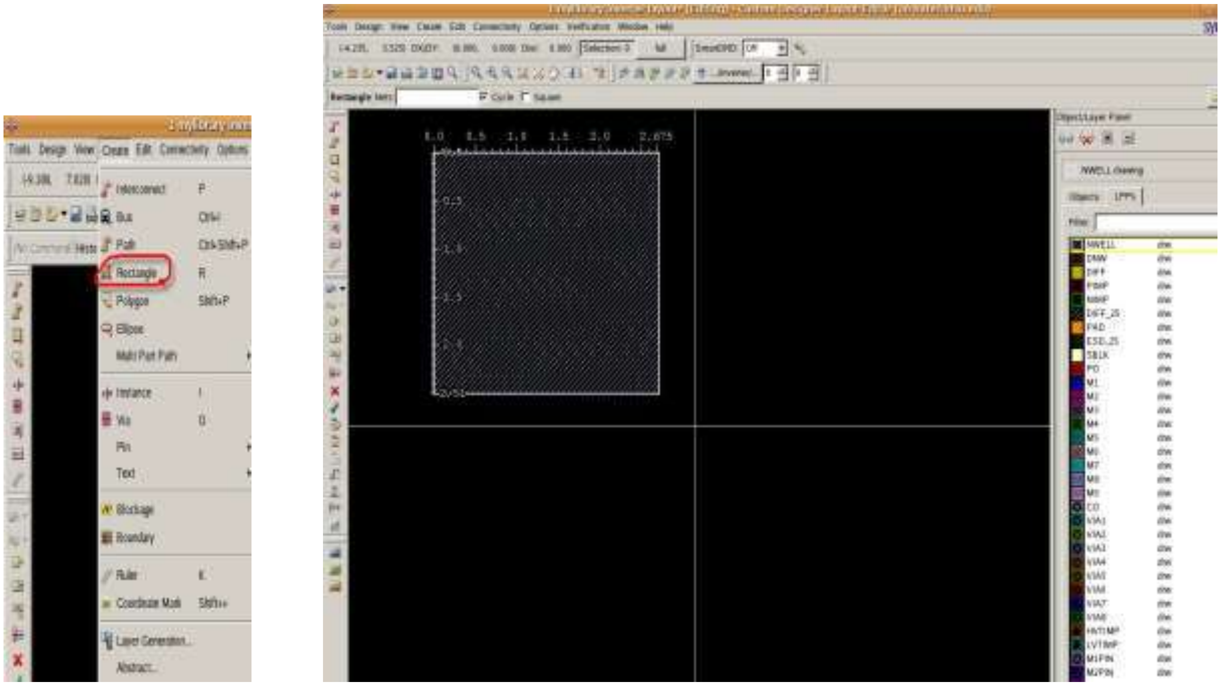


Figure 42: Drawing Rulers and NWell Layer

After you have created an N-well, move your mouse over it. Notice at the top left there are two fields that tell you what layer you are on and the coordinates. This is useful information if you are having trouble figuring out what a layer is and when you are fixing errors found in DRC.

Now we are going to make diffusion areas for PMOS, NMOS and body connections. From our schematic, we know that the width of the PMOS should be 0.5um and the width of NMOS is 0.25um (Fig.43). The location of the diffusion should be similar to the ones in Fig. 44. There are two horizontal diffusion areas that are the NMOS and PMOS devices, and two vertical rectangles that will be the body connections. Place rulers down to help you make sure the width of the diffusion areas for the NMOS and PMOS match our schematic area exactly. Select the "DIFF" layer and again use the "Create Rectangle" tool to draw the diffusion area. Use rulers to check the width of the rectangles. If the widths are different than the widths of the devices in the schematic, you will not pass LVS. You can also use the property editor in **Edit → Property Editor** to change the dimensions of the rectangle to exact values.

Figure 43: Drawing Diffusion

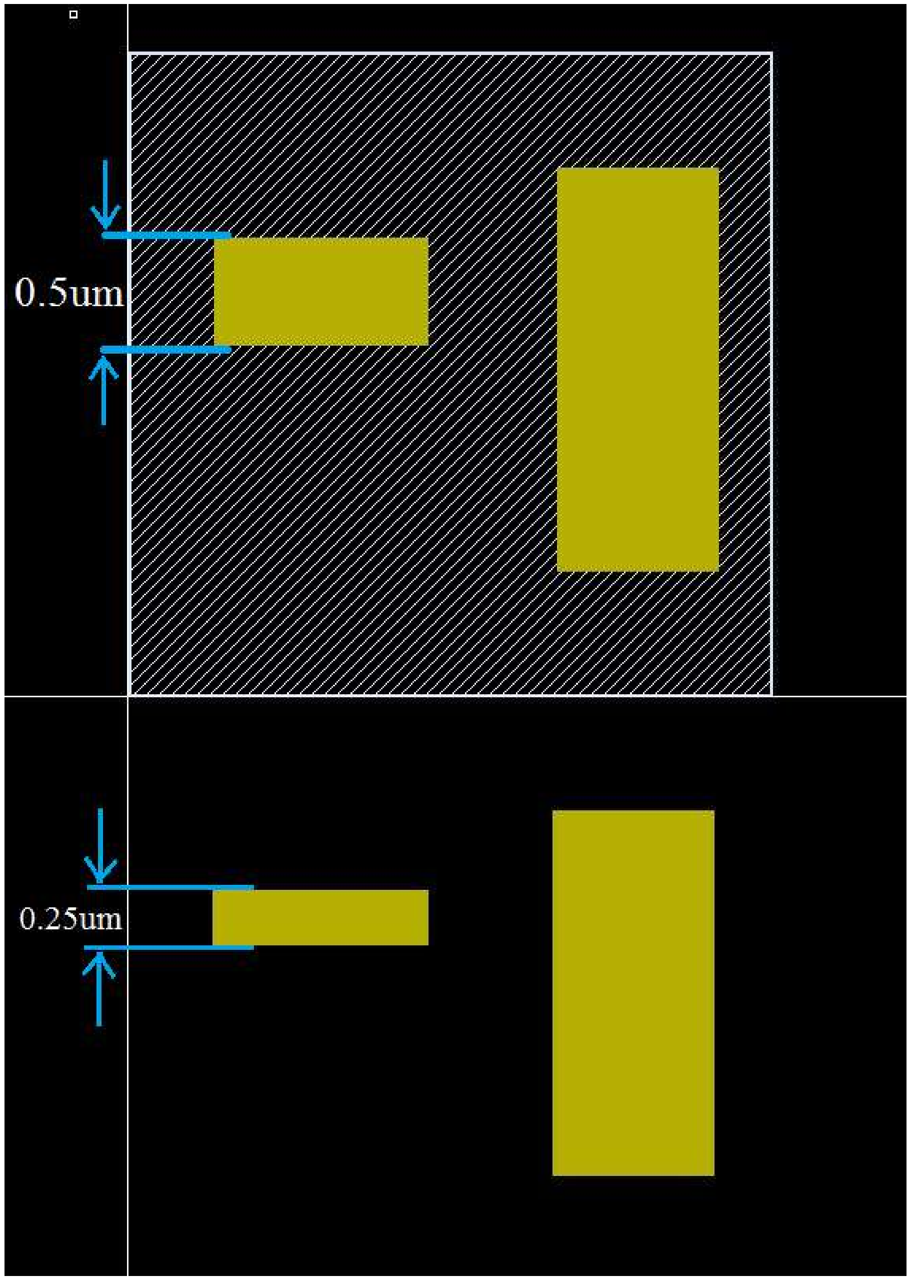


Figure 44: Layout with NWell and Diffusion

Now we will add the P-implant and N-implant areas. When manipulating layers on top of each other sometimes it is useful to “hide” a layer, like you would do in a program like Photoshop. You can do hide or reveal layers in Cosmos by clicking the .

Use the P-implant and N-implant layers with the “Create Rectangle” to cover and surround the diffusion areas. It is important to note that the P-IMP is drawn to the edge of the NWELL where the NWELL meets the NIMP. This can be seen in Fig. 45. The PMOS area should be covered with P-IMP and the NMOS with N-IMP, except for the body connections which have the opposite implantation.

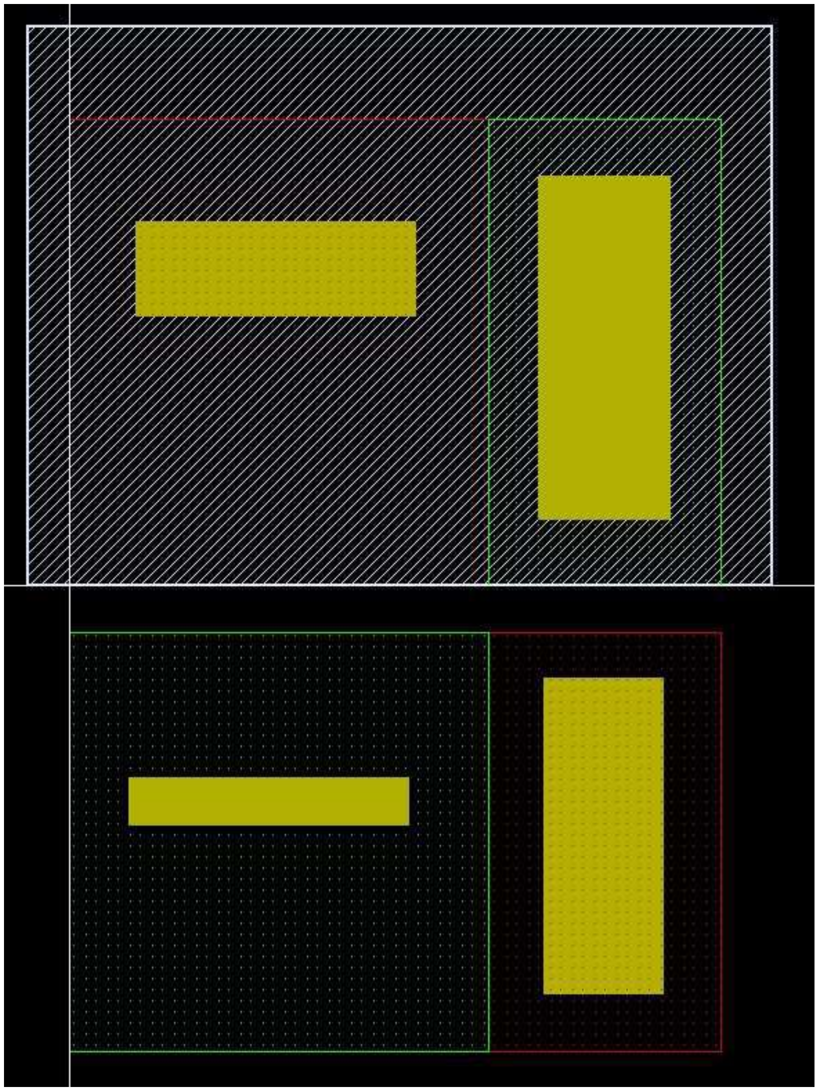


Figure 45: Drawing PIMP and NIMP Layers

Now select the POLY layer and use the "**Create → Path**" tool (Fig.46) to draw a strip of poly through both PMOS and NMOS diffusion areas. Make sure the poly is sticking out past the diffusion areas by at least the amount specified in the design rule manual. When drawing the Poly path, be sure to make sure the thickness is 0.1um to match the transistor lengths in the schematic, see figure 49 to set width/thickness for the draw path tool. Create a rectangle of poly in the center of the strip that would be used for the input signal, see fig. 47.

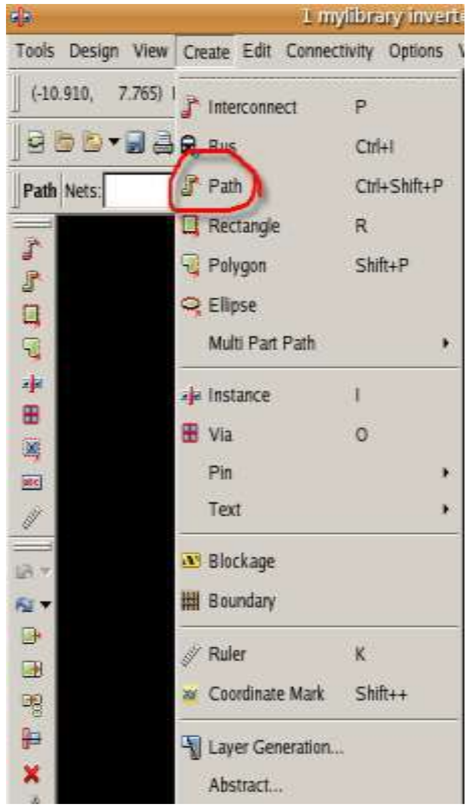


Figure 46: Drawing a Path

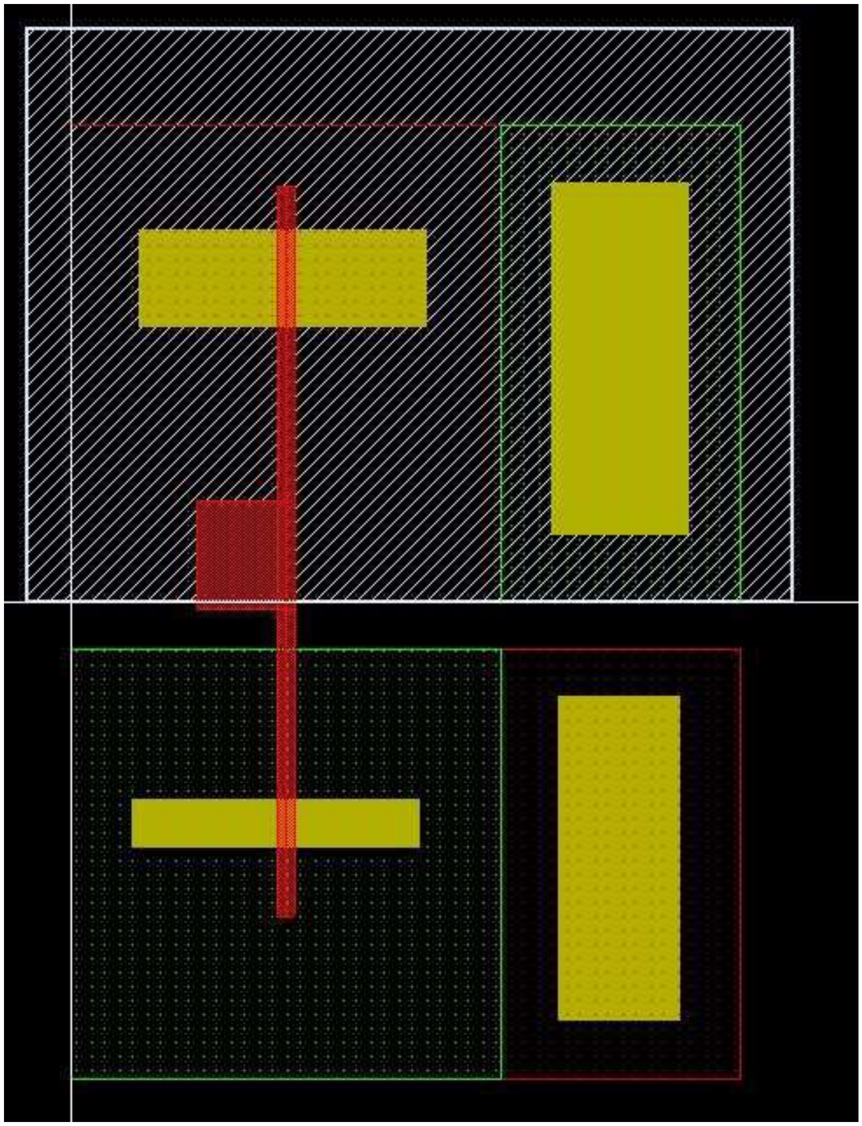


Figure 47: Drawing Poly

Select the “CO” (contact) layer and use the Create Rectangle or Create Polygon tool in conjunction with rulers to make a contact 0.13 by 0.13. You can also use the property editor to get these exact values. After you have created one contact click on  and the contact and make a copy to place the other contacts. Contact placements are shown in Fig. 48. Check to see that your contact placements meet the design rules.

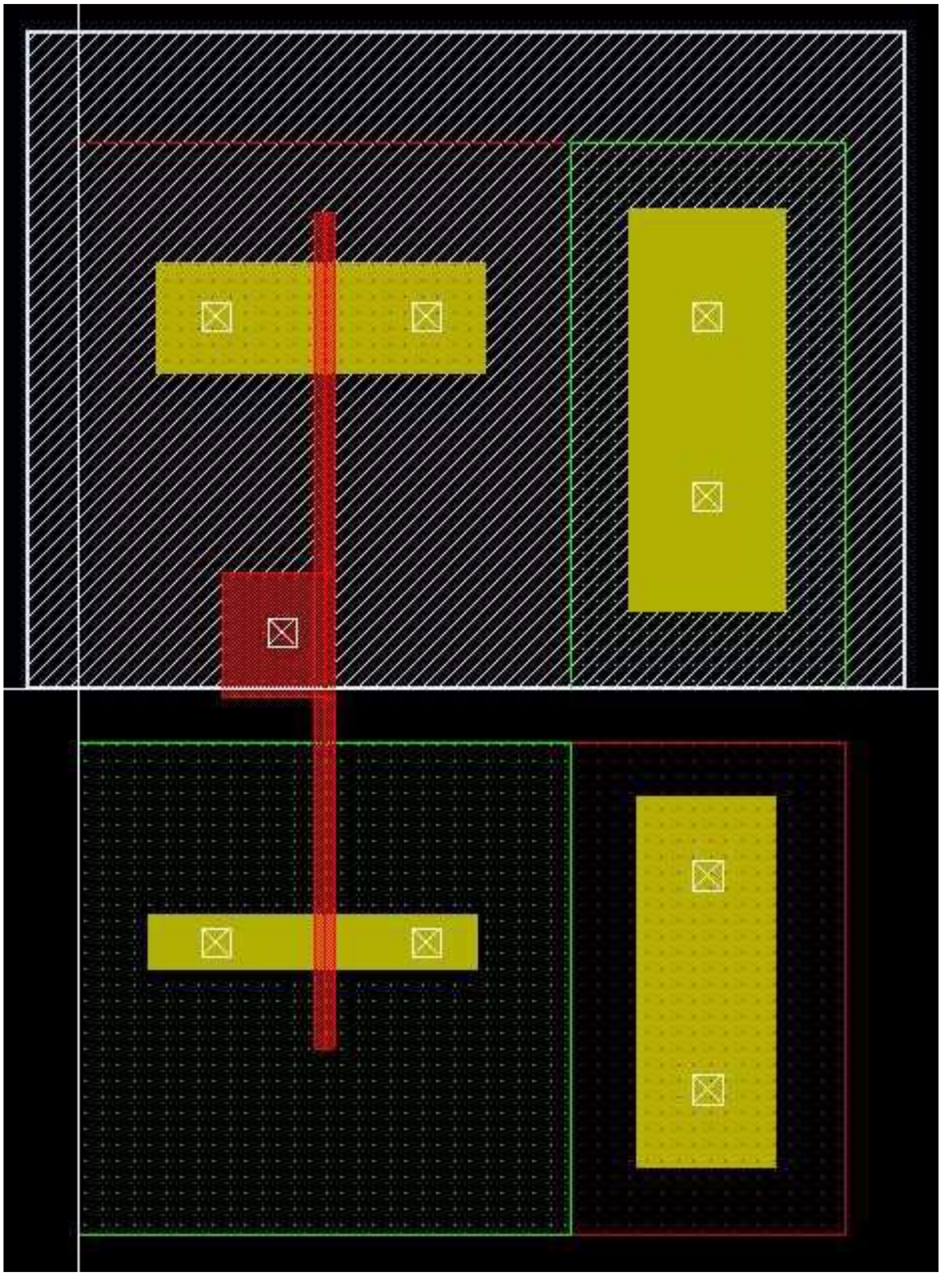


Figure 48: Drawing Contacts

Select the M1 layer and again select the "**Create → Path**" tool. This time in the "Create Path" window that pops ups, click on the width box to then enter 0.16 in the Width field as shown in fig.49.

Figure 49: Modifying Width

Draw the M1 layer the way it is shown in Fig.50. Make sure the metal is covering the contacts by the amount specified in the design rule manual. You can also draw rectangles over the contacts to cover them more.

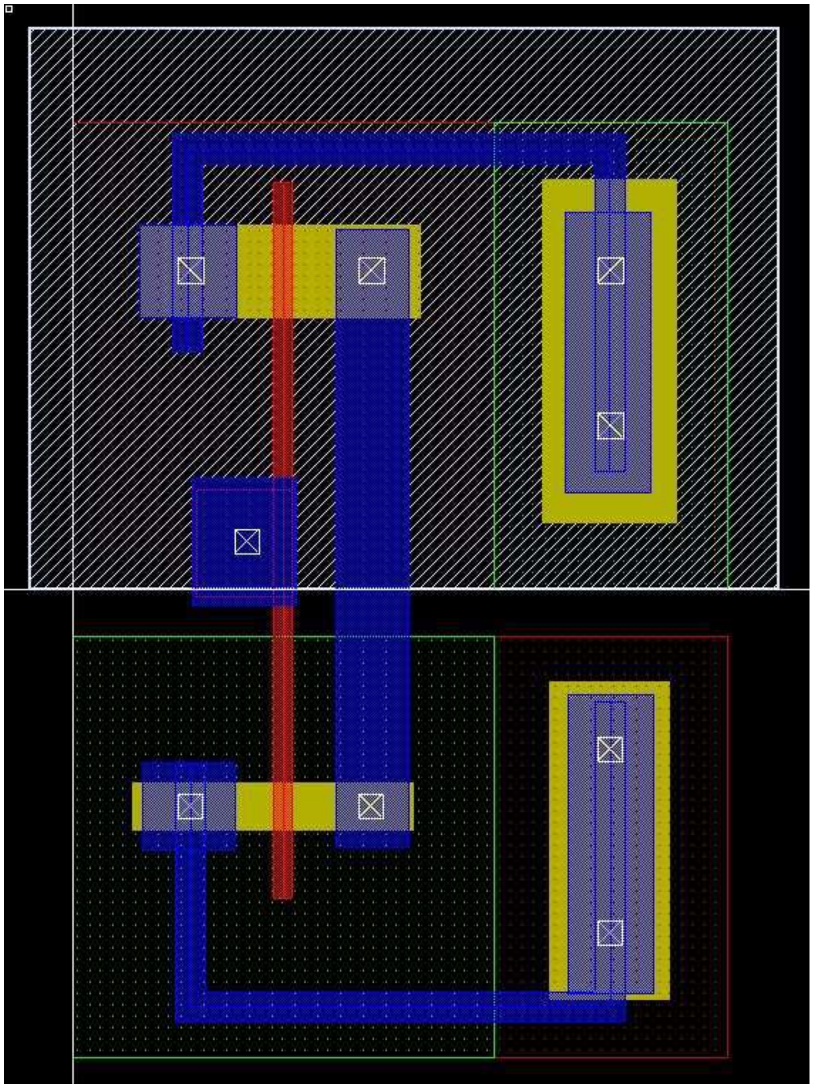


Figure 50: Drawing Metal Connections

Select the **M1PIN** layer. Select the "**Create → Text → Label**" tool and place text labels labeled as AVDD, AVSS, VIN, and VOUT (see Fig. 51). Note that you need to match the label names in layout as the labeled pins in the schematic in order to pass LVS (Layout vs Schematic) later. You have now completed the initial layout and can move onto DRC. Save your layout by going to **Design → Save.**

As a general convention, use uppercase letters for labels instead of lowercase letters.

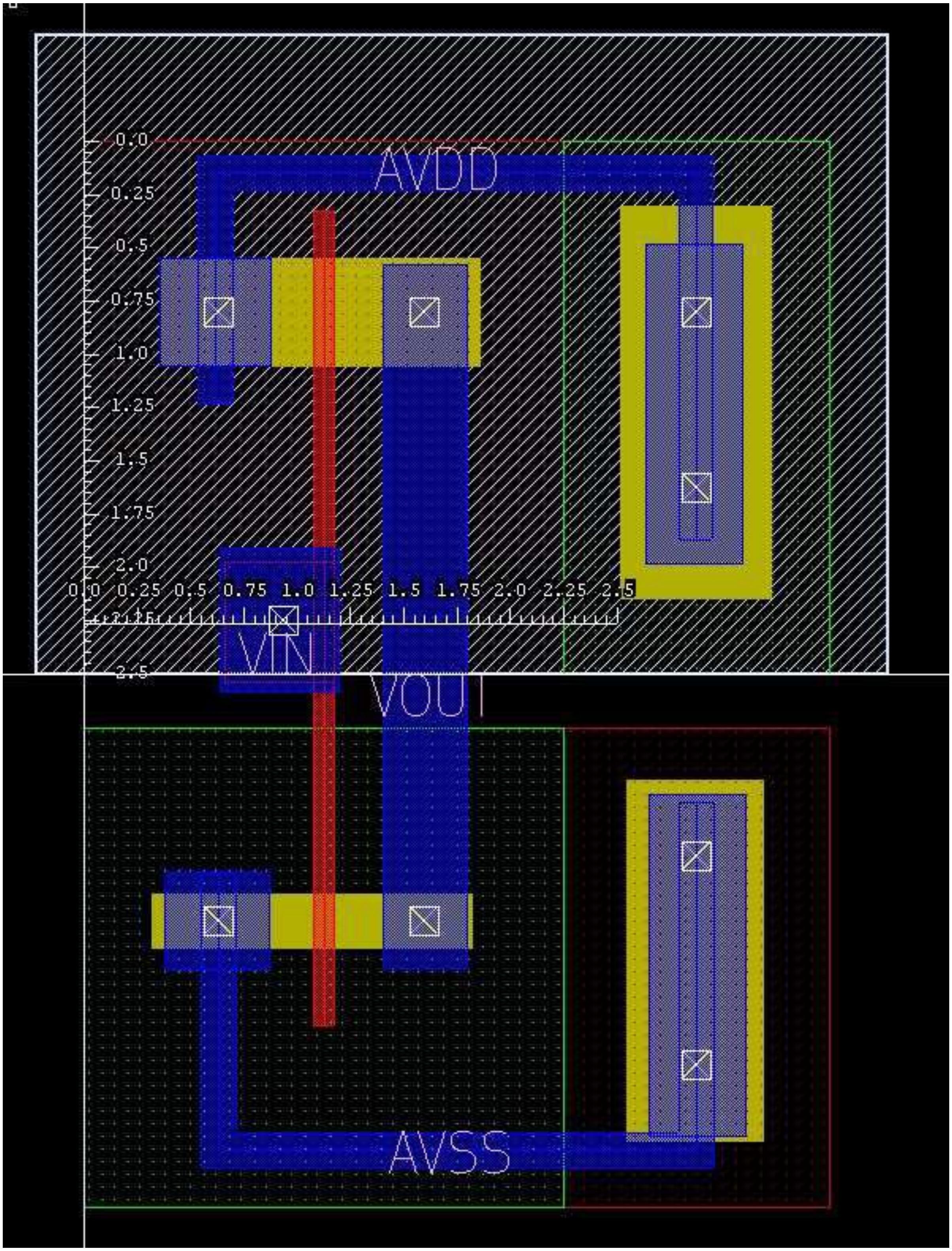


Figure 51: Labeling Connections

Running DRC

After the inverter layout has been drawn to accurately represent the schematic, to verify that the layout meets all the basic design rules, we need to run a DRC (Design Rule Check). Save the layout cell by clicking on **Save**. In Custom Designer Editor, go to **Verification→ DRC → Setup and Run**.

Locate the runset file **rules.drc.9m_saed90ev** from the following directory and click **Ok** (Fig 52).

Directory:
/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/hercules/drc

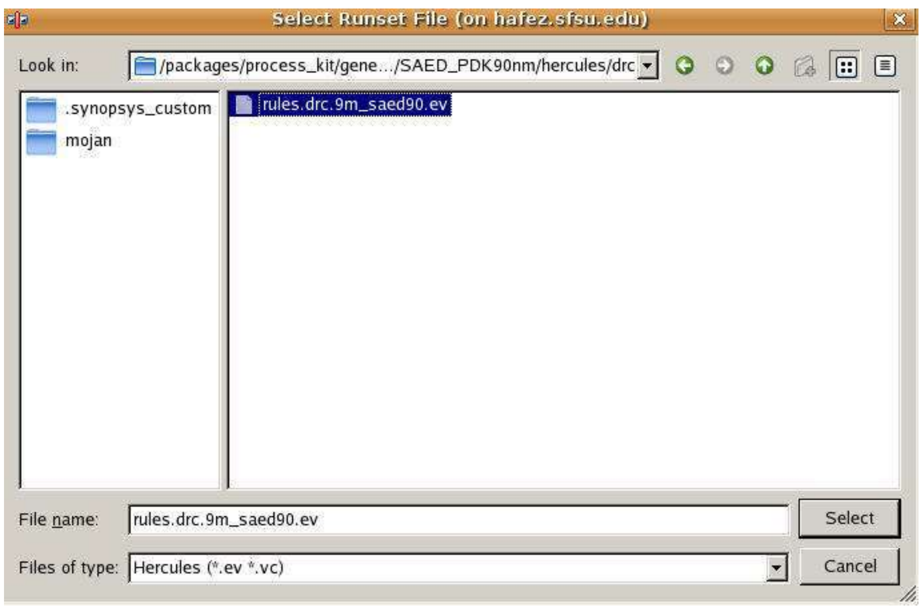


Figure 52: Runset File

For Run Dir, choose your current directory and for tools chose Hercules. Make sure that you select OpenAccess as a format Fig. 53.

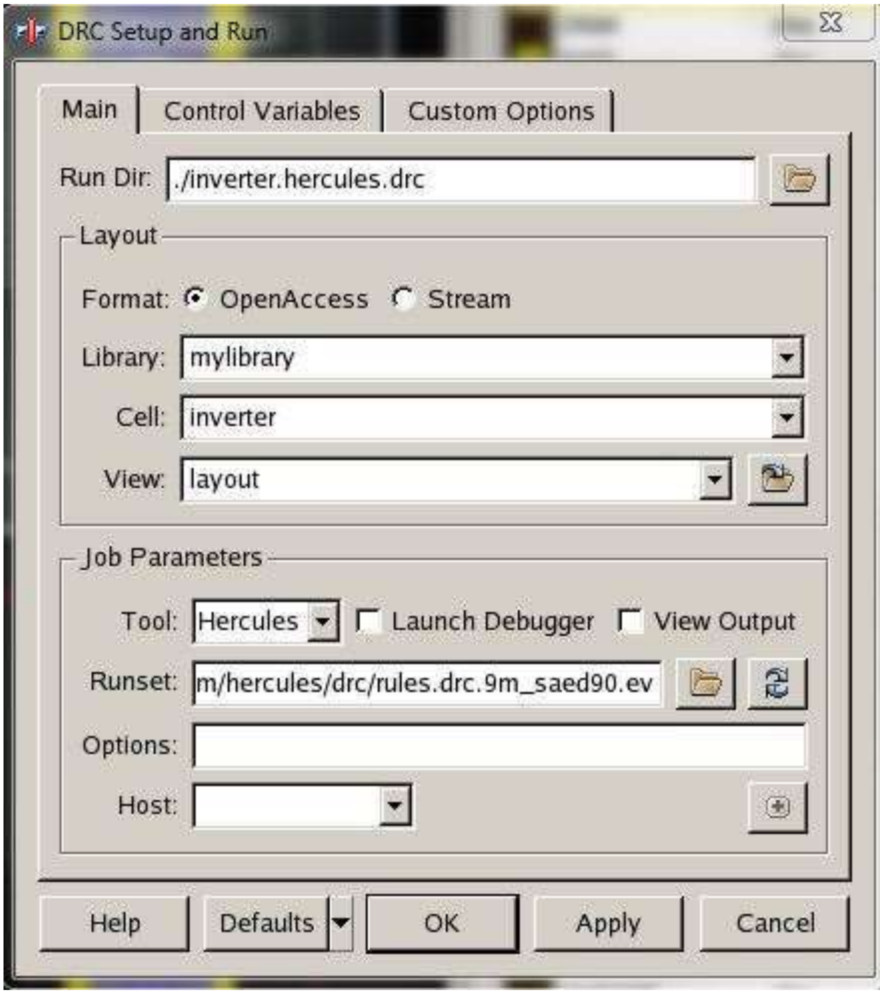


Figure 53: DRC Setup

Click **OK**. After DRC is done, you will see a message on your Console indicating that DRC is done, see Fig. 54.

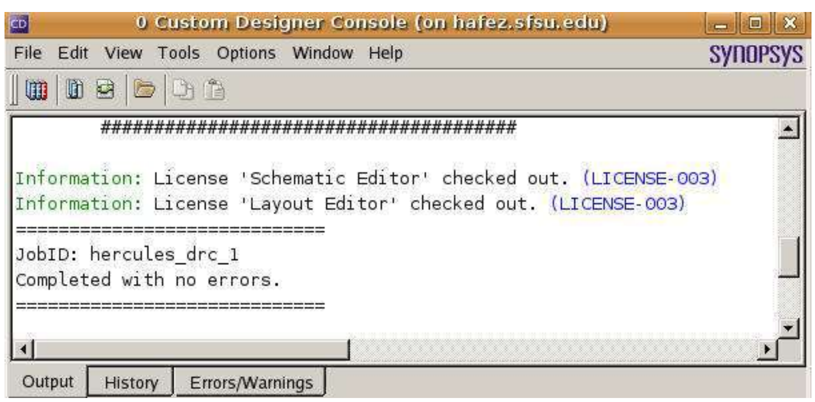


Figure 54: DRC Confirmation

When the layout is free from all the errors and meets all the design rules, the output file inverter.LAYOUT_ERRORS will say CLEAN as In Fig.55 and 56. If there are some errors in the layout, it will say ERROR and the error details are specified in this file. If you have errors in your DRC, check the location and type of errors and correct them. This may take multiple iterations. In the beginning it is advisable to try to correct only a few errors and run the DRC again to check if you corrected them properly. It is important to pass the DRC check before you proceed to LVS and parasitic extraction.

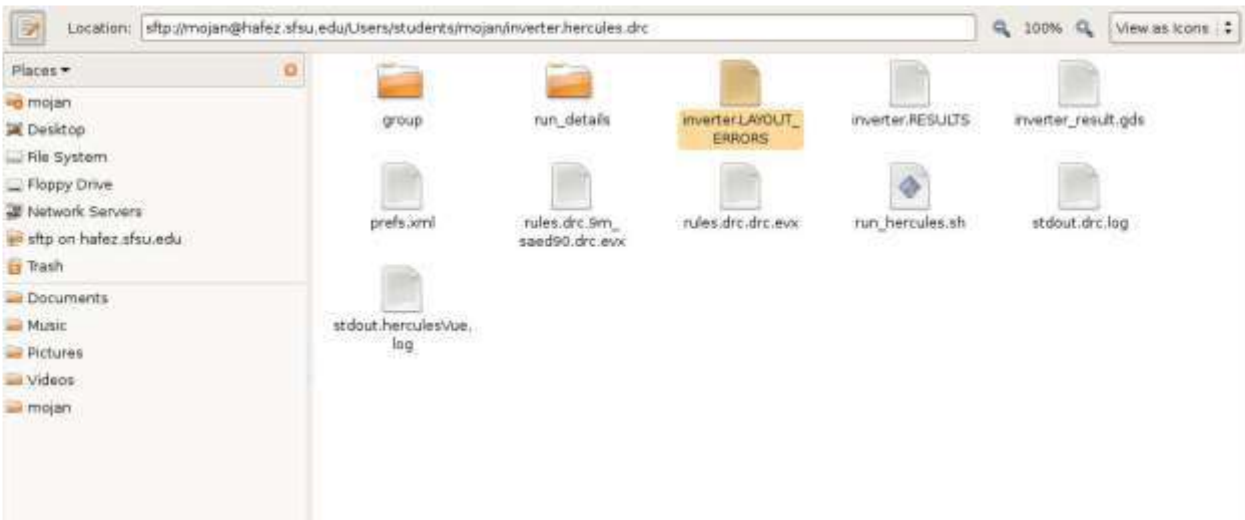


Figure 55: DRC Error File

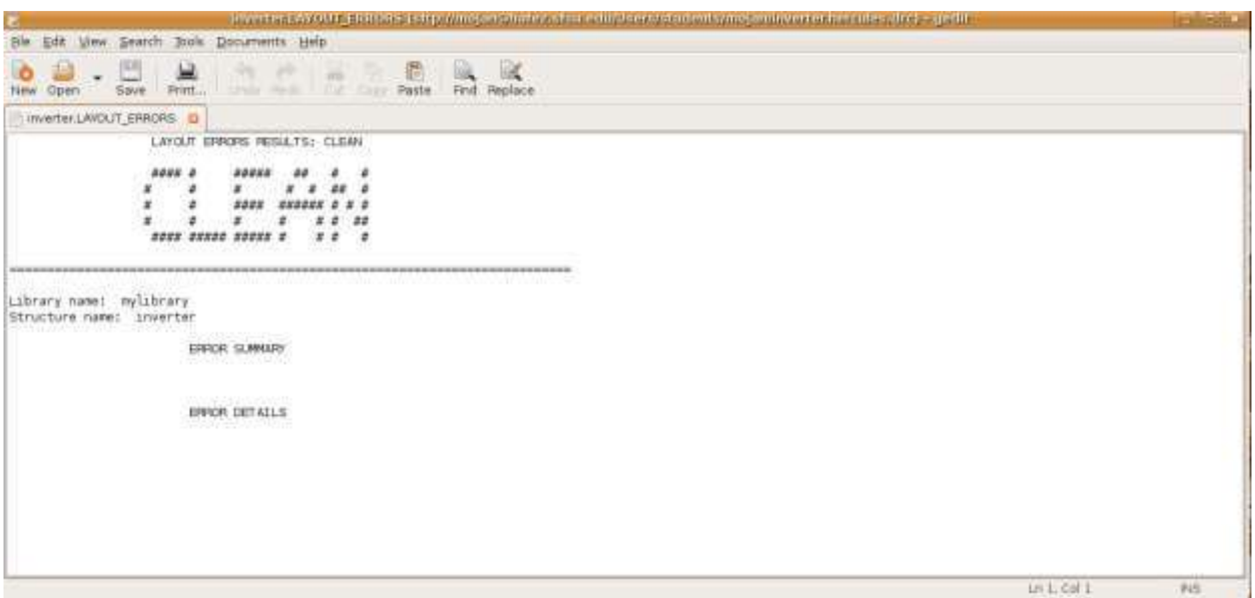


Figure 56: DRC Passed Confirmation

Also for debugging, you can go to **Verification → Debug,** see Fig. 57.

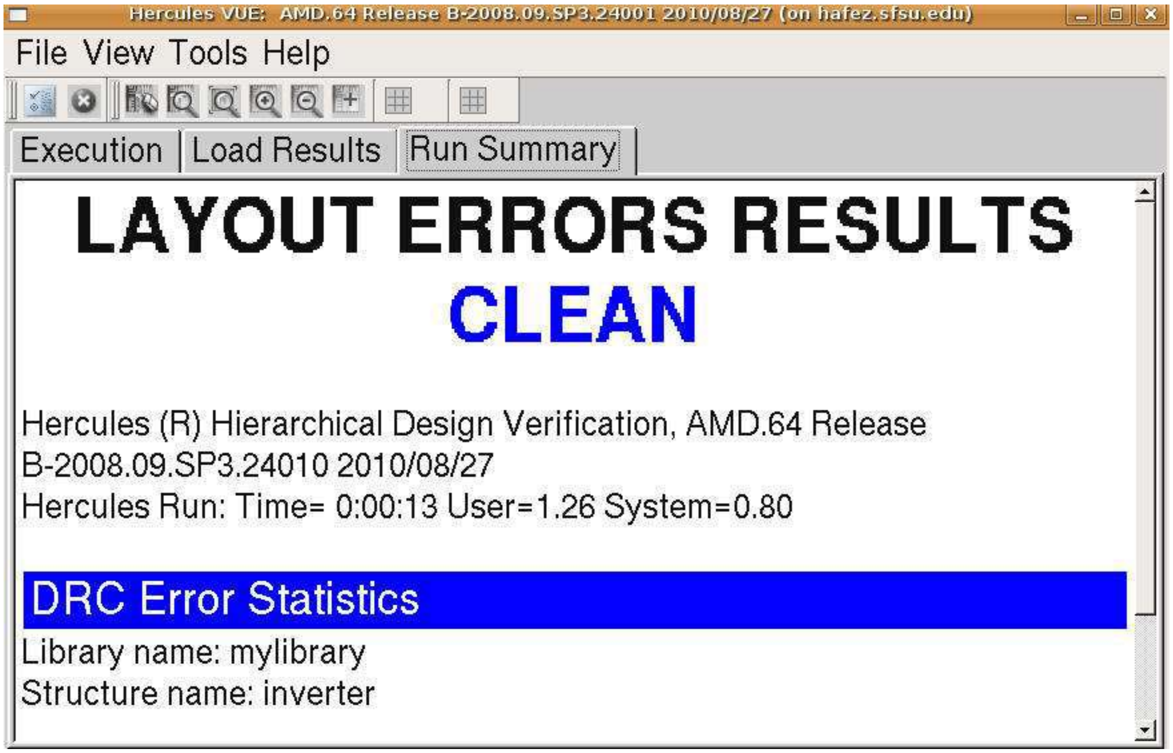


Figure 57: DRC Errors

Running LVS

The LVS (Layout versus Schematic) check performs LVS comparison to verify that the design layout accurately represents the electronic equivalent of the design schematic. Hercules LVS verifies whether the physical design design matches the schematic by: extracting the devices, verifying the connectivity between the devices and comparing the extracted information with the schematic netlist.

Notice that in order to pass LVS, **schematic names and layout names must match** one to one. Make sure the names for labels and pins are using uppercase letters instead of lowercase letters. Also transistor dimensions for **gate width and length in layout and schematic must match**. See figure 58 for reference.

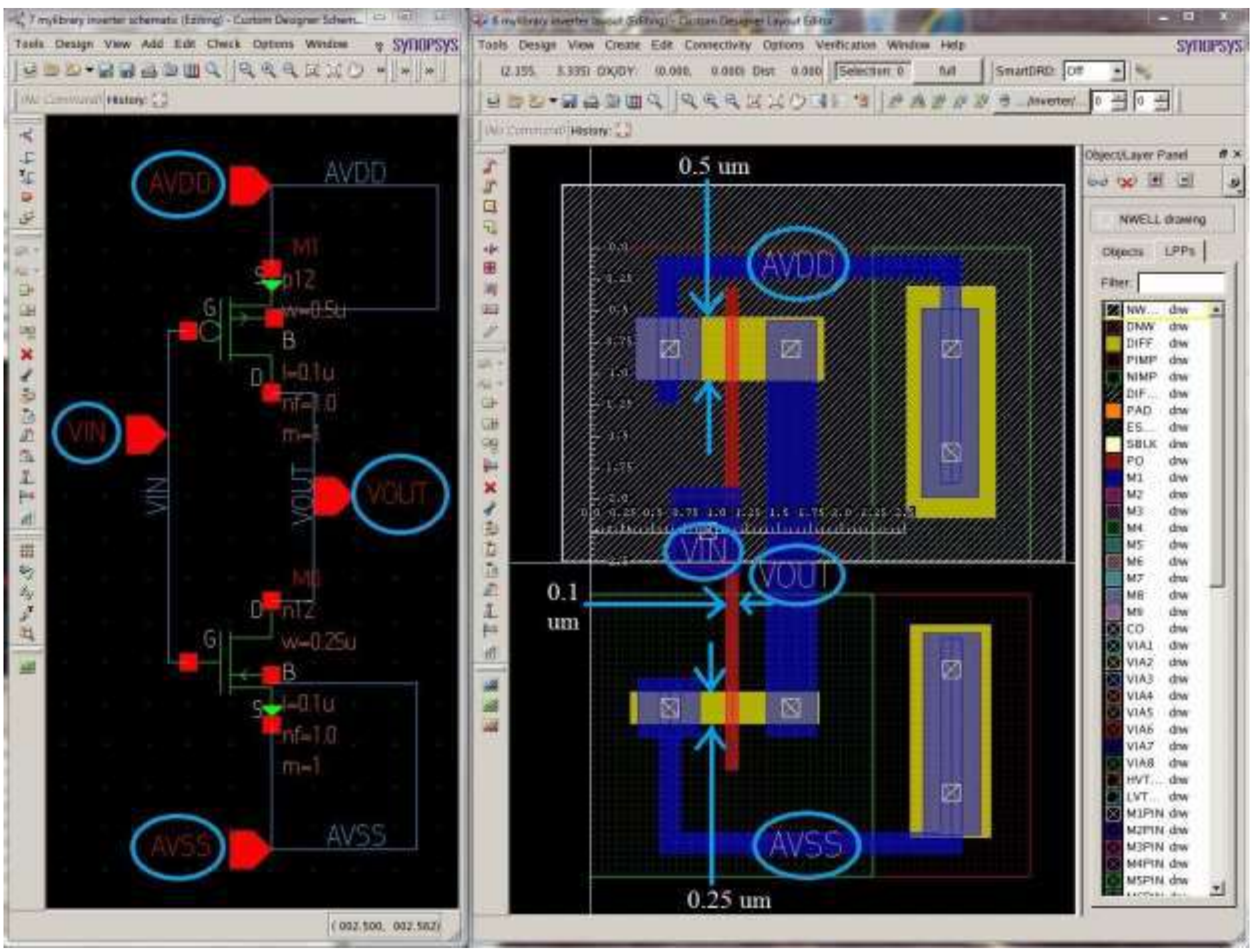


Figure 58: Layout versus Schematic

In the layout window go to **Verification→ LVS → Setup and Run**.
Please make sure your setup mirrors the Fig.59.

Under main option select the file "rules.lvs.9m_saed90.ev" as the Runset File in the following directory:
/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/hercules/lvs/rules.lvs.9m_saed90.ev

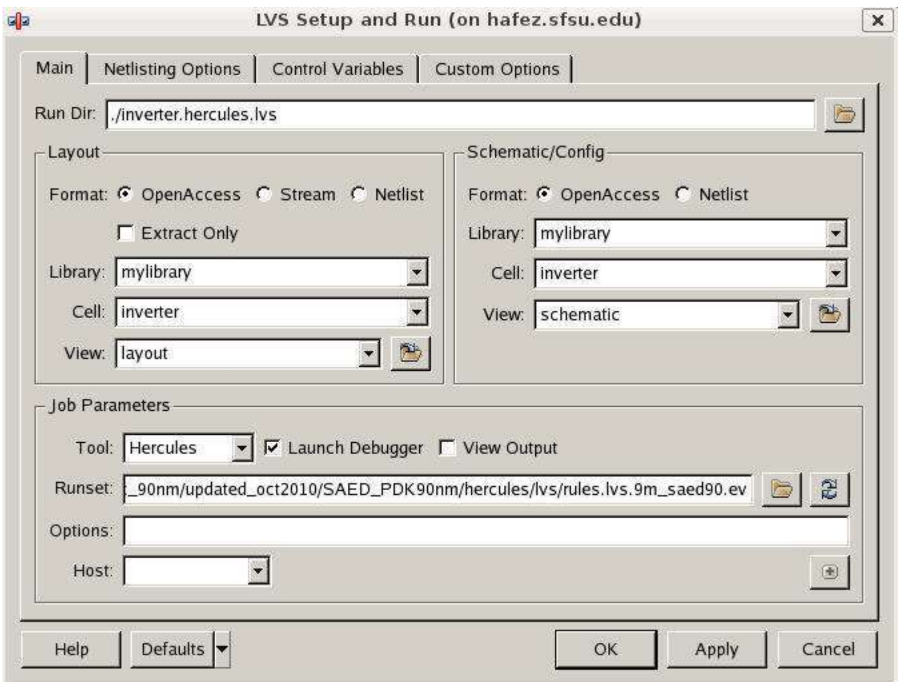


Figure 59: LVS Setup Main Tab

Under Netlisting Option tab, for the netlister, select: CDL if not already selected. See Fig. 60.

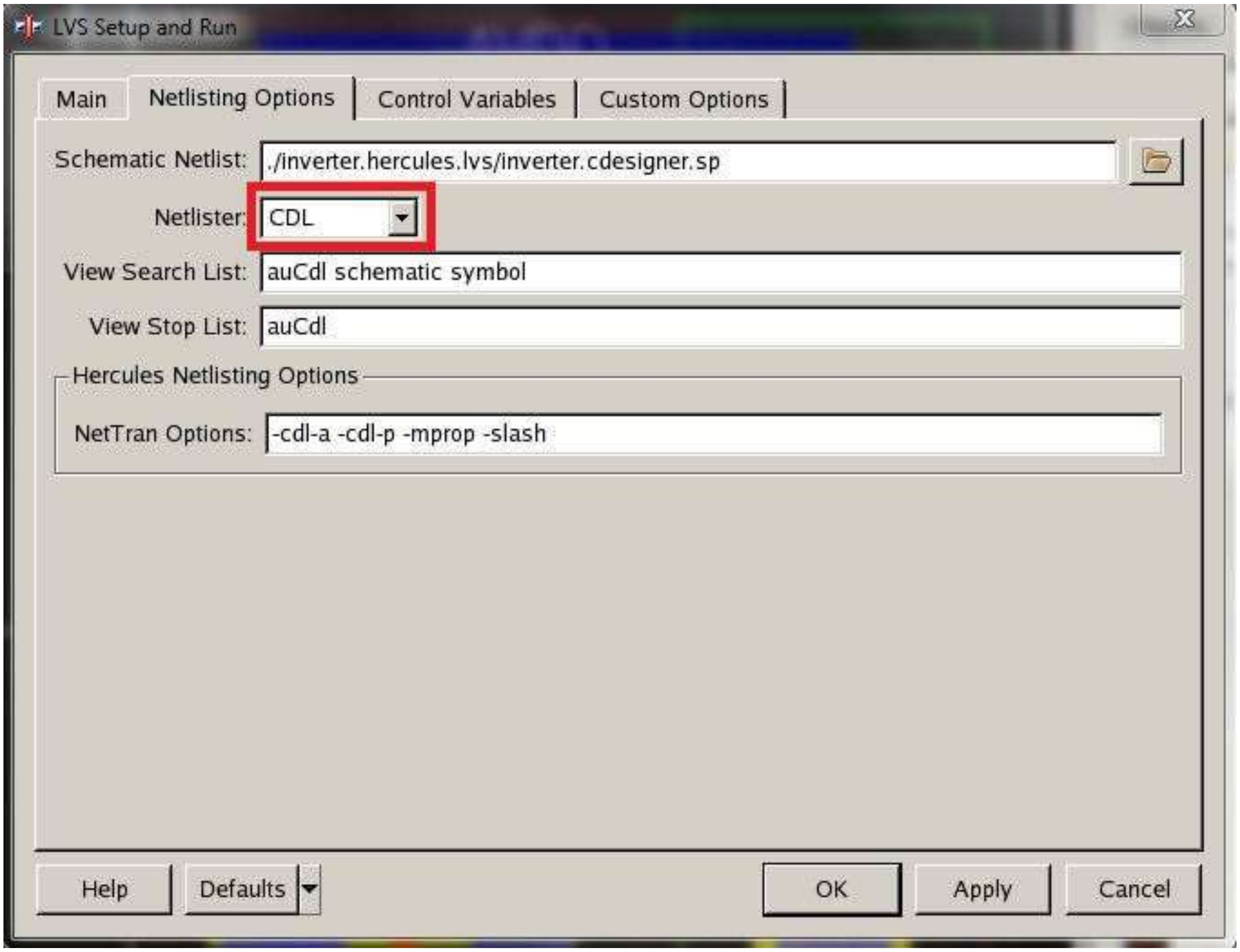


Figure 60: LVS Setup Netlisting Options Tab

Under Control Variables tab set DEC_TYPE Value to PEX_DECK, see fig. 61.

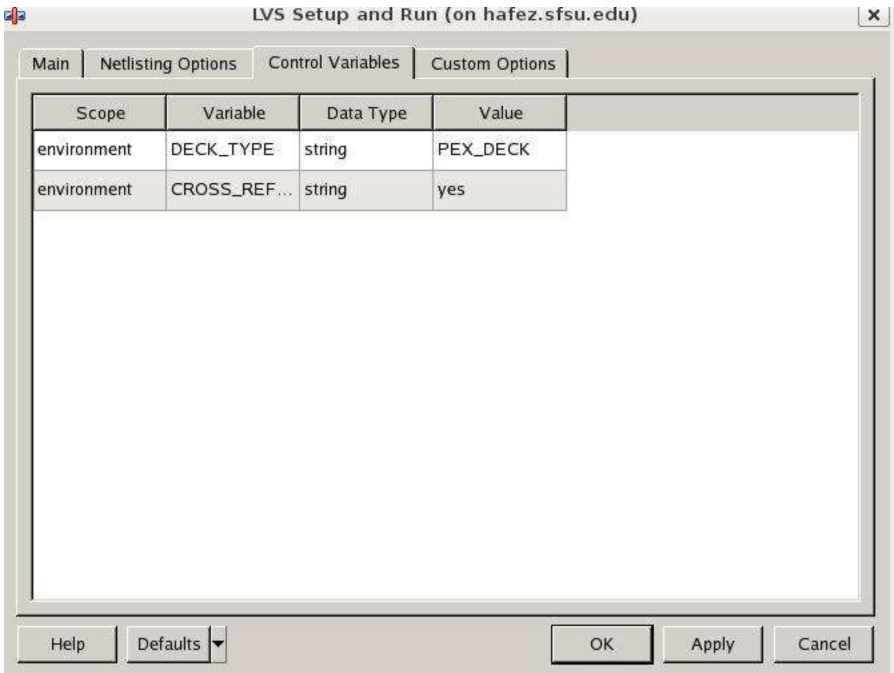


Figure 61: LVS Setup Control Variables Tab

Under the Custom Options tab, leave the defaults as shown in figure 62. Click **OK** when done.

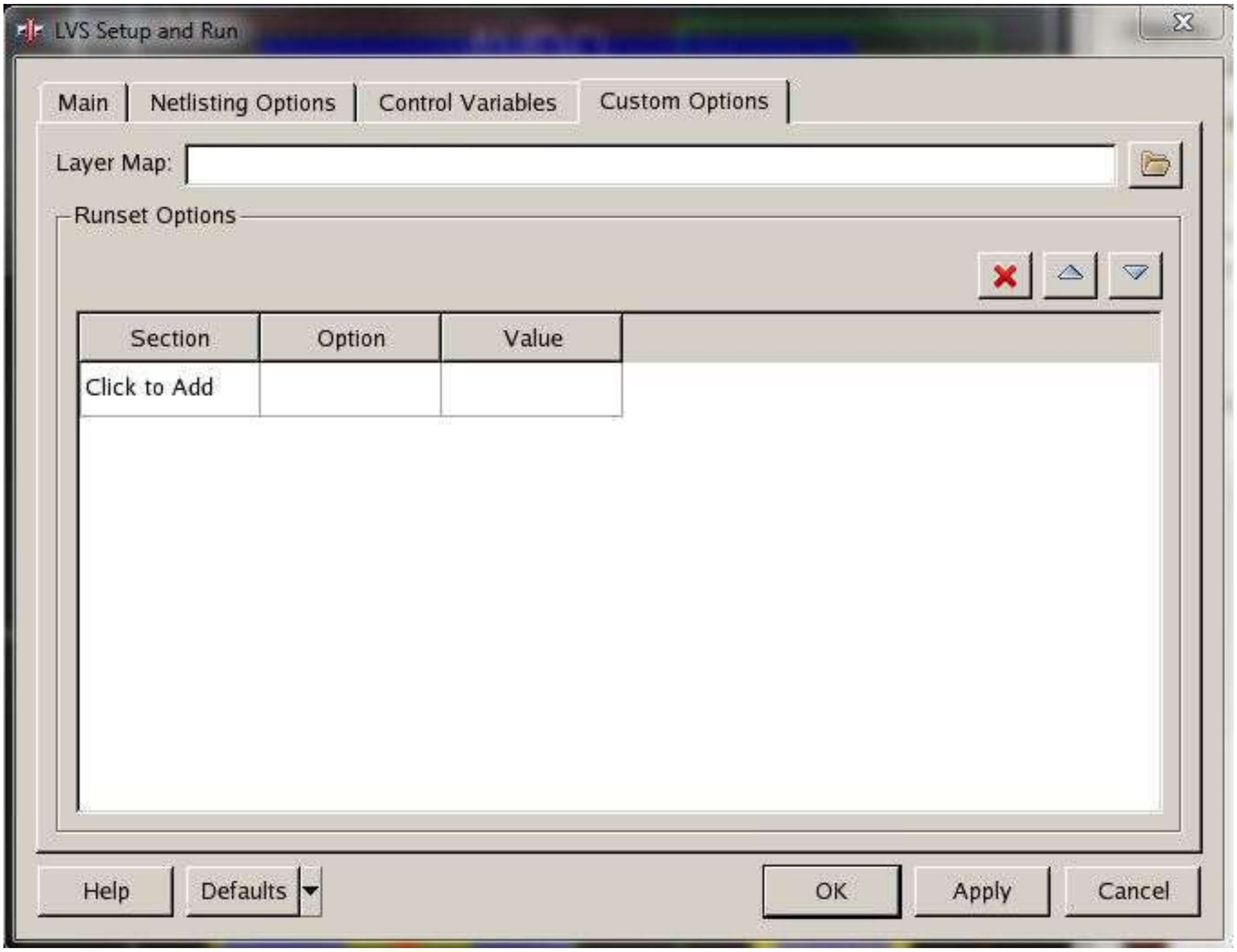


Figure 62: LVS Setup Custom Options Tab

On your Console window you should get the following message as in Fig. 63 if you passed LVS.

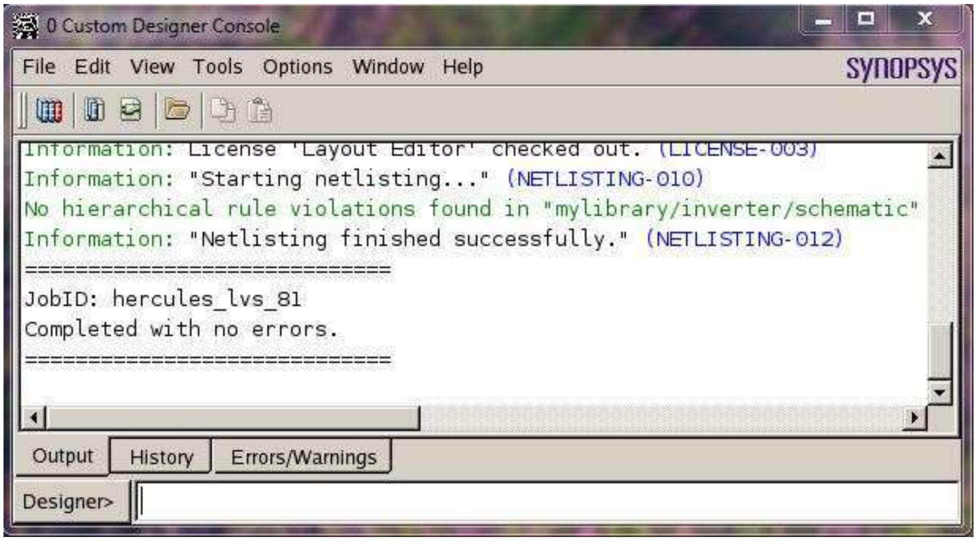


Figure 63: LVS Done Confirmation

Also you can go to LVS Debugger to see the result of you LVS Fig. 64.

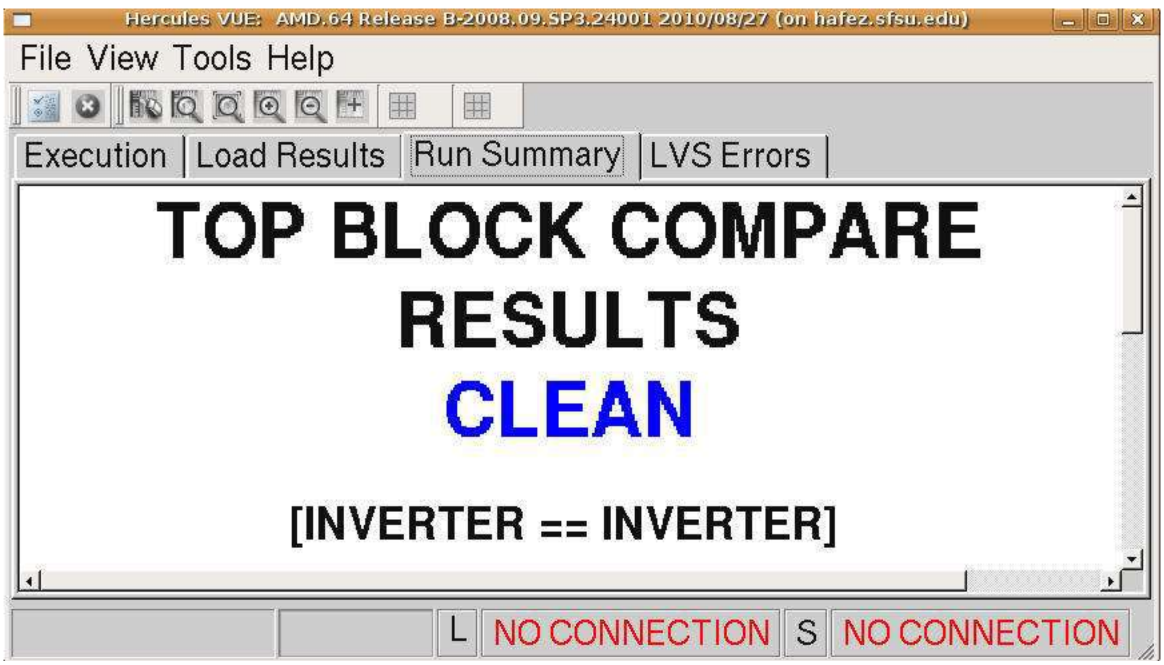
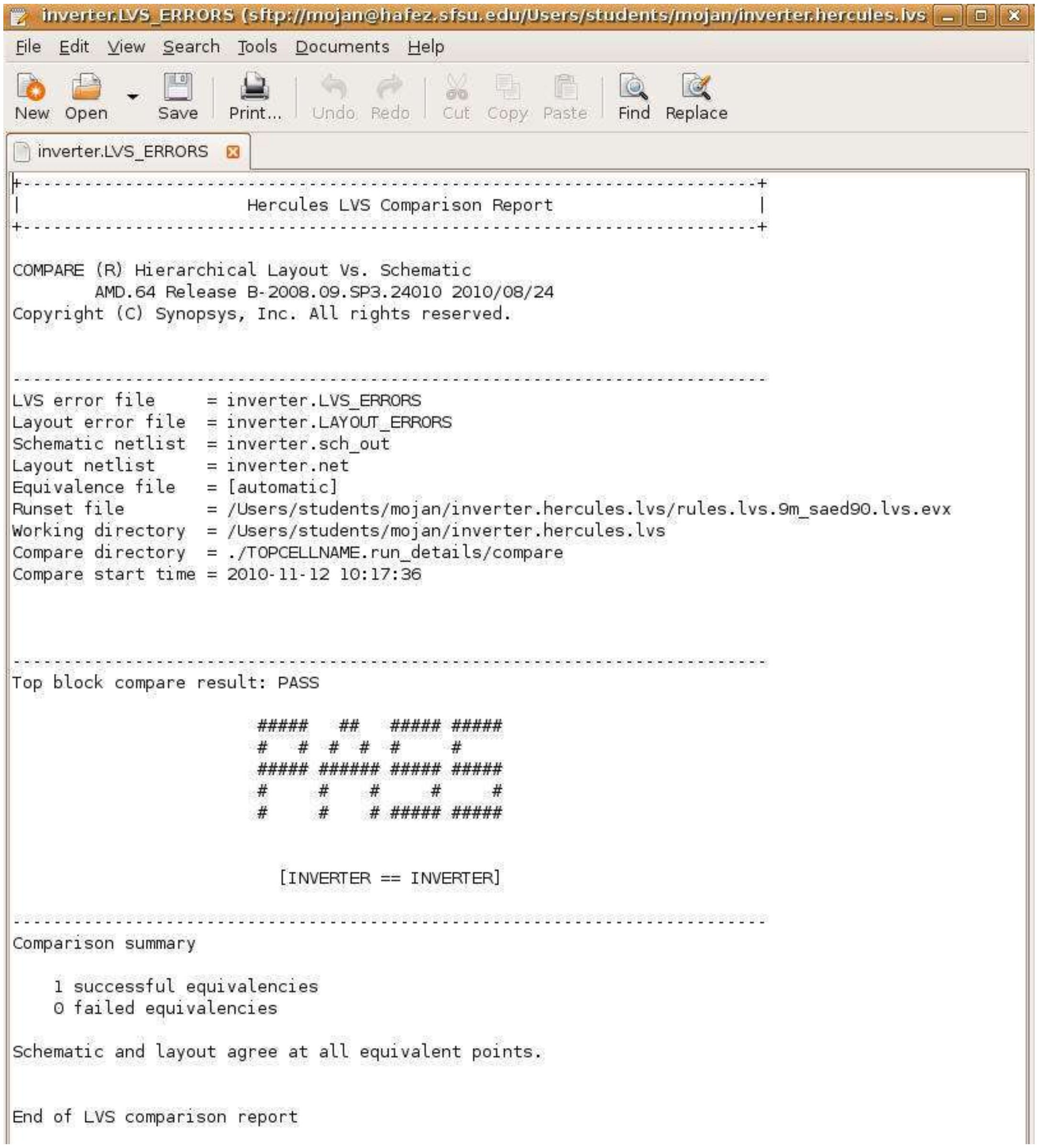


Figure 64: LVS Debugger

Now open the directory that you specified as the Hercules Run directory, in this case ./inverter.hercules.lvs. There should be many new files created by Hercules there now. Open inverter.LVS_ERRORS If you have done everything correctly You should see a "PASS" in the inveter.LVS_ERRORS file. If it says "FAIL" read the errors it reports and try to fix them on the schematic or layout. If the error is in the schematic, make sure to rebuild the spice netlist. Run Hercules again and see if inverter.LVS_ERRORS now says "PASS" Fig. 65.




```
+-----------------------------------------------------------------------+
|                      Hercules LVS Comparison Report                   |
+-----------------------------------------------------------------------+

COMPARE (R) Hierarchical Layout Vs. Schematic
        AMD.64 Release B-2008.09.SP3.24010 2010/08/24
Copyright (C) Synopsys, Inc. All rights reserved.


------------------------------------------------------------------------
LVS error file     = inverter.LVS_ERRORS
Layout error file  = inverter.LAYOUT_ERRORS
Schematic netlist  = inverter.sch_out
Layout netlist     = inverter.net
Equivalence file   = [automatic]
Runset file        = /Users/students/mojan/inverter.hercules.lvs/rules.lvs.9m_saed90.lvs.evx
Working directory  = /Users/students/mojan/inverter.hercules.lvs
Compare directory  = ./TOPCELLNAME.run_details/compare
Compare start time = 2010-11-12 10:17:36



------------------------------------------------------------------------
Top block compare result: PASS

                        #####   ##   ##### #####
                        #    # #  #  #     #
                        ##### ###### ##### #####
                        #     #    #     #     #
                        #     #    # ##### #####


                          [INVERTER == INVERTER]

------------------------------------------------------------------------
Comparison summary

    1 successful equivalencies
    0 failed equivalencies

Schematic and layout agree at all equivalent points.


End of LVS comparison report
```

Figure 65: LVS Errors File

Extracting Parasitics

After passing DRC and LVS you can now move on to LPE (Layout Parasitic Extraction). In this phase, resistive and capacitive components will be extracted from the layout. In layout window go to **Verification → LPE → Setup and Run**. Please make sure your setup mirrors the Fig. 66 for the Main tab.

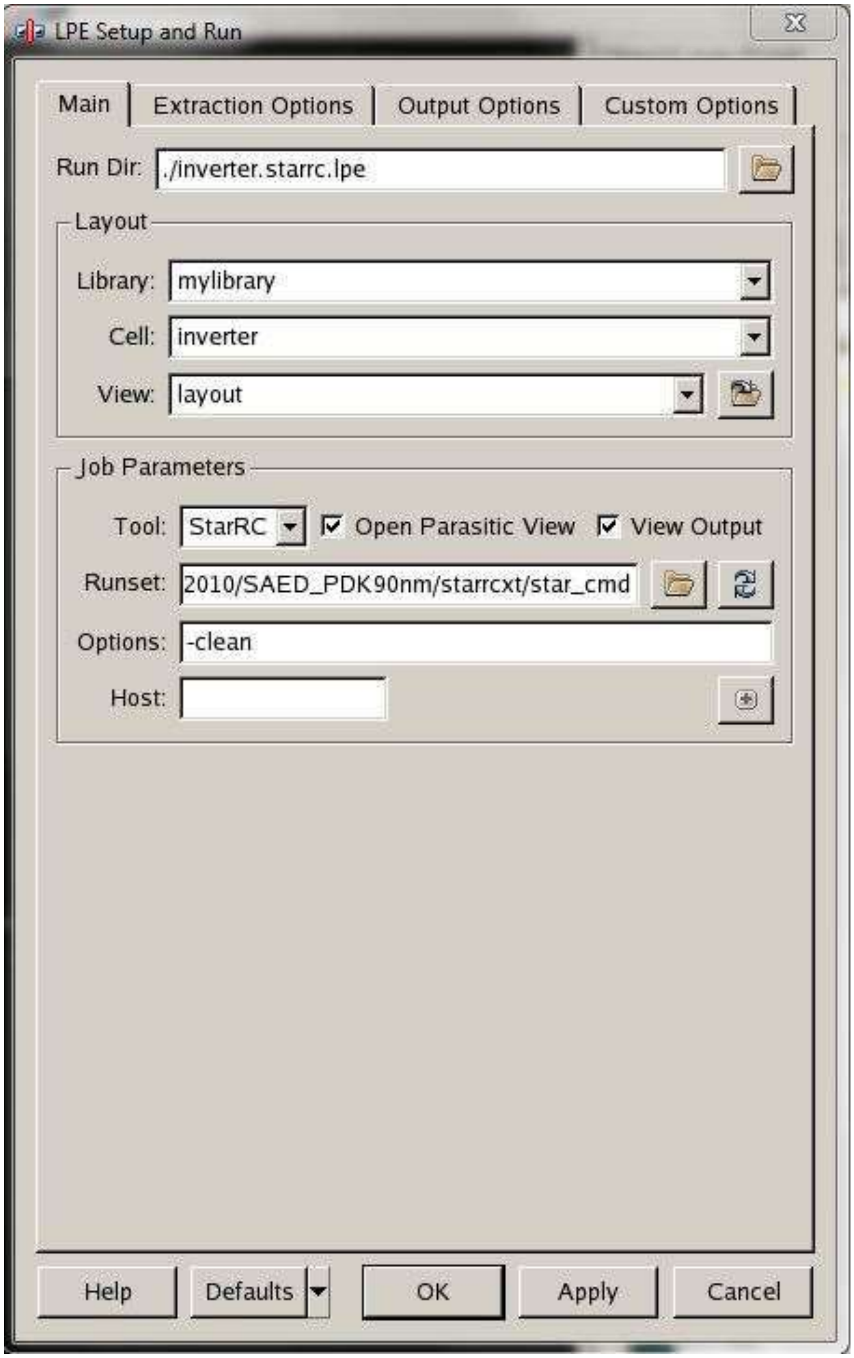


Figure 66: LPE Setup Main Tab

Under the Extraction Option tab select the following file for Mapping File.
/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/starrcxt/saed90nm.map

For Milkyway XTR View select the hercules.lvs folder created from running LVS, then select the TOPCELLNAME_MILKWAY folder. See fig 67 below.

Make sure the other options in this tab match up with figure 67.

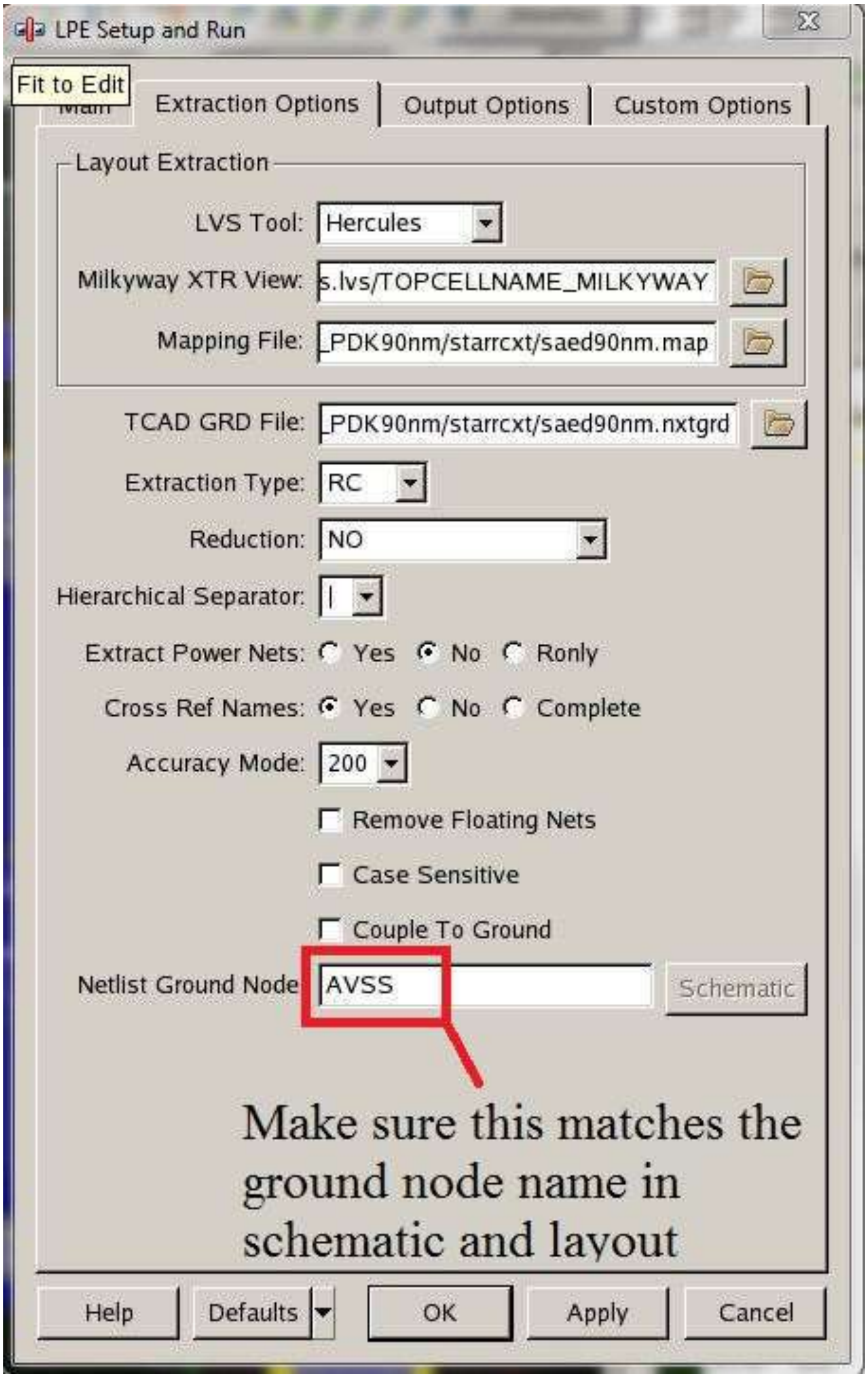


Figure 67: LPE Setup Extraction Options

Under Output Options tab type make sure that you have the same setup as shown in Fig. 68. Make sure the following map files are set as noted below if not already set by default.
Device Map:
/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/starrcxt/device_map
Layer Map:
/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/starrcxt/output_layer_map

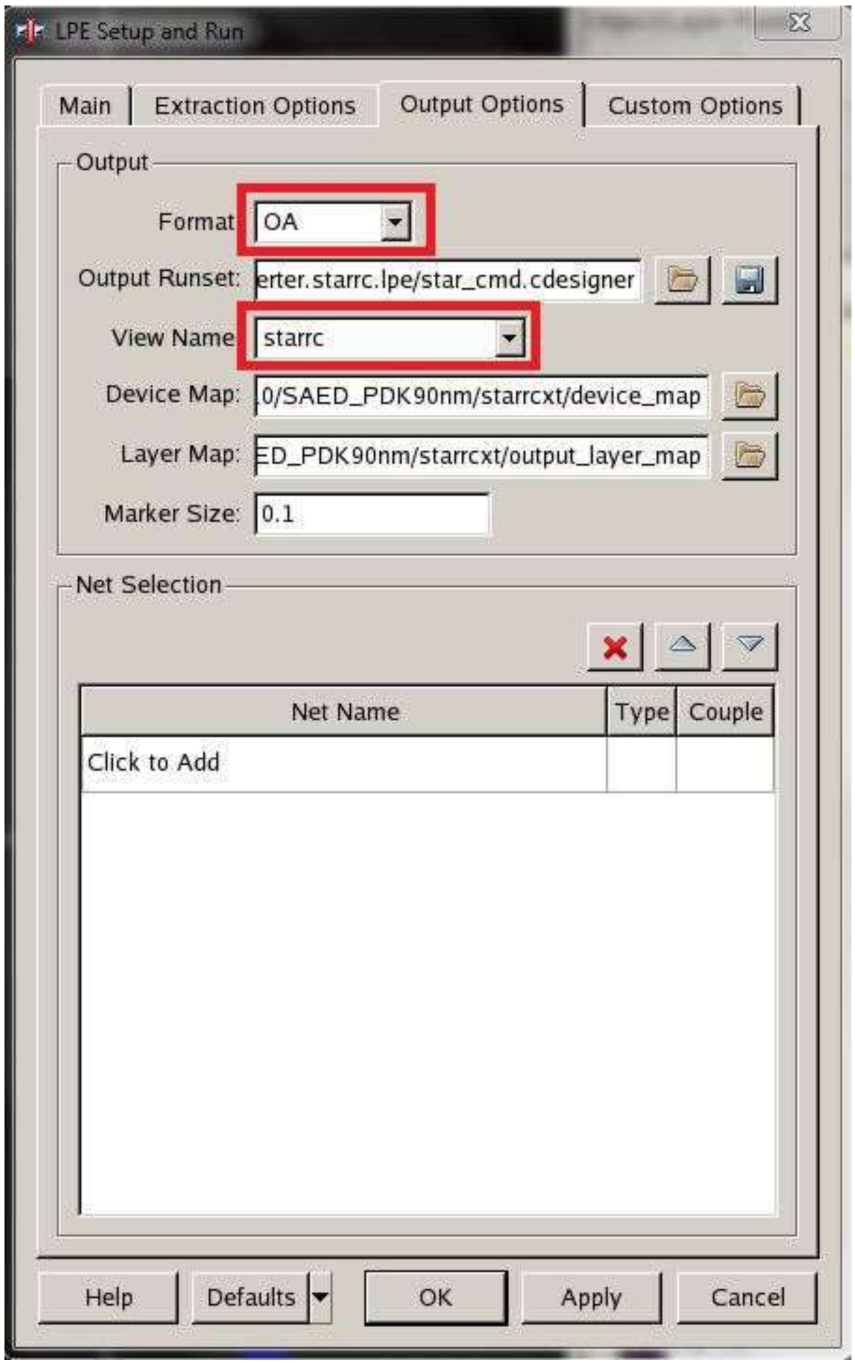


Figure 68: LPE Setup

There is nothing you need to change under the custom options tab so you can leave that as default.

Now click on **OK**. Then you will see Customer Designer Text Viewer as Fig. 69.

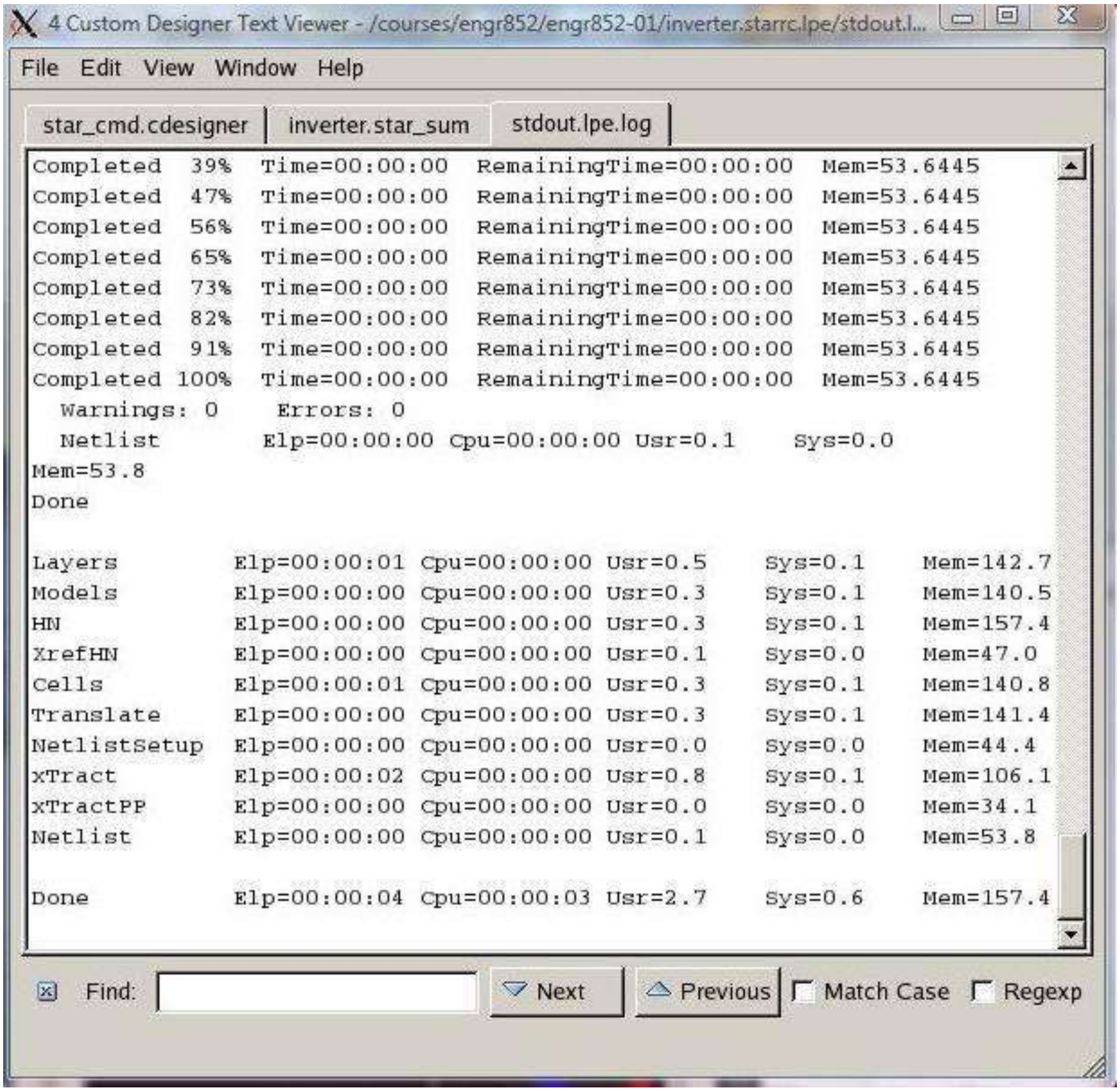


Figure 69: Console Output

After some time you should able to see the following message as shown in the console window if the parasitics were generated correctly. See figure 70.

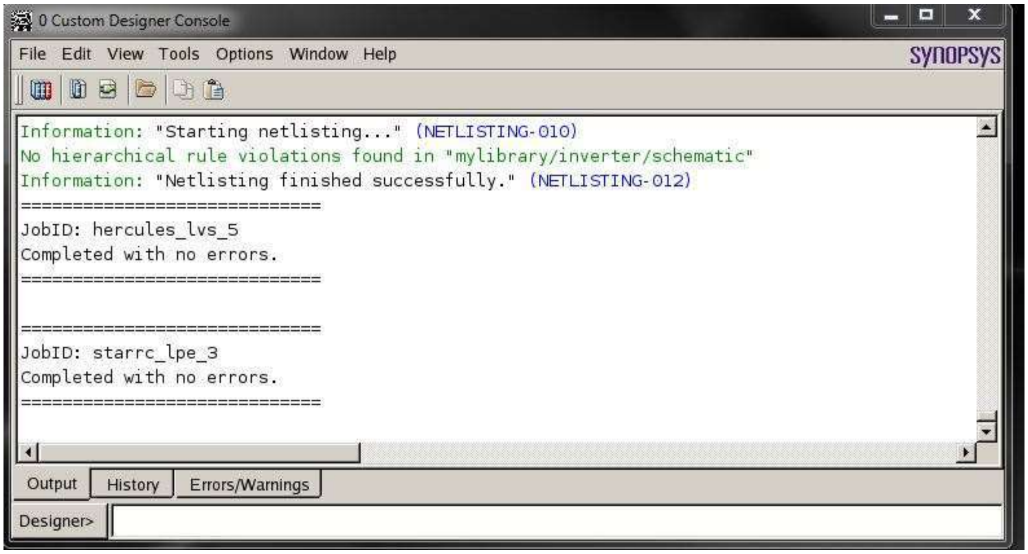


Figure 70: Passing LPE

After LPE has successfully run, a parasitic view window should open. See figure 71 and 72 for reference. The RC components are very small in the window that opens up (fig 71) so you may need to zoom in to see the details (fig 72). It may help to drag a box around the RC components using the mouse cursor to highlight them, then zoom in to see them. Also note that your parasitic view may not match exactly as shown below which is fine since this depends on differences in layout

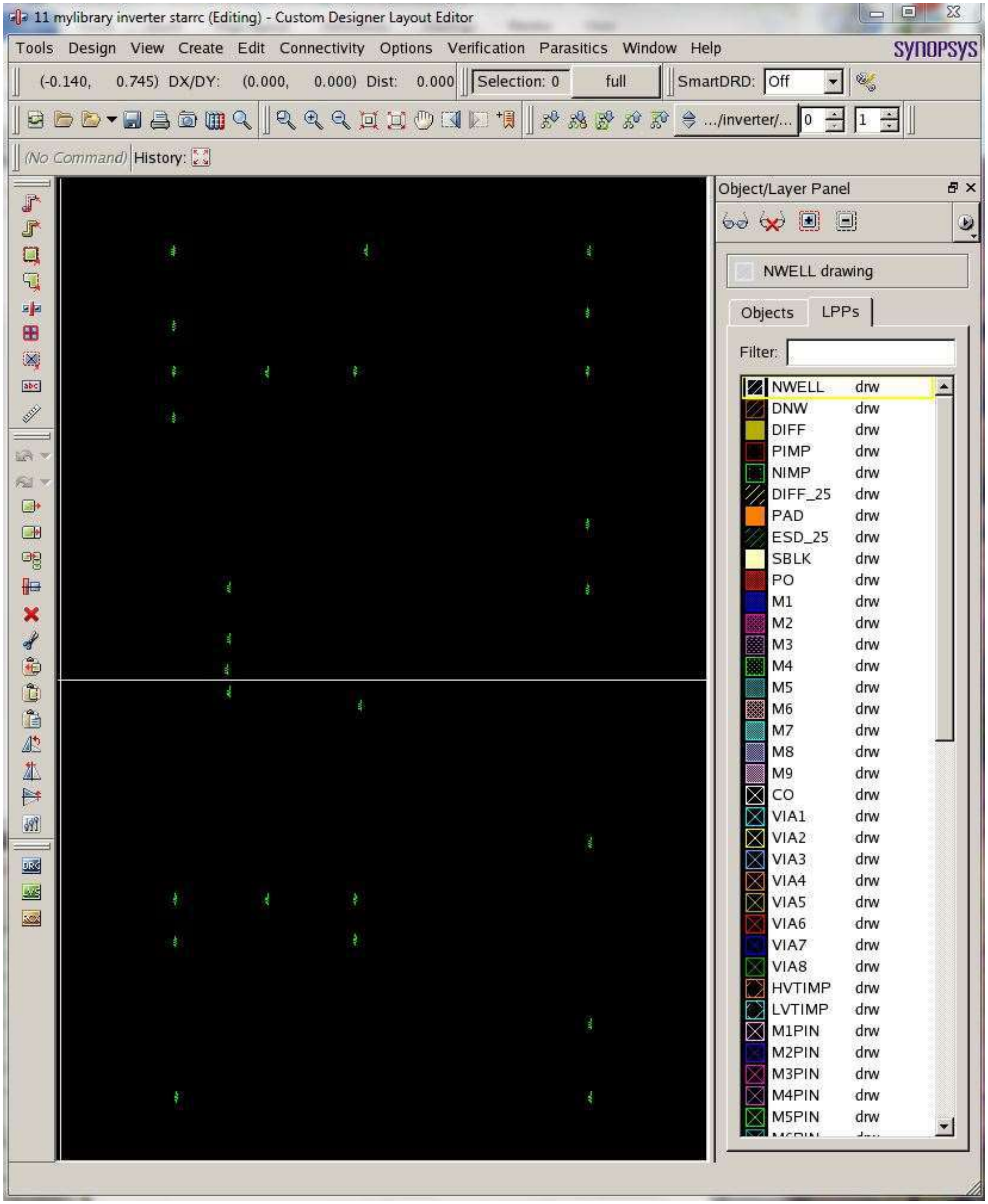


Figure 71: Parasitic View

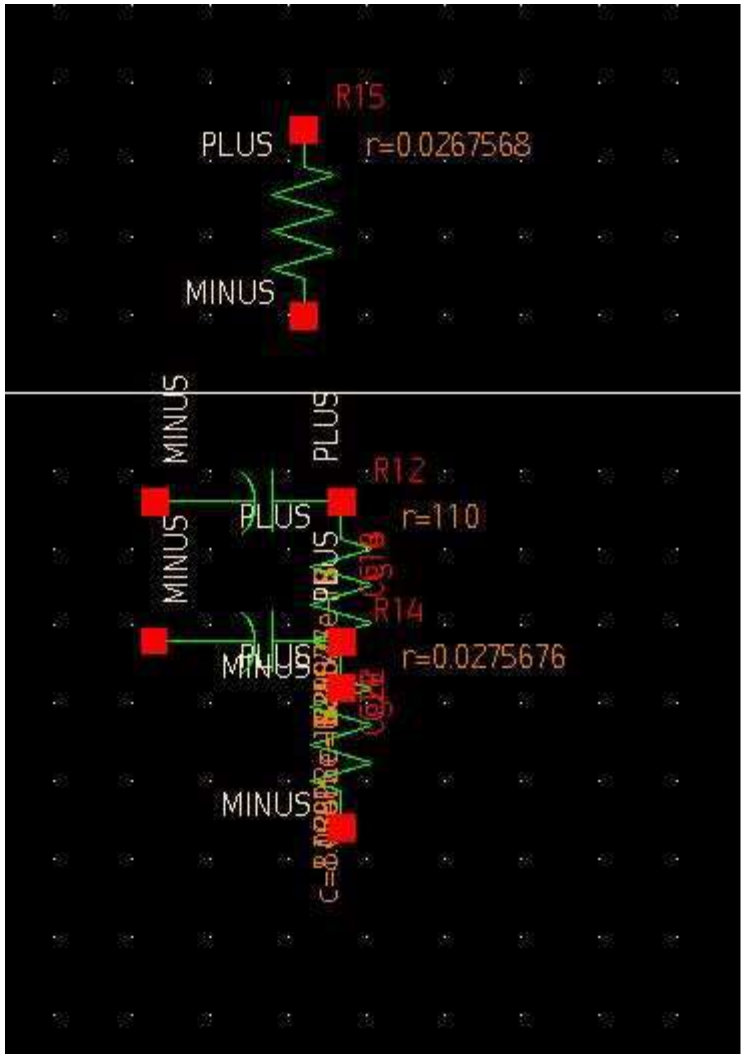


Figure 72: Parasitic View Zoom

You have successfully generated the parasitic view for the inverter and are ready to run post layout simulation. Save the parasitic view with **Design → Save**. The parasitic view will be saved as “starrc” for the view name.

Part 6: Post Layout Simulation

After parasitic extraction we want to apply the parasitics to the schematic for a more accurate representation of our inverter and test it. From the Custom Designer Console window go to **File → Open Design** and open the inverter circuit schematic created earlier.

Select the following options for the following columns:
Libraries: mylibrary
Cells: inverter_schematic
Views: schematic

See figure 73 for reference. Once all options are highlighted, click **Open** and the schematic that was created earlier for testing the inverter will open, see figure 74 for reference.

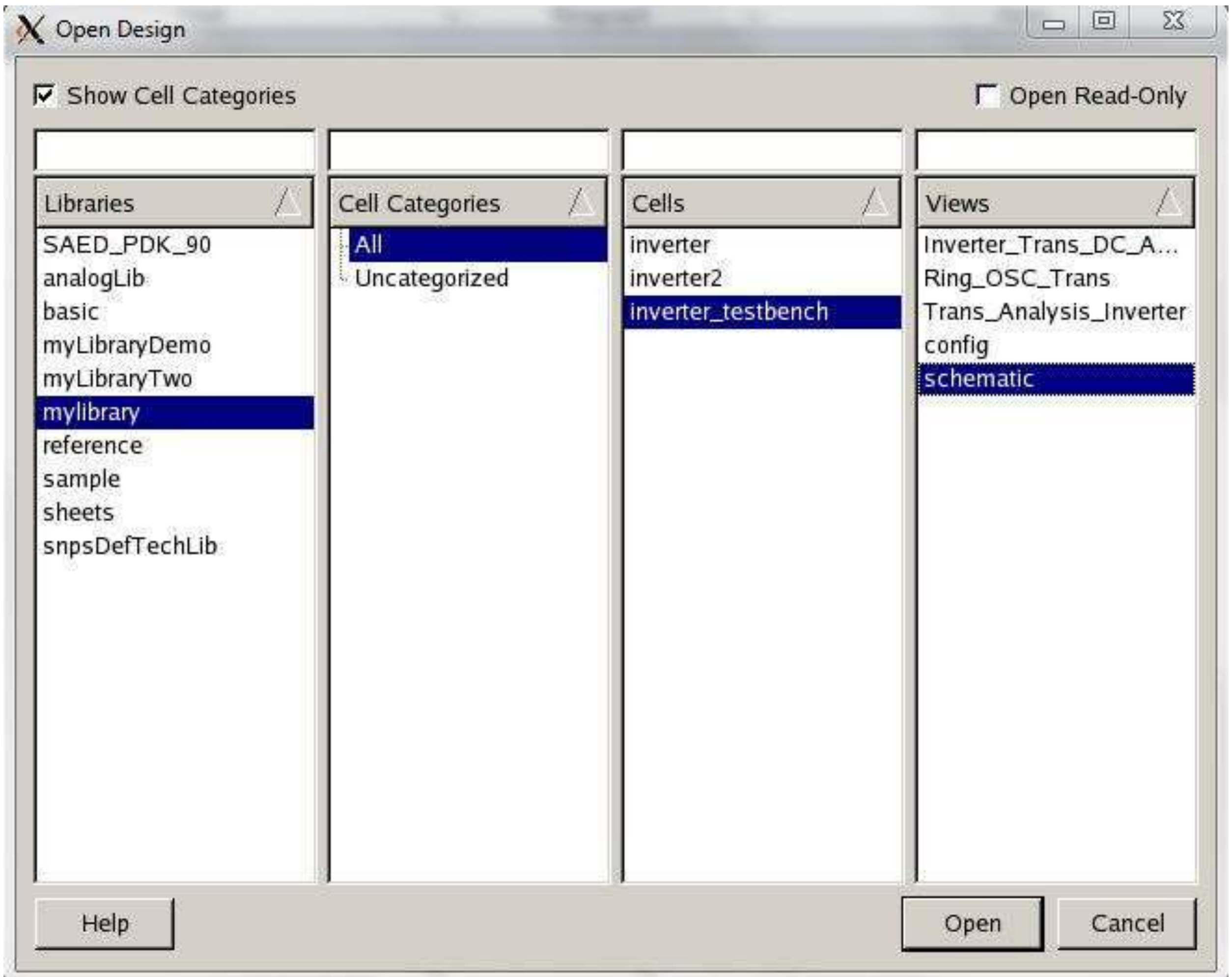


Figure 73: Opening Circuit with Inverter Schematic

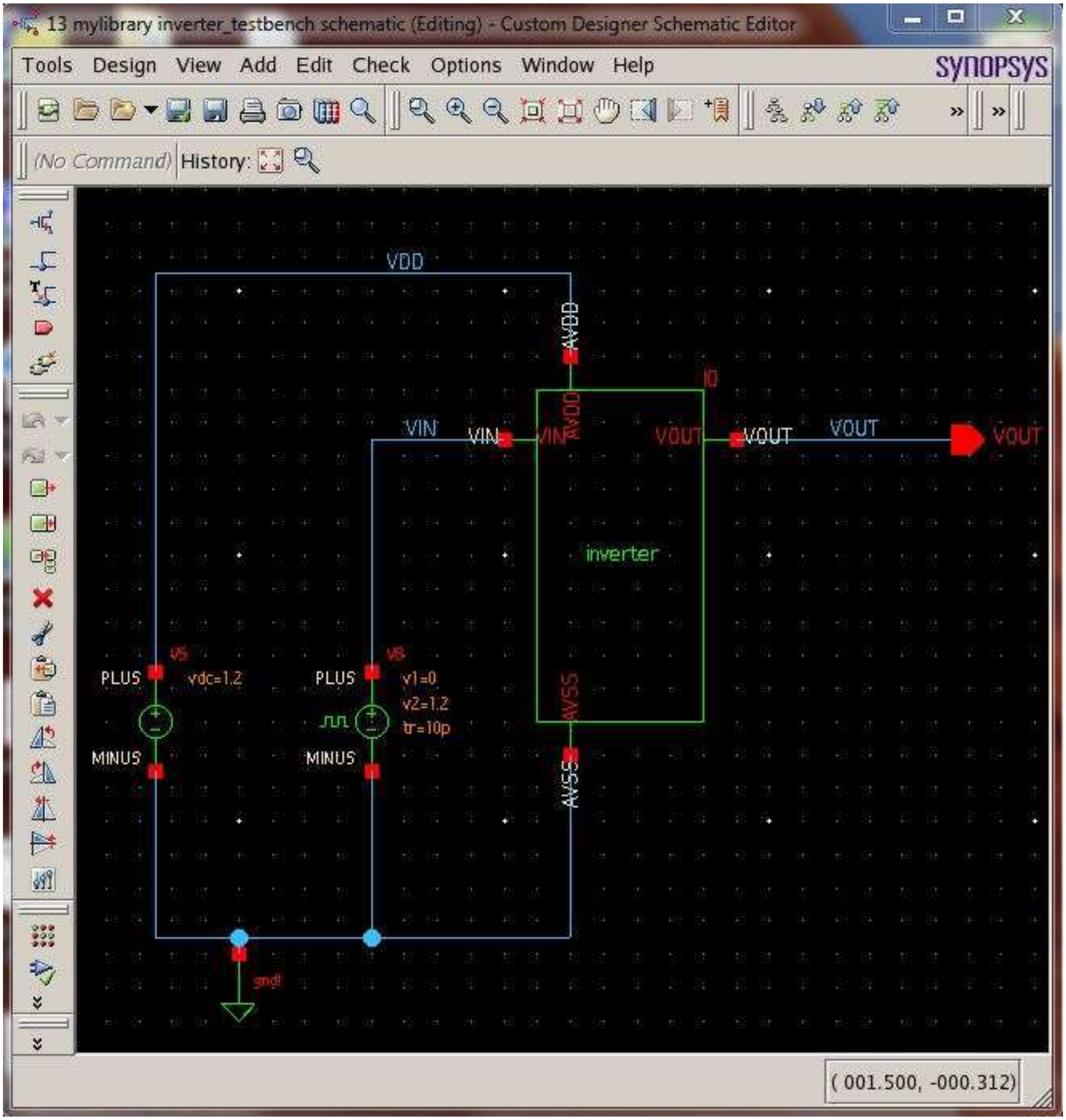


Figure 74: Testbench Circuit for Inverter

After opening the testbench circuit, make sure the circuit is similar to the one shown in figure 74. Now the parasitics need to be loaded into the inverter cell used in the circuit. From the Custom Designer Console window, go to **File → New → CellView.**

In the New CellView window, create a new configurations file as follows:

- Select **inverter_testbench** for the cell.
- Set view name to **config**

- Set editor to **HE – config**

See figure 75 below for reference and click **OK**.

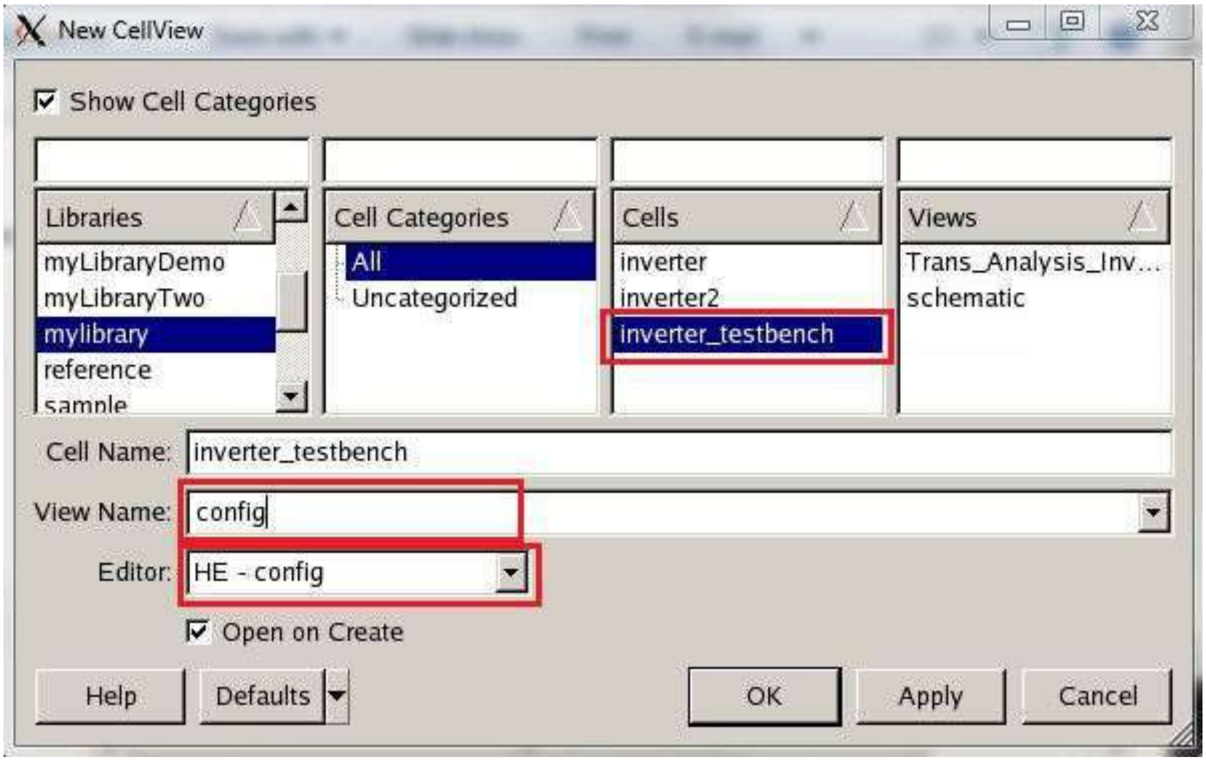


Figure 75: Configurations File Setup

A Hierarchy Editor window will display. Setup the view and list options as noted in figure 76 below.
View: **schematic**
View Search List: **schematic hspice symbol**
View Stop List: **symbol**

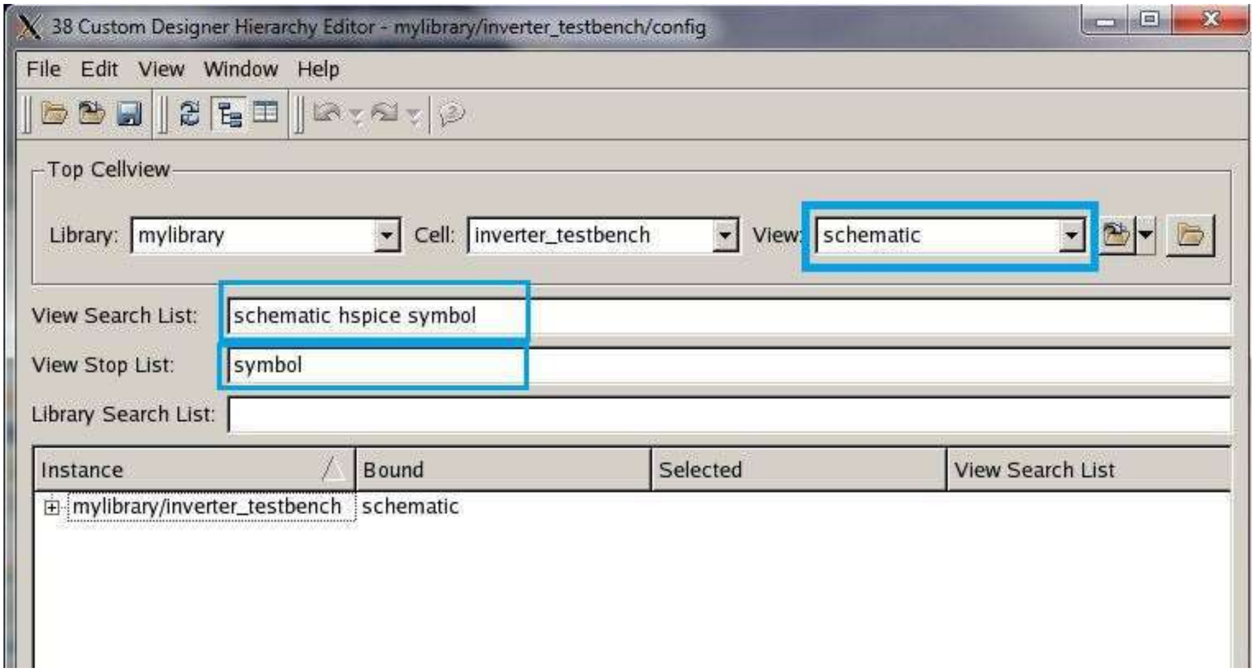


Figure 76: Hierarchy Editor

In order to map the parasitics generated from LPE to your inverter cell, select **starrc** under the "Selected" column for the inverter instance, see figure 77 for reference. The format for your inverter instance name in the instance column is *schematic_instance_name* (mylibrary, inverter). Notice that as starrc is selected, resistor and capacitor instances will show up under the instance column, this substitution replaces the inverter cell with its equivalent schematic containing its resistive and capacitive components. Afterwards save the settings by going to **File → Save**.

For future reference, to apply parasitics for a general case, a schematic must have its equivalent schematic symbol and layout created since a layout is used to generate the parasitics and a schematic symbol is used as a vessel to hold the parasitics. The testbench schematic is created to test the symbol containing the schematic with its applied parasitics.

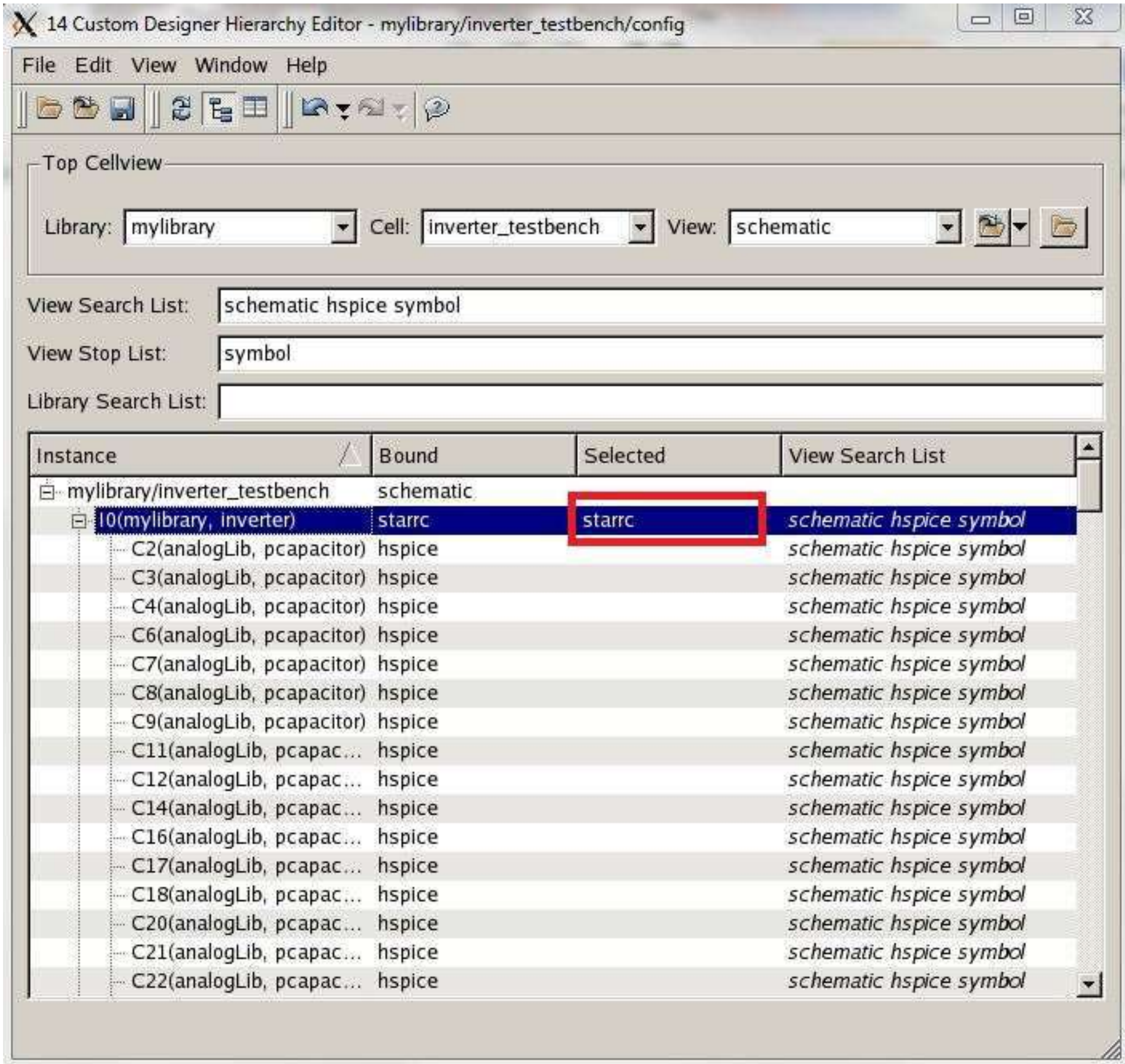


Figure 77: Loading Parasitics into the Inverter

To start simulation with parasitics, go to **File → Open Design** from the Custom Designer Console window. In the Open Design window that opens, select **inverter_testbench** under the cells column and **config** under the views column and right click on config under views. Select **Open Design** from the drop down menu. See figure 78 below for reference.

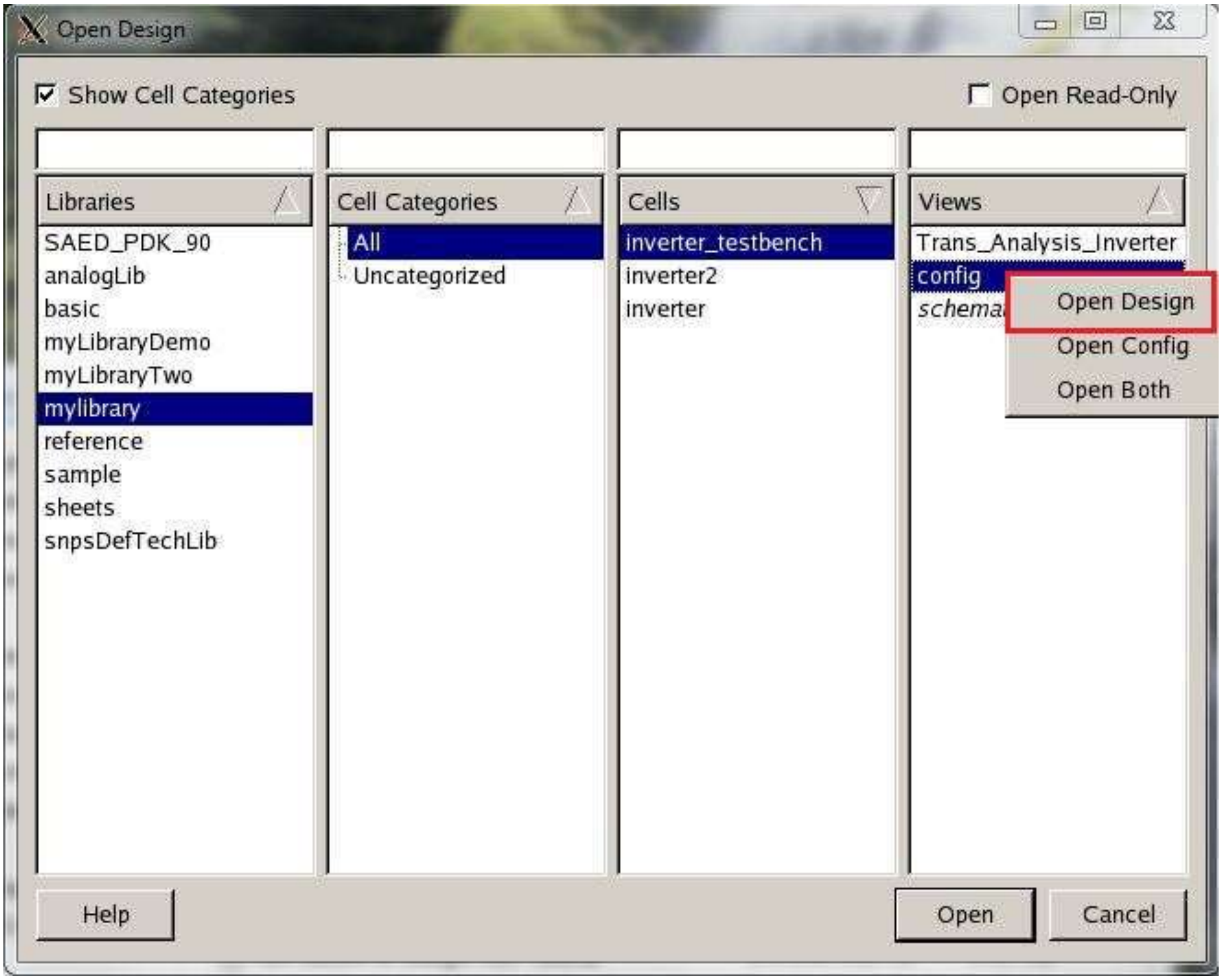


Figure 78: Open Design with Parasitics

Afterwards, a schematic view should open up with parasitics applied, see figure 79 below for reference. To check if the parasitics were applied, you can double click on the inverter symbol/cell and it should display the same parasitics view that was generated from running LPE. To change between parasitic and schematic views, select the desired view in the red box noted in figure 80.

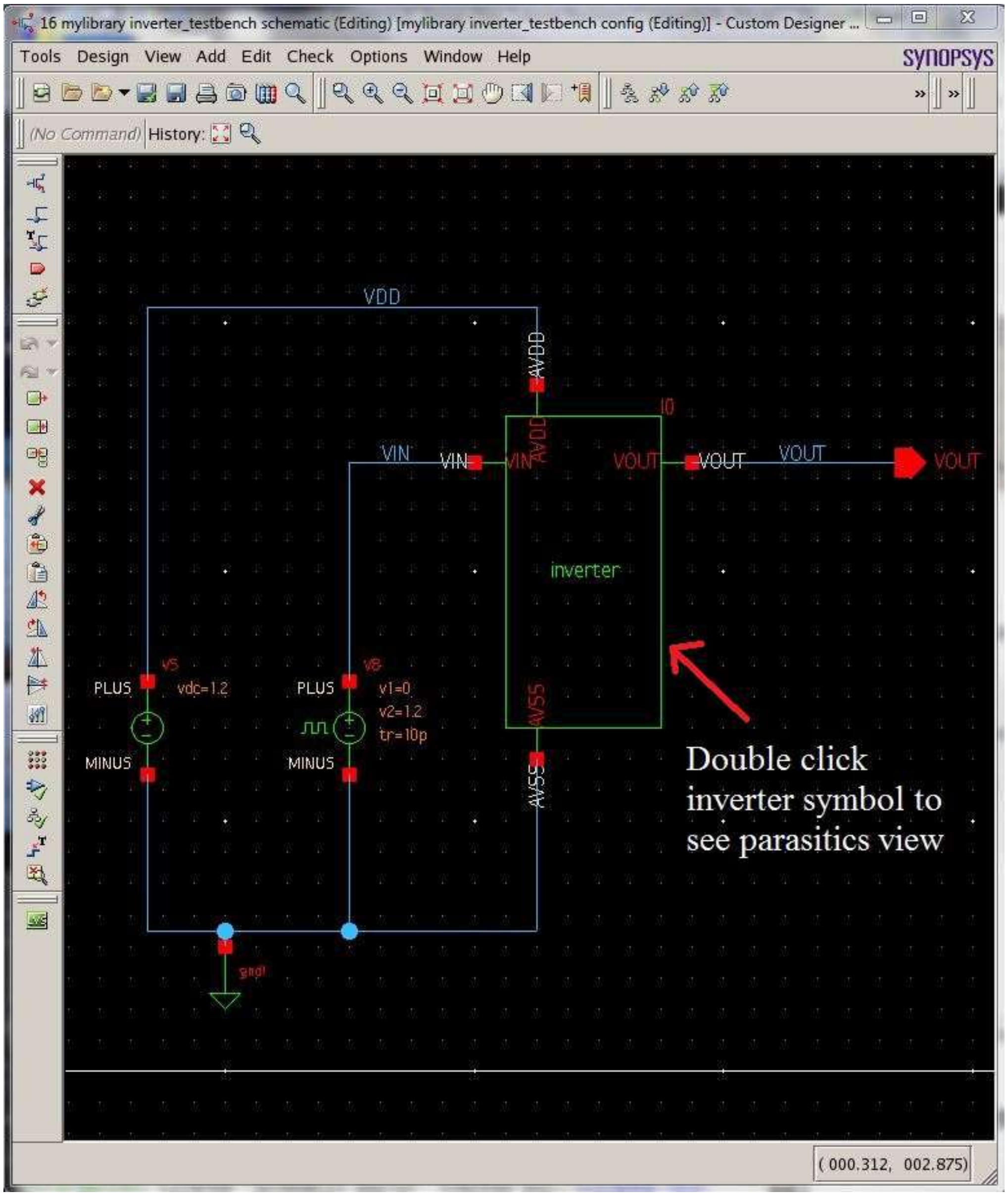


Figure 79: Schematic with Parasitics Applied to the Inverter

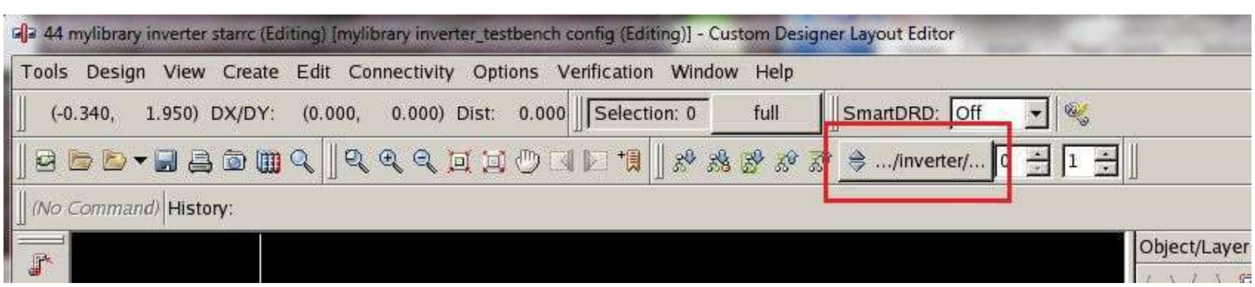


Figure 80: Changing Between Views

Now go ahead and simulate your circuit as you did previously in Part 3 of the tutorial. From the schematic window, go to **Tools → SAE** to open a new SAE window. When comparing the two waveforms (inverter parasitics to inverter without parasitics) take note of the difference between delays from VIN to VOUT for transient waveforms.

Tip:
If there is a mismatch error in the console regarding mismatched nets that are uppercase and lowercase between parasitics (starrc) and the symbol when running the SAE simulation, set all pin names to use **uppercase letters** in schematic and set all layout labels to use uppercase letters as well. After this change, you need to run LVS, LPE, regenerate the inverter symbol, redo the configurations file for the testbench, and rerun the SAE.

Part 7: Hierarchical Design

Using smaller instances of circuits to create a larger design is what hierarchical design is all about. In this section, we use an inverter we created earlier and use several instances of it to create a five stage oscillator in schematic and layout views.

Create a new schematic for the ring oscillator by going to **New → CellView** from the Custom Designer Console and setup the options as shown in figure 81 below. The setup is as follows:

Library: **mylibrary**
Cell Name: **ringOscillator**
View Name: **schematic**
Editor: **SE - schematic**

Click **OK** when done.

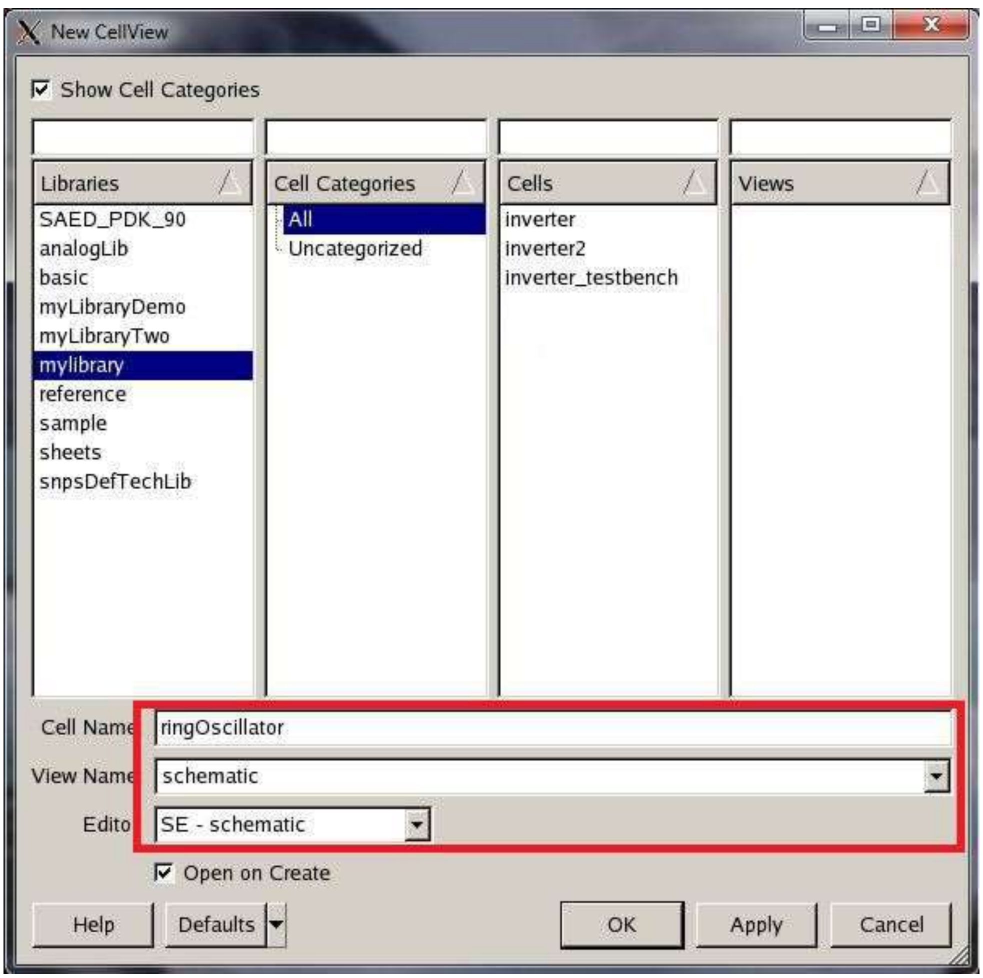


Figure 81: Ring Oscillator Schematic Setup

In the schematic window, building a ring oscillator circuit with pins as shown below in figure 82.

For the inverter instances, look for them in **Add → Instance** to open the add instance window. In the add instance window, choose **mylibrary** for library, **inverter** for cell, and **symbol** for the view and place five instances of the inverter on the schematic.

Add wires with **Add → Wire.**

For the pins, go to **Add → Pins** and place two **input** pins for the AVDD and AVSS signals, and place five **input/Output** pins at each inverter output. For the five input/output pins, I called them VIO1-5 in the schematic. Feel free to give the wires the same names as the pins using **Add → Wire Name**. Also as a convention, use uppercase letters for pin names instead of lowercase letters.

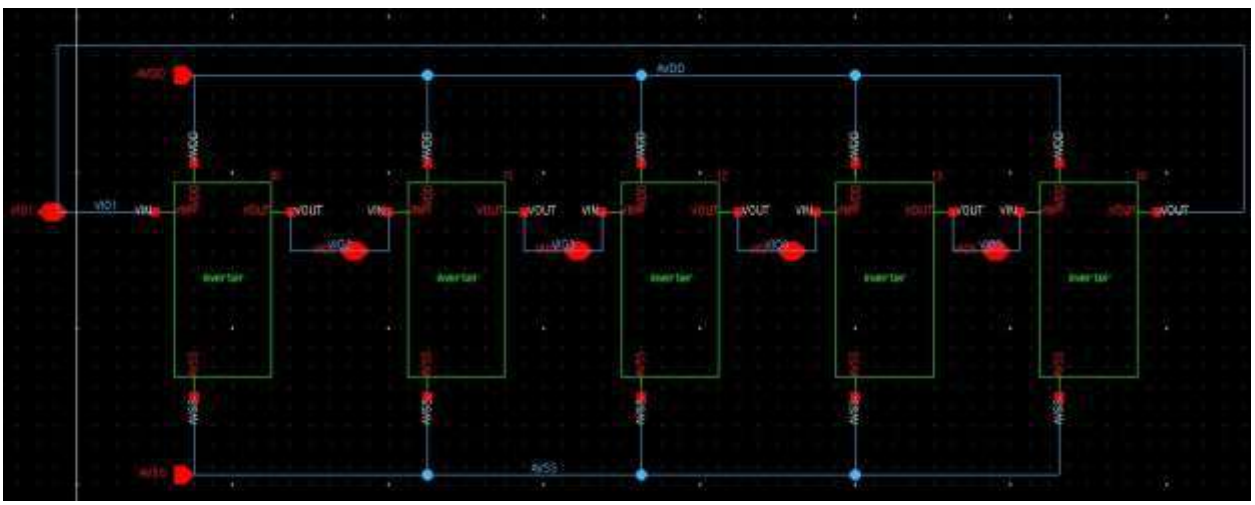


Figure 82: Ring Oscillator Schematic

Save your schematic using **Design → Save**. Now create a symbol of your inverter using **Design → New CellView →From CellView**. Make sure your options match up as shown below in figure 83 and click **OK**.

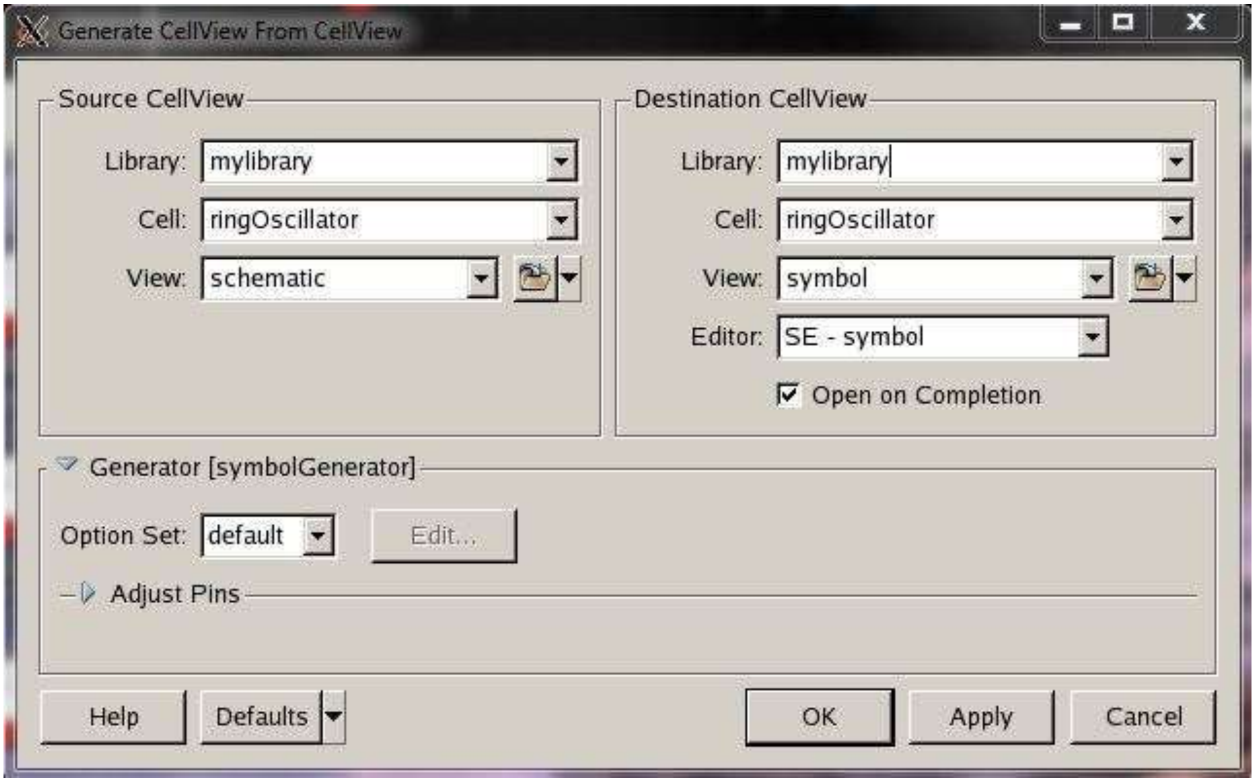


Figure 83: Ring Oscillator Symbol Setup

After clicking OK, a new schematic window opens up with the ring oscillator symbol. Feel free to move around the pin placements for a better pin organization. See figure 84 below for reference. Save the symbol when done with **Design → Save**.

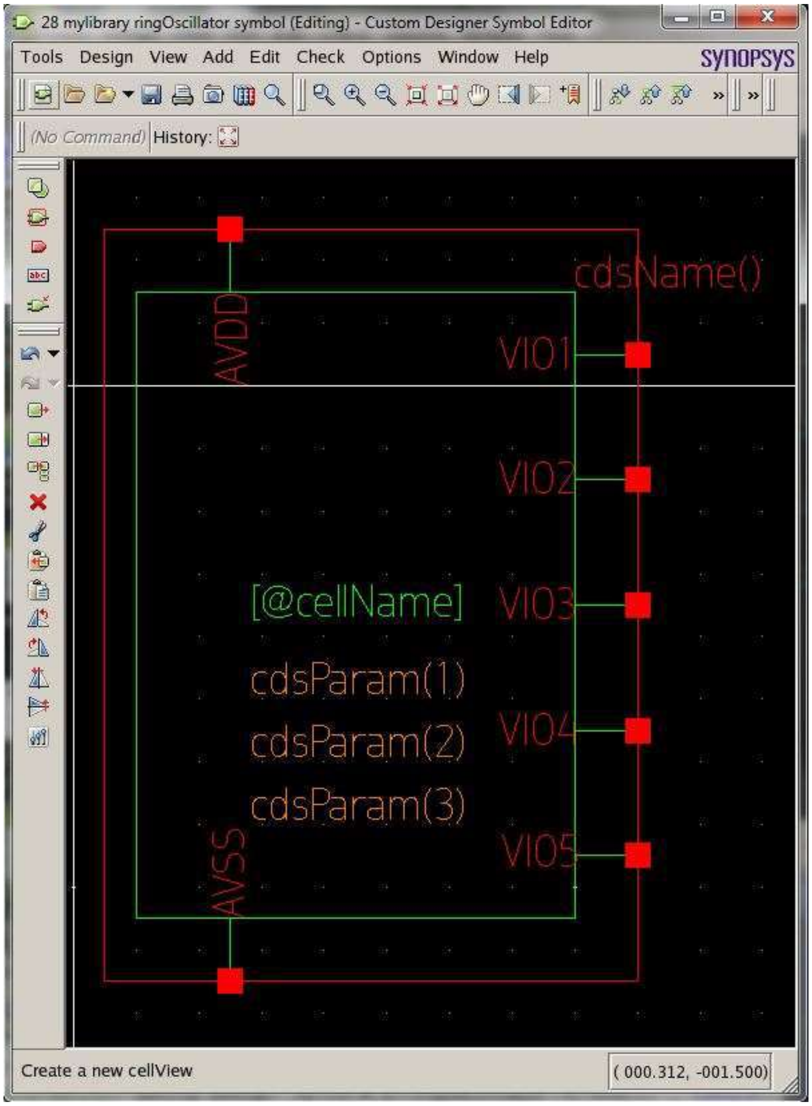


Figure 84: Ring Oscillator Symbol

Now create a new schematic to use as a testbench for the ring oscillator by going to **New → CellView** from the Custom Designer Console and setup the options as shown in figure 85 below. The setup is as follows:

Library: **mylibrary**
Cell Name: **ringOscillator_testbench**
View Name: **schematic**
Editor: **SE - schematic**

Click **OK** when done.

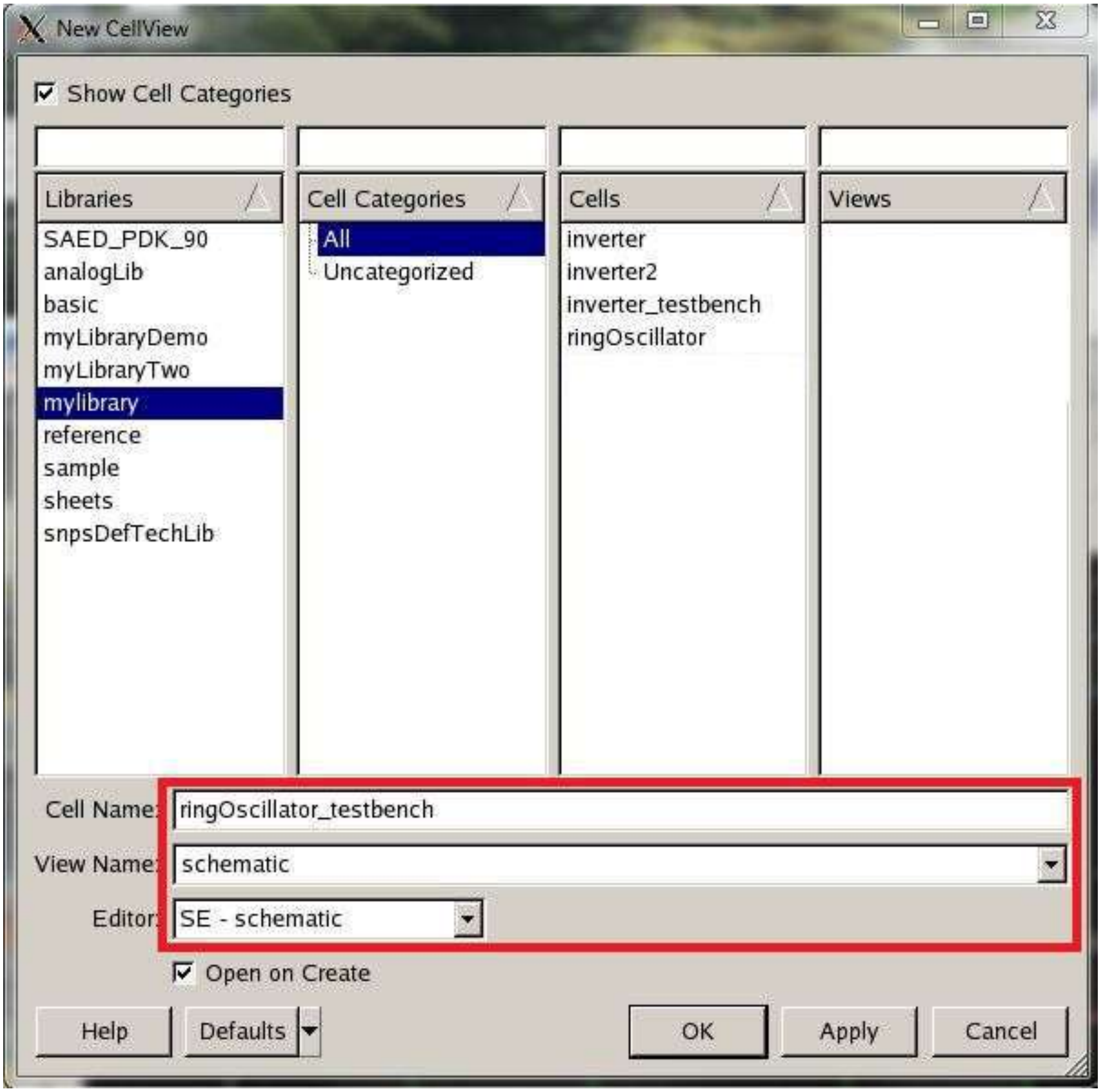


Figure 85: Ring Oscillator Testbench Setup

Afterwards a new schematic window should open. In the new schematic window, setup the ring oscillator testbench circuit as shown in figure 86.

To place a ringOscillator instance, look for them in **Add → Instance** to open the add instance window. In the add instance window, choose **mylibrary** for library, **ringOscillator** for cell, and **symbol** for the view and place an instance of your ring oscillator on the schematic. Also place an instance of ground and a voltage source in the schematic. You can find these instances under library: **analogLib** and cell: **gnd** and cell: **vsource** respectively. For the voltage source, set the voltage to 1.2 volts.

Add wires with **Add → Wire.**

For the pins, go to **Add → Pins** and place five output pins for each of the five VIO# pins. Feel free to give the wires the same names as the pins using **Add → Wire Name**.

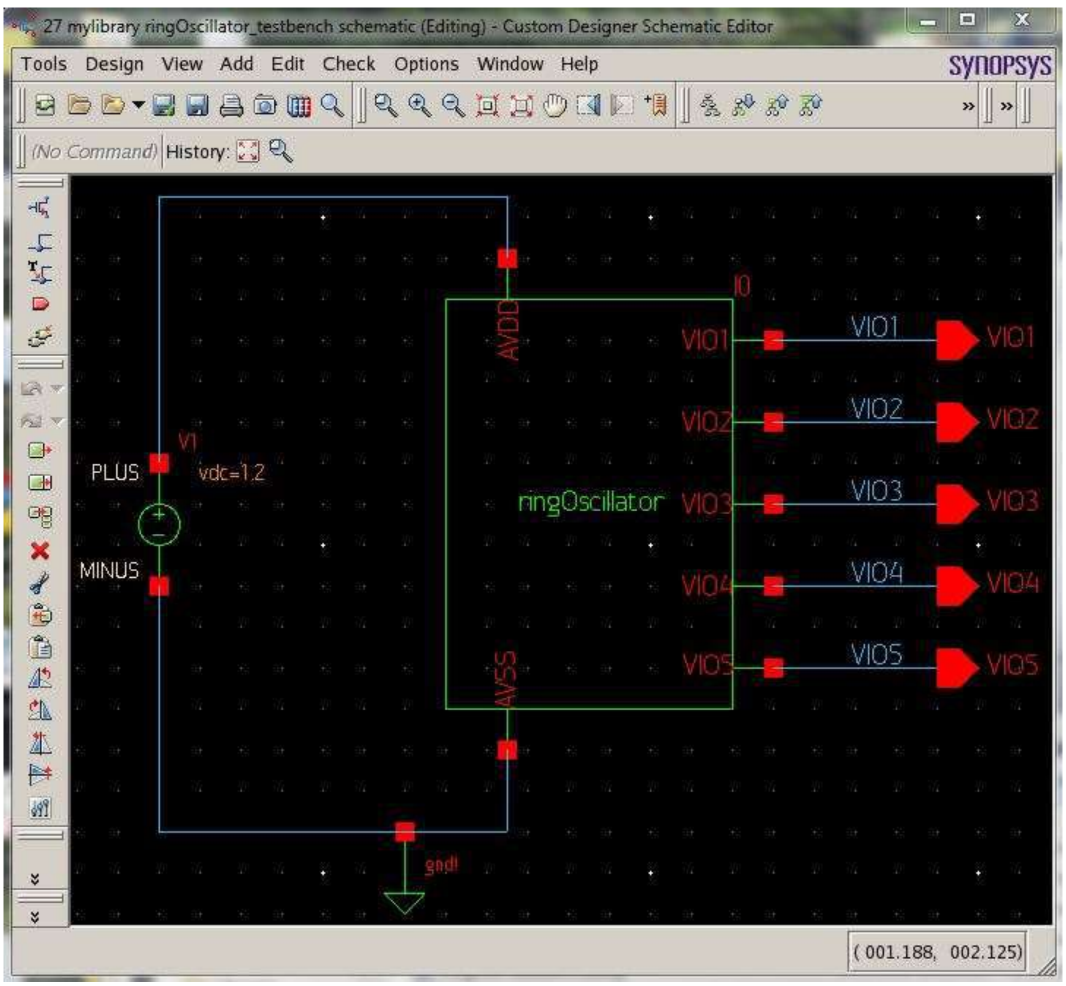


Figure 86: Ring Oscillator Testbench Circuit

Save with **Design → Save** once your testbench circuit is done. Now we need to create a new layout so go to **New → CellView** from the Custom Designer Console and setup the options as shown in figure 87 below.

Library: **mylibrary**
Cell Name: **ringOscillator**
View Name: **layout**
Editor: **LE - layout**

Click **OK** when done.

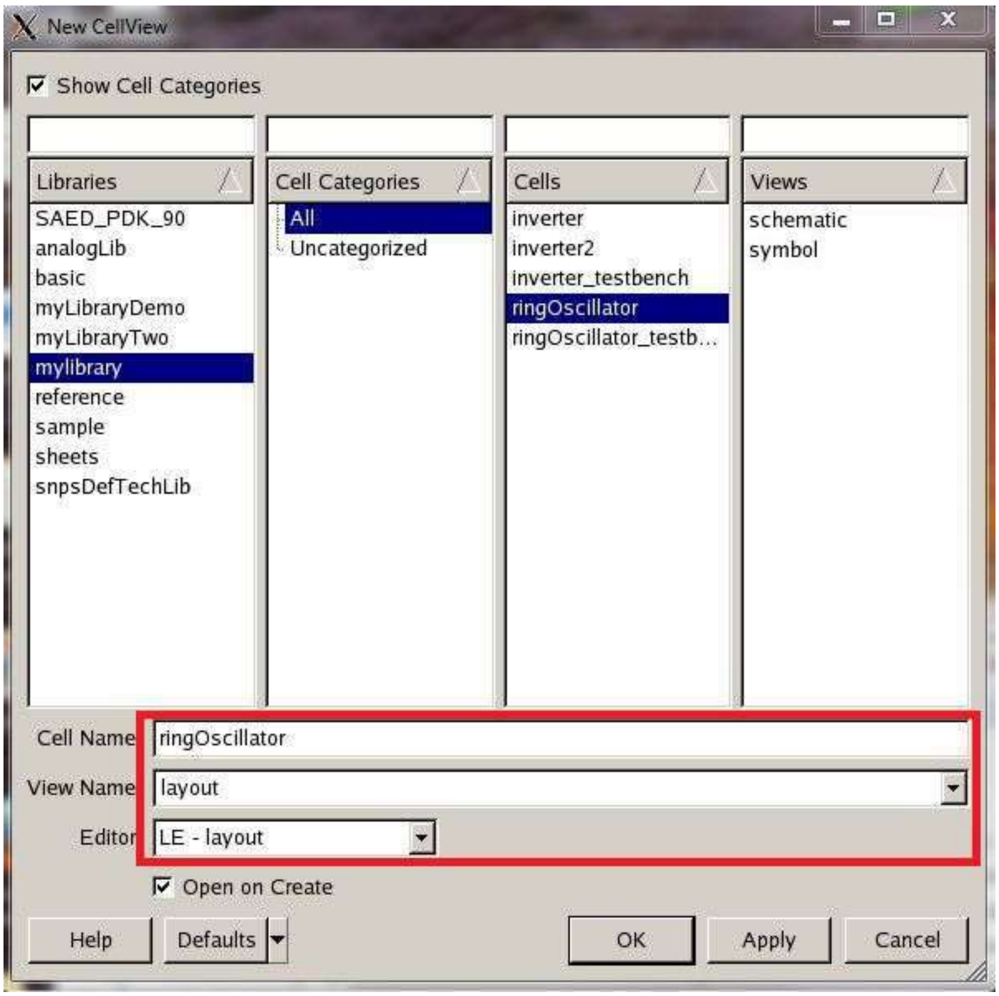


Figure 87: Ring Oscillator Layout Setup

In the new layout window, we can use the layout of the inverter created earlier to build a ring oscillator circuit. Go to **Create → Instance** to open up a new create instance window. In the window select **mylibrary** for library, **inverter** for cell, and **layout** for the view and place five instances of the inverter layout on the layout screen. See figure 88 for reference.

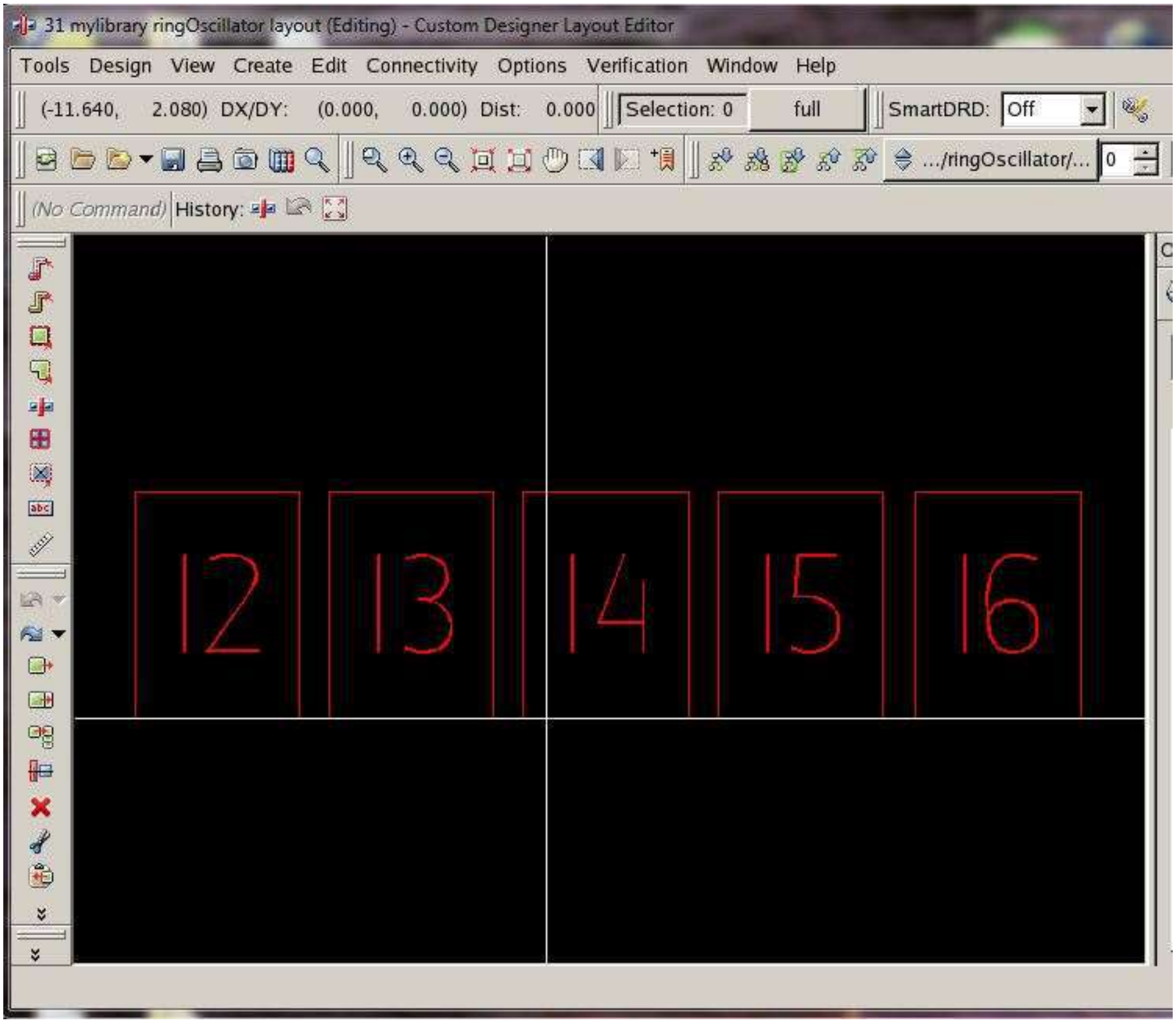


Figure 88: Placing Five Inverter Layout Instances

Notice that the layout components for the inverter layouts don't display. This is because the inverter layouts are hiding one level up in the hierarchy. In order to view them, change the hierarchy bounds as shown in figure 89 below. The numbers represent a range of hierarchy levels that are displayed where the left number is the lower limit and the right number is the higher limit. Afterwards the inverter layouts are viewable as shown in figure 90.

Figure 89: Placing Five Inverter Layout Instances

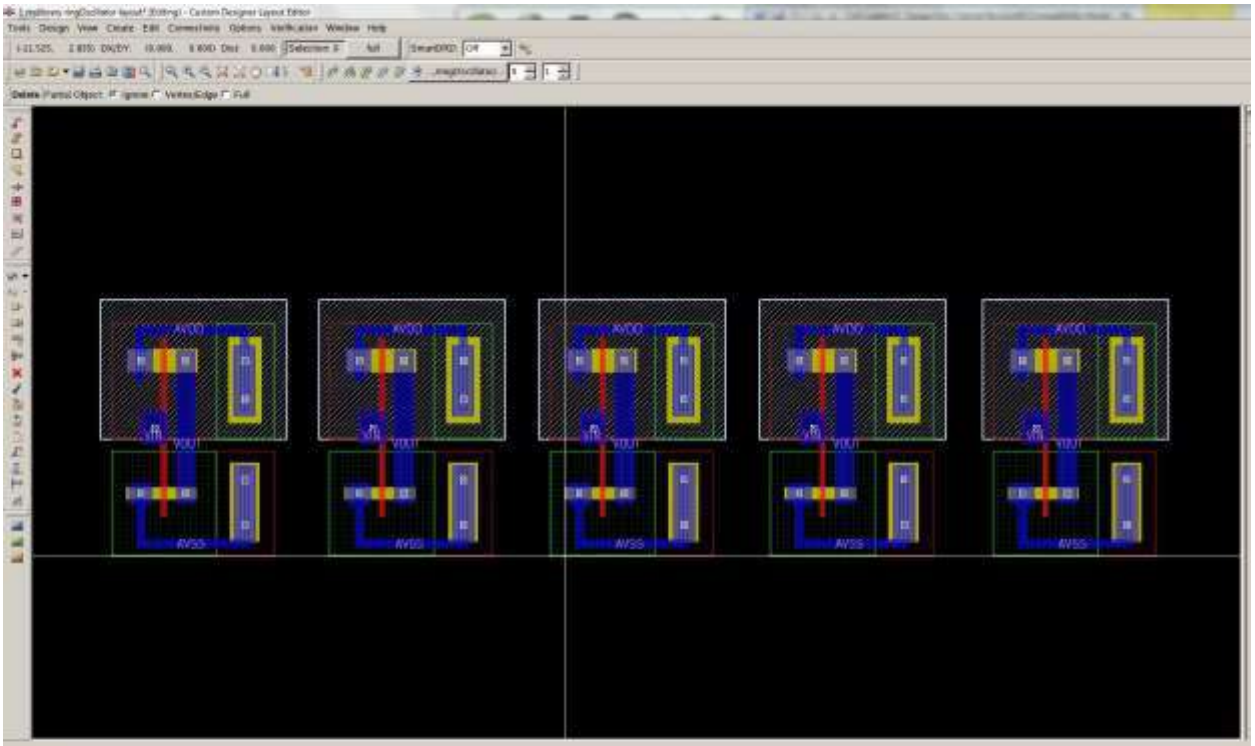


Figure 90: Viewing Inverter Layout Instances

Now draw metal paths with **Create → Path** using the **M1** layer under the LPP panel. Connect all AVDD signals with a single M1 connection and all AVSS signals with a single M1 connection. Also connect the output of an inverter to the input of the next inverter using the **M1** layer. See figure 91 for M1 connections.

In addition you need to add labels for the metal connections just added. To add labels, select the **M1PIN** layer in the LPP panel and go to **Create → Text → Label**. Enter a name for each label in the box noted in figure 92 and place the text labels as noted by the red boxes in figure 91. Label names used are: **AVDD, AVSS, VIO1, VIO2, VIO3, VIO4, and VIO5**. Remember that in order to pass LVS, your M1PIN label names in layout need to match up with the pin names from your ring oscillator schematic. Also names for schematic pins and names for layout labels should use uppercase letters. Save the layout.

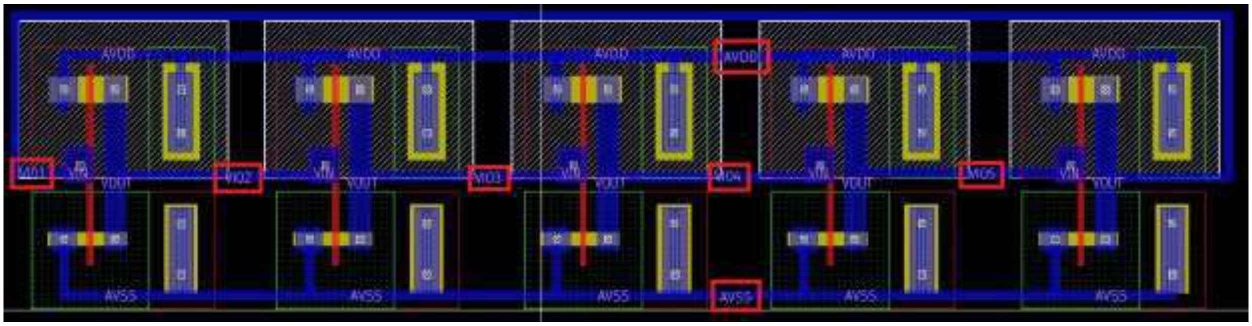


Figure 91: Ring Oscillator Layout

Figure 92: Label Text

After your layout matches figure 91, go to **Verification → DRC → Setup and Run** to setup and run DRC (as done earlier in part 5 of the tutorial). Your options for DRC should match figure 93. Leave the options on the custom tab as their defaults. Click **OK** when done.

Runset file for main tab:
/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/hercules/drc/rules.drc.9m_saed90.ev

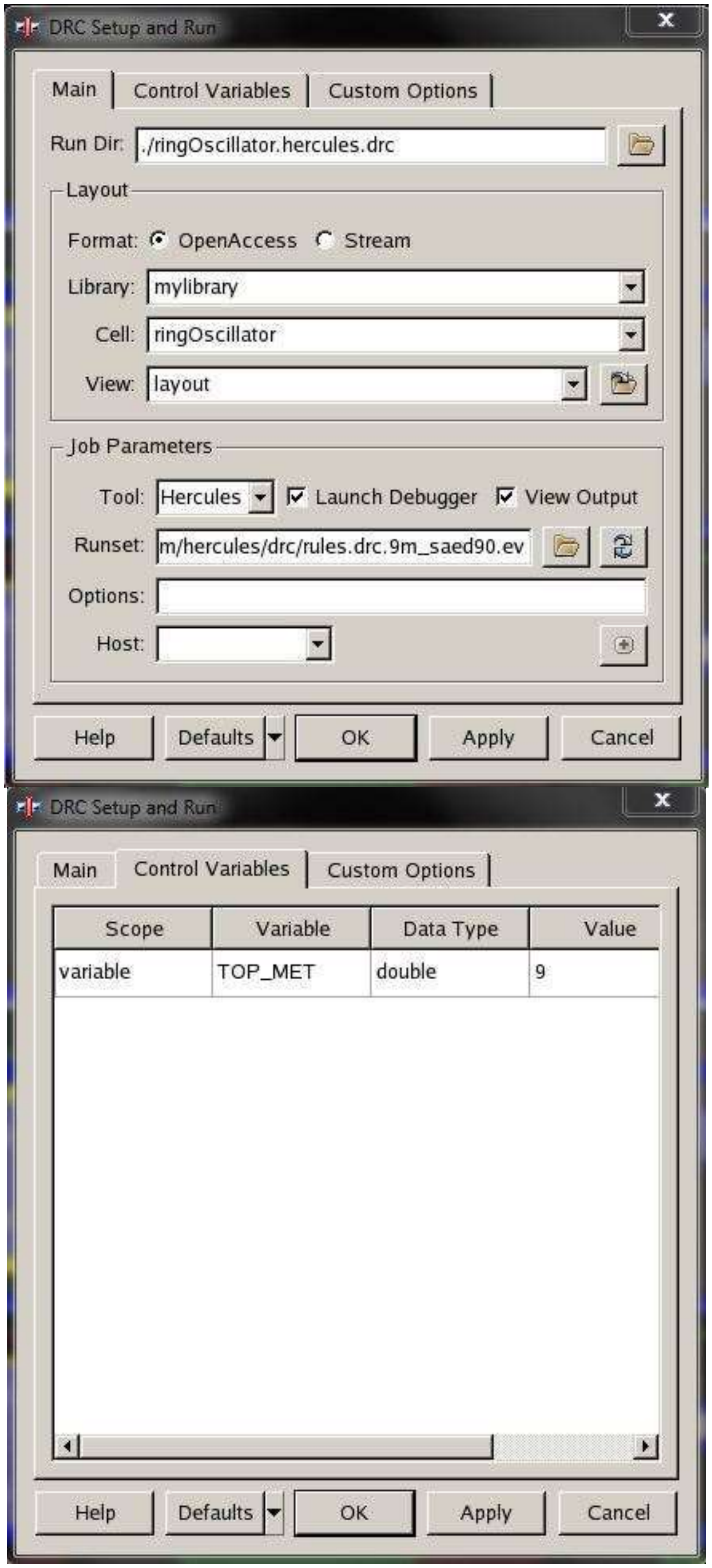


Figure 93: DRC Setup

Debug any DRC errors that come up. When DRC is passed, continue on to **Verification → LVS → Setup and Run** to run LVS. In LVS, setup the options as shown in figure 94 and figure 95 and leave the defaults for the custom options tab. Click **OK** when done.

Runset file under Main tab:
/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/hercules/lvs/rules.lvs.9m_saed90.ev

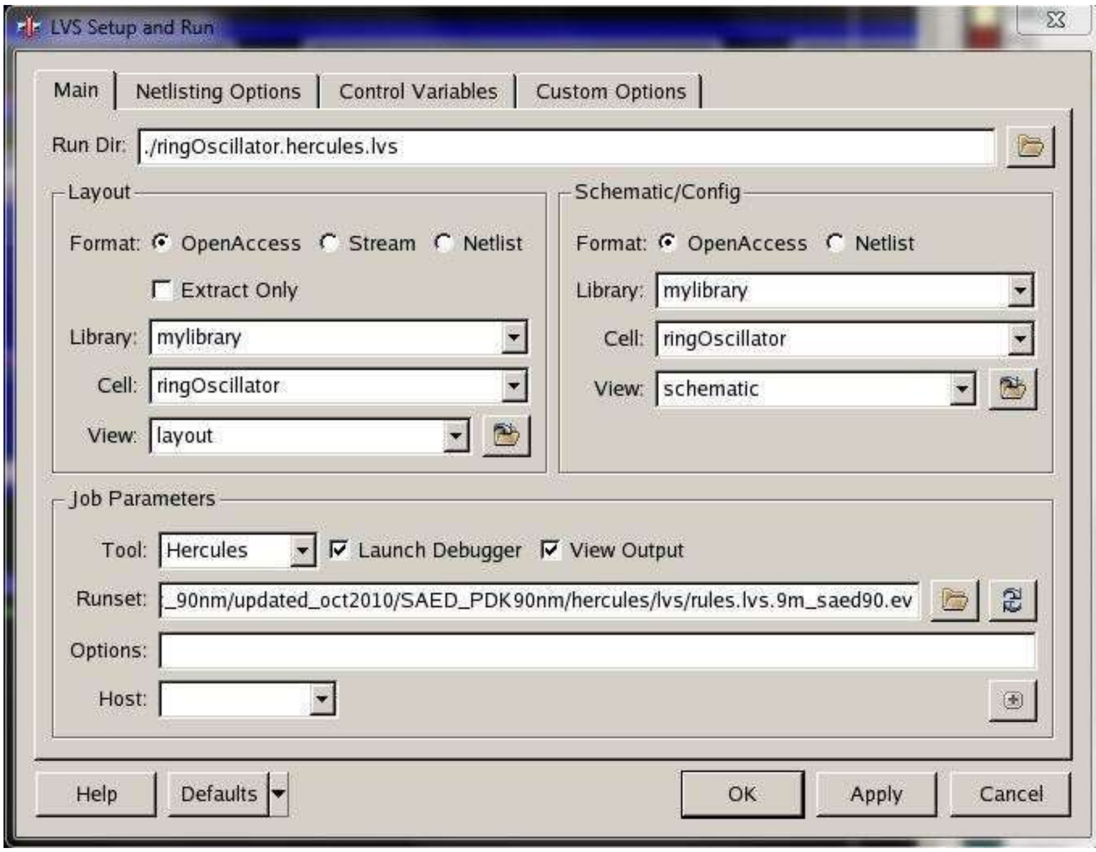


Figure 94: LVS Setup

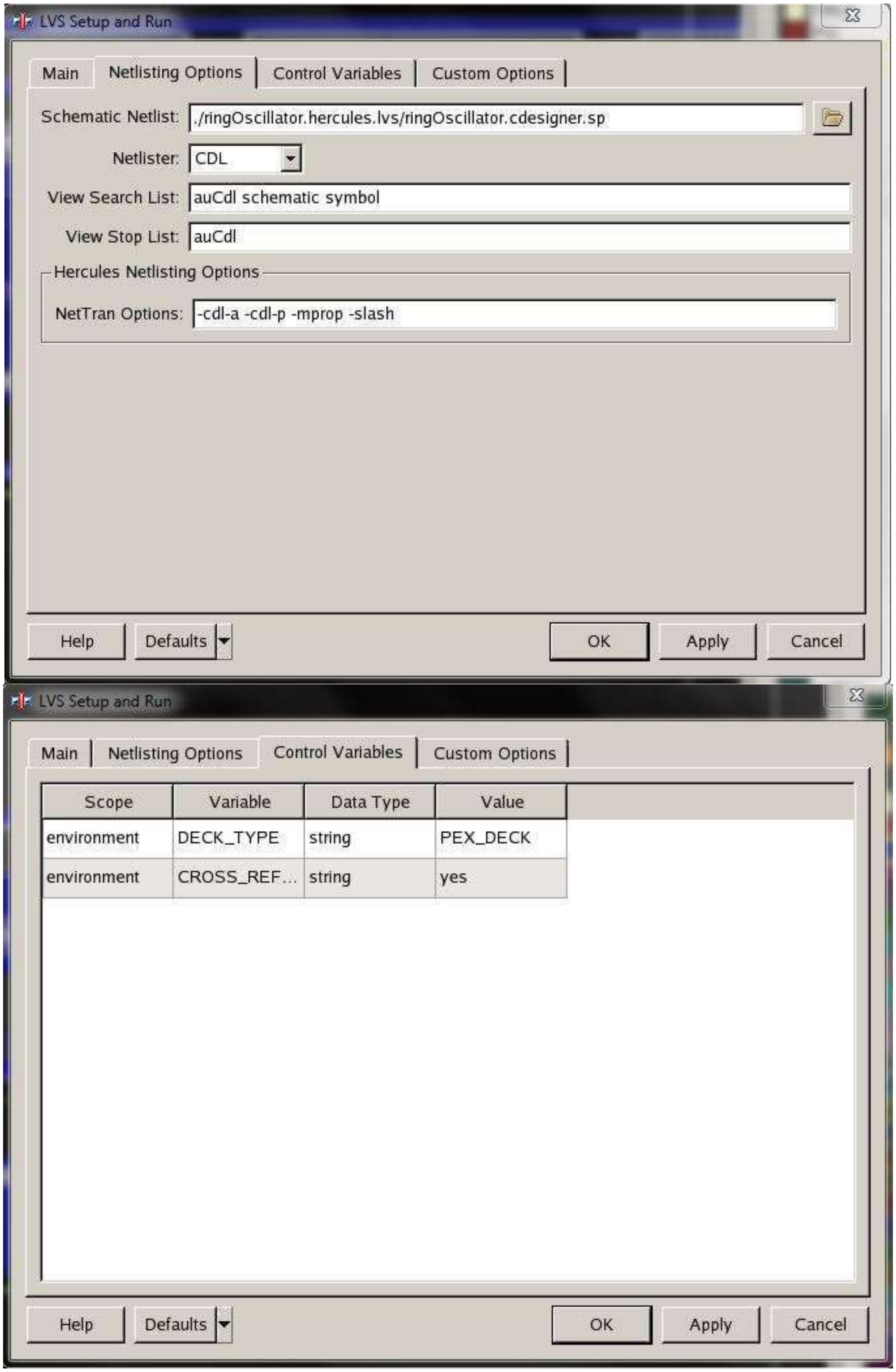


Figure 95: LVS Setup

At this point if there are any LVS errors, an error window will show up. Debug any errors you have and rerun LVS until you pass it. After running LVS successfully, go to **Verification → LPE → Setup and Run** to run parasitic extraction.

Under Extraction Option tab select the following file for Mapping File. /packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/starrcxt/saed90nm.map

For Milkyway XTR View select the hercules.lvs folder created from running LVS, then select the TOPCELLNAME_MILKWAY folder. See figure 96 below for reference on setups.

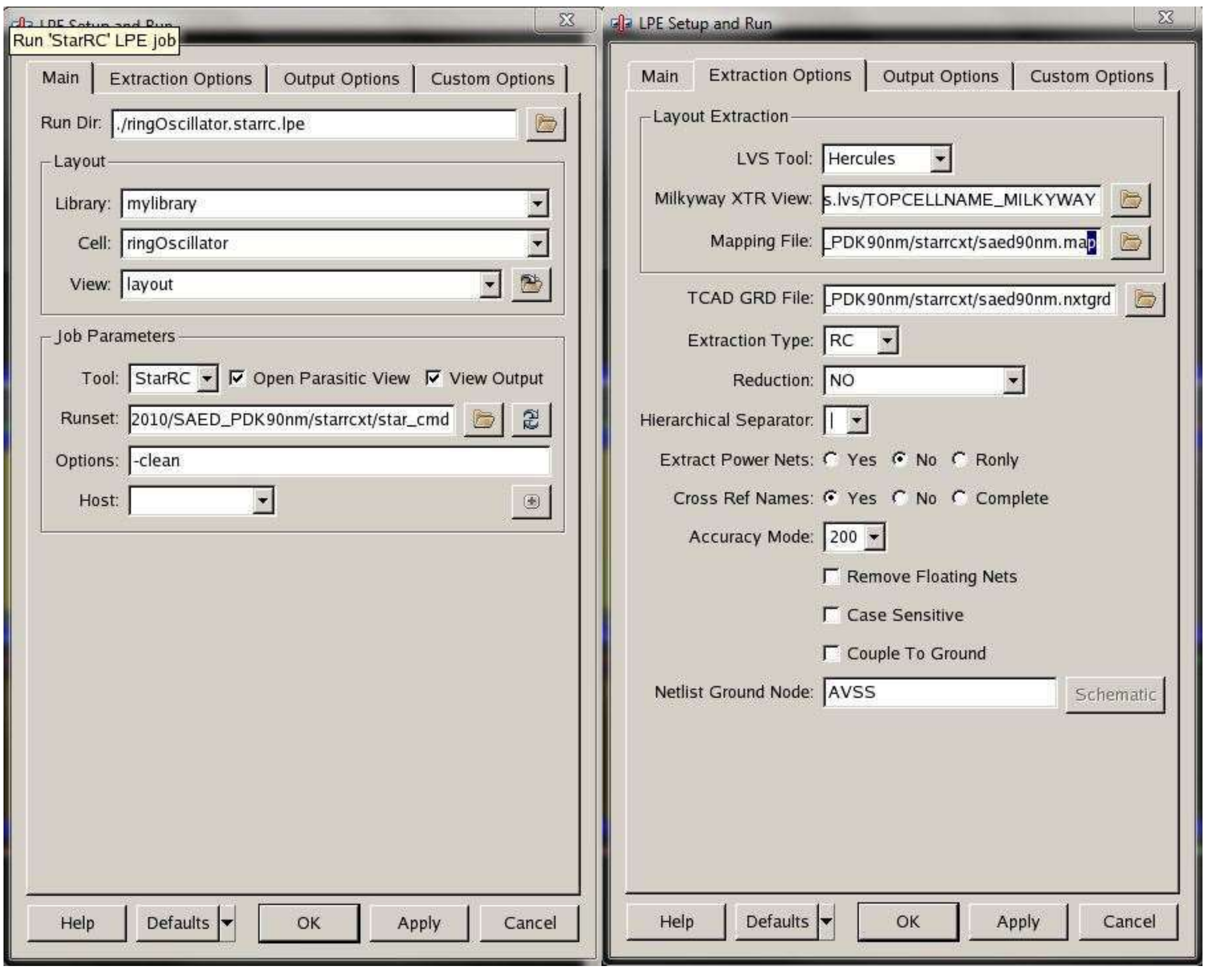


Figure 96: LPE Setup

Under the Output Options tab make sure that you have the same setup as shown in Fig. 97. Make sure the following map files are set as noted below if not already set by default.

Device Map:
/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/starrcxt/device_map

Layer Map:
/packages/process_kit/generic/generic_90nm/updated_oct2010/SAED_PDK90nm/starrcxt/output_layer_map

Leave the custom options tab with their set defaults. Click **OK** when done.

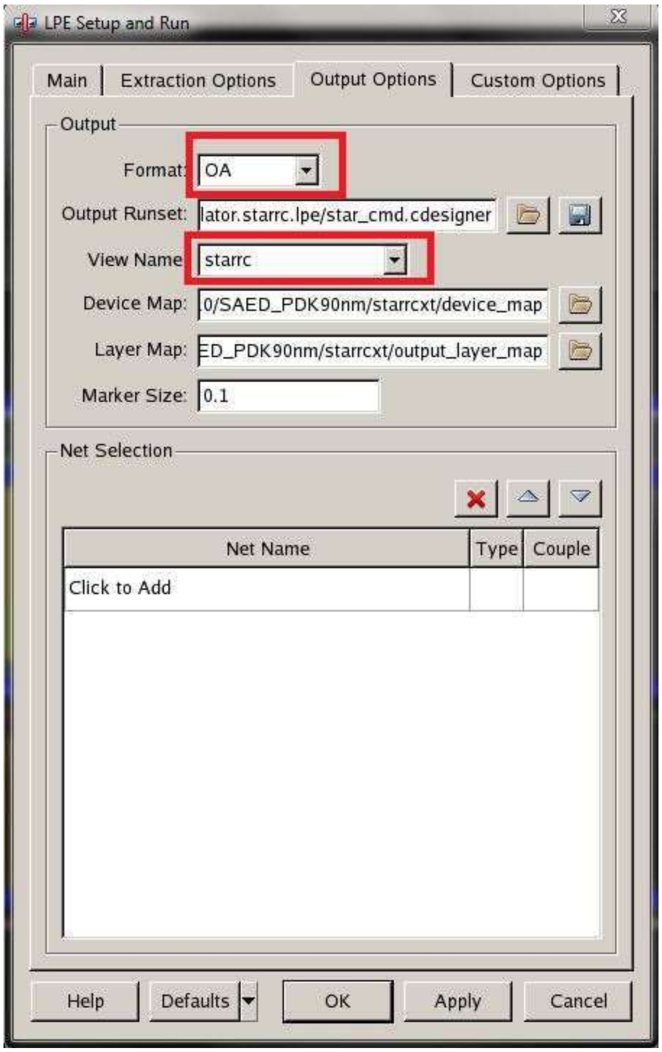


Figure 97: LPE Setup

If LPE ran successfully, a parasitics view will open up. The parasitics are small so drag a box over it with a mouse cursor and zoom in to see individual components if you don't see it at first. See figure 98 below for reference. Afterwards save the parasitics view with **Design → Save**.

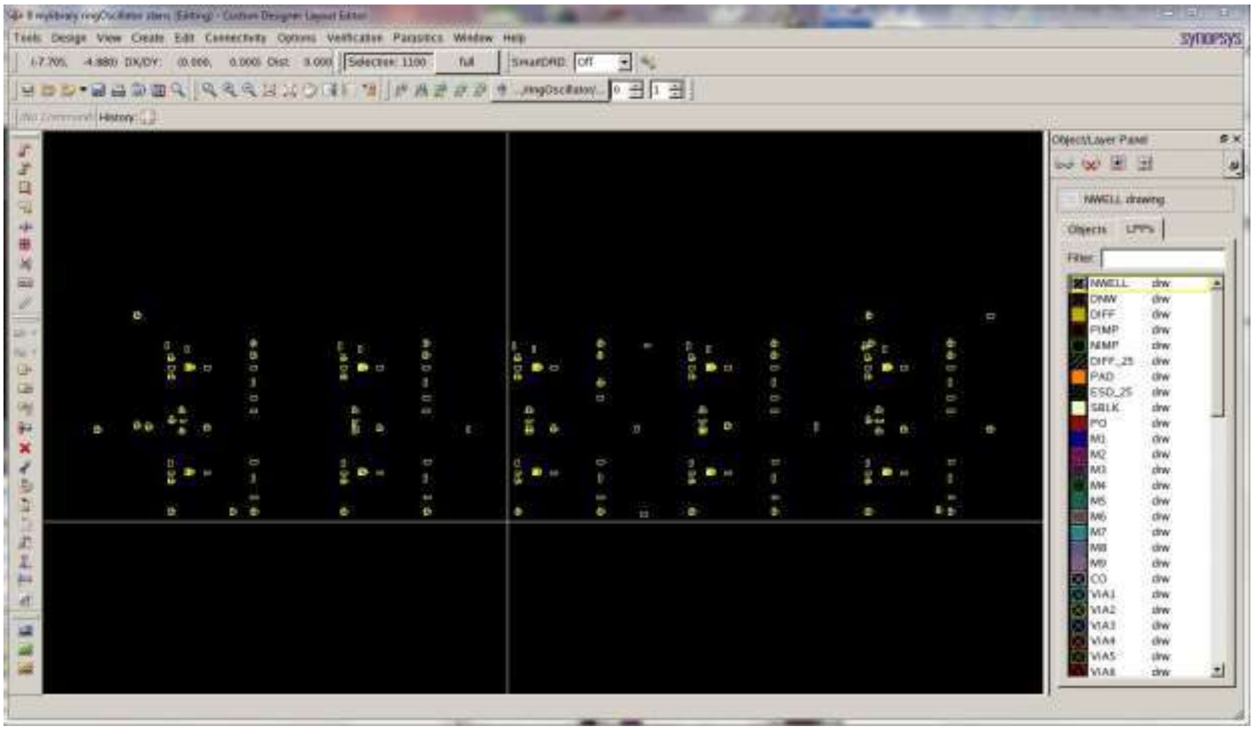


Figure 98: Parasitics View

After parasitic extraction, create a new configuration files by going to **File → New → CellView** in the custom designer console. Setup the options as noted in figure 99 to setup a configurations file for the ring oscillator testbench. Click **OK** when done.

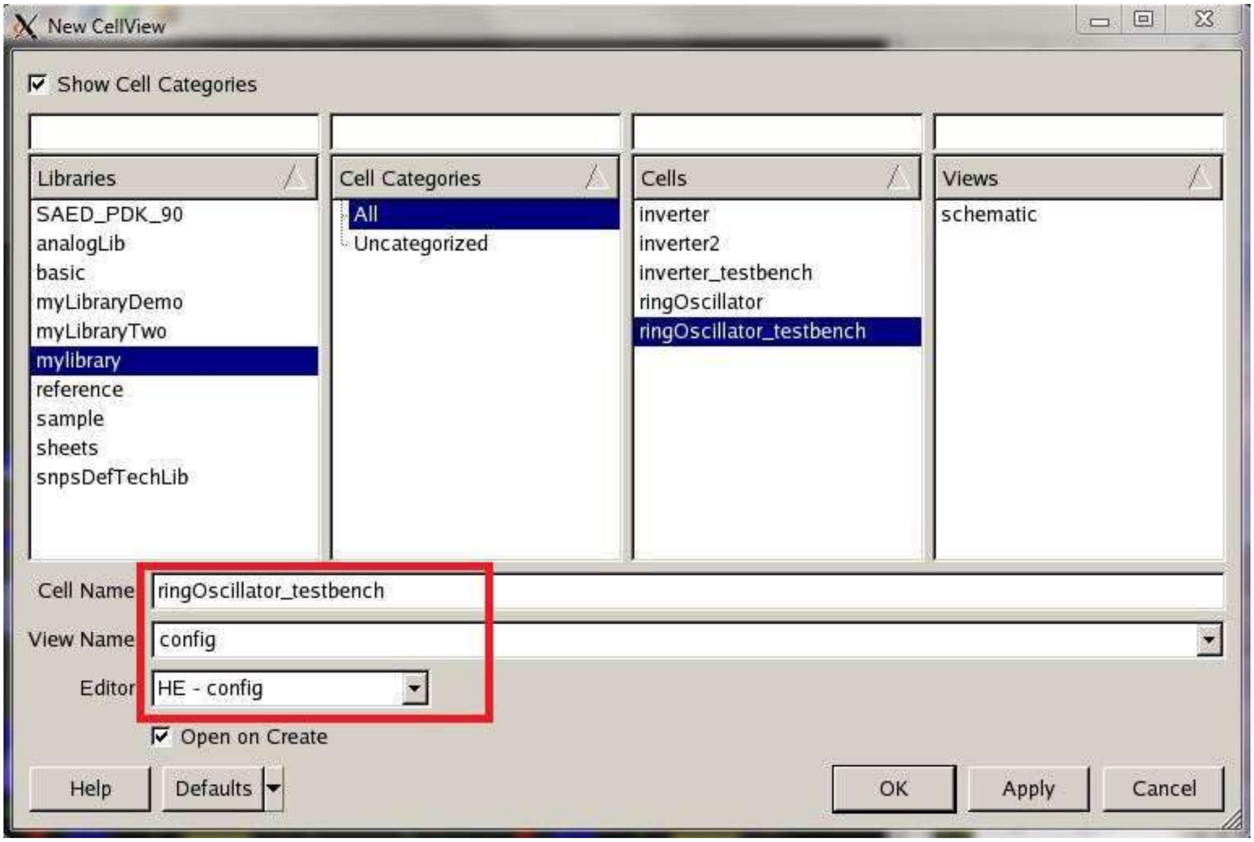


Figure 99: Configurations File Setup

A new configurations file will open up. From here, setup the options as noted in figure 100 to load the ring oscillator parasitics into the ring oscillator symbol. Save with **File → Save** when done.

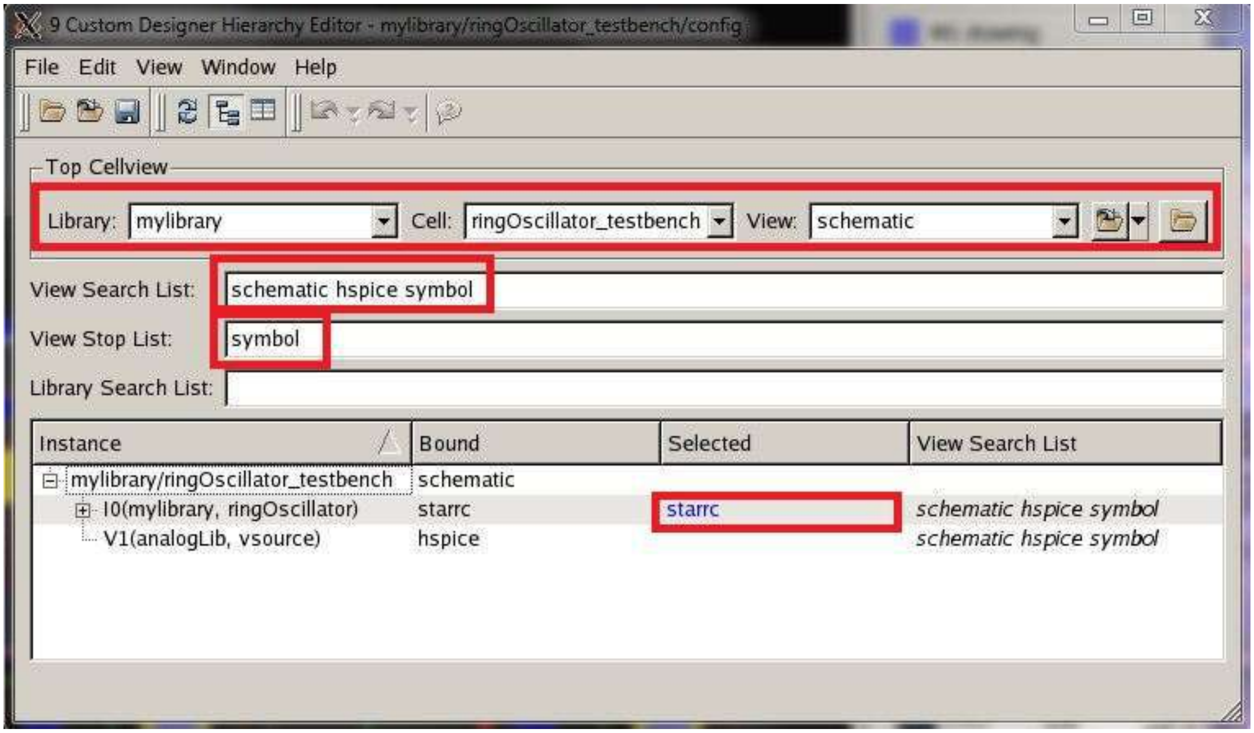


Figure 100: Configurations File Setup

To start simulation with parasitics, go to **File → Open Design** from the Custom Designer Console window. In the Open Design window that opens, select **ringOscillator_testbench** under the cells column and **config** under the views column and right click on config under views. Select **Open Design** from the drop down menu. See figure 101 below for reference.

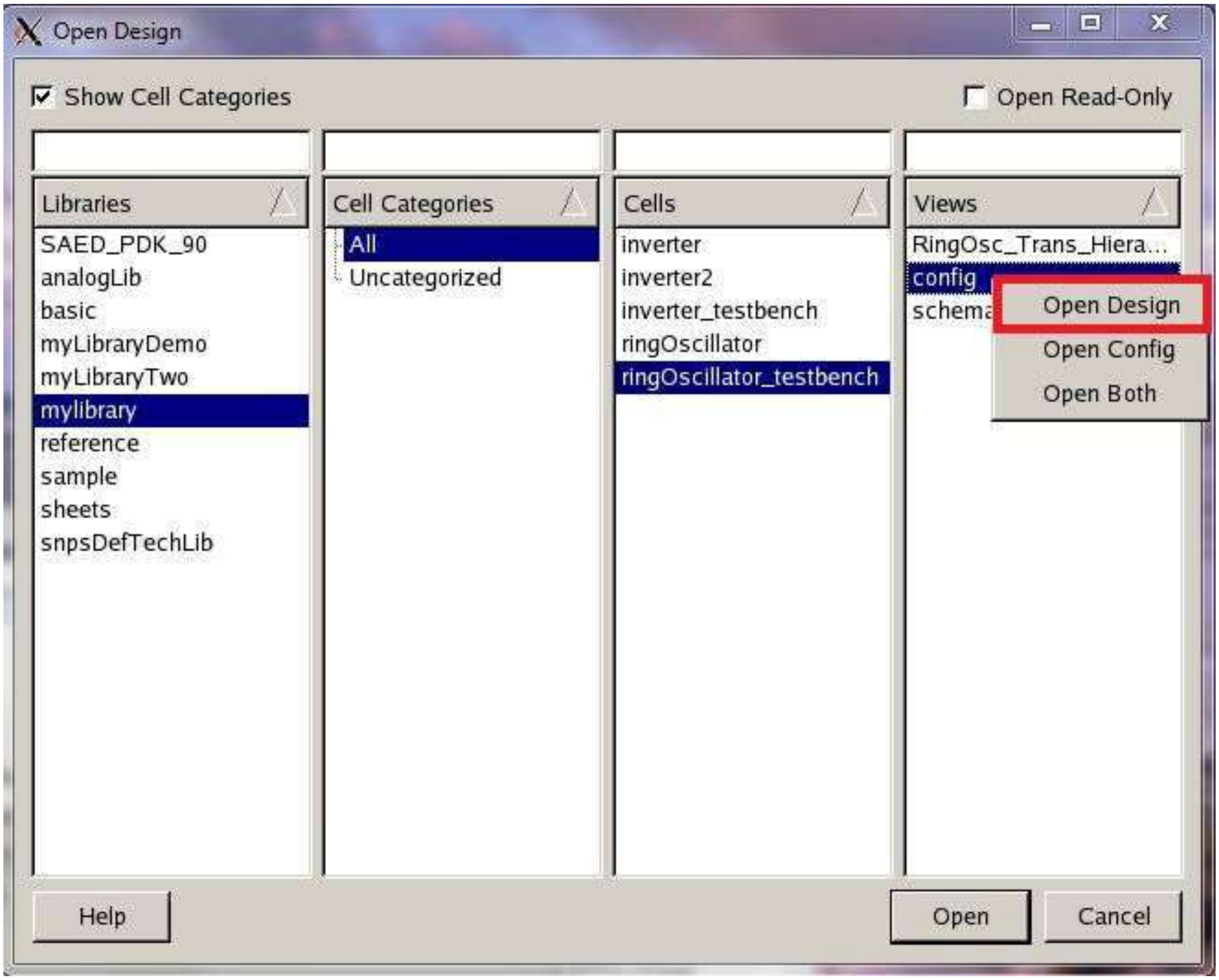


Figure 101: Open Design with Parasitics

Afterwards a schematic window opens up with the ring oscillator testbench circuit created earlier, see figure 102 for reference. To check if the parasitics were properly loaded into the ring oscillator, double click the ring oscillator symbol and the parasitics view generated from LPE earlier should display.

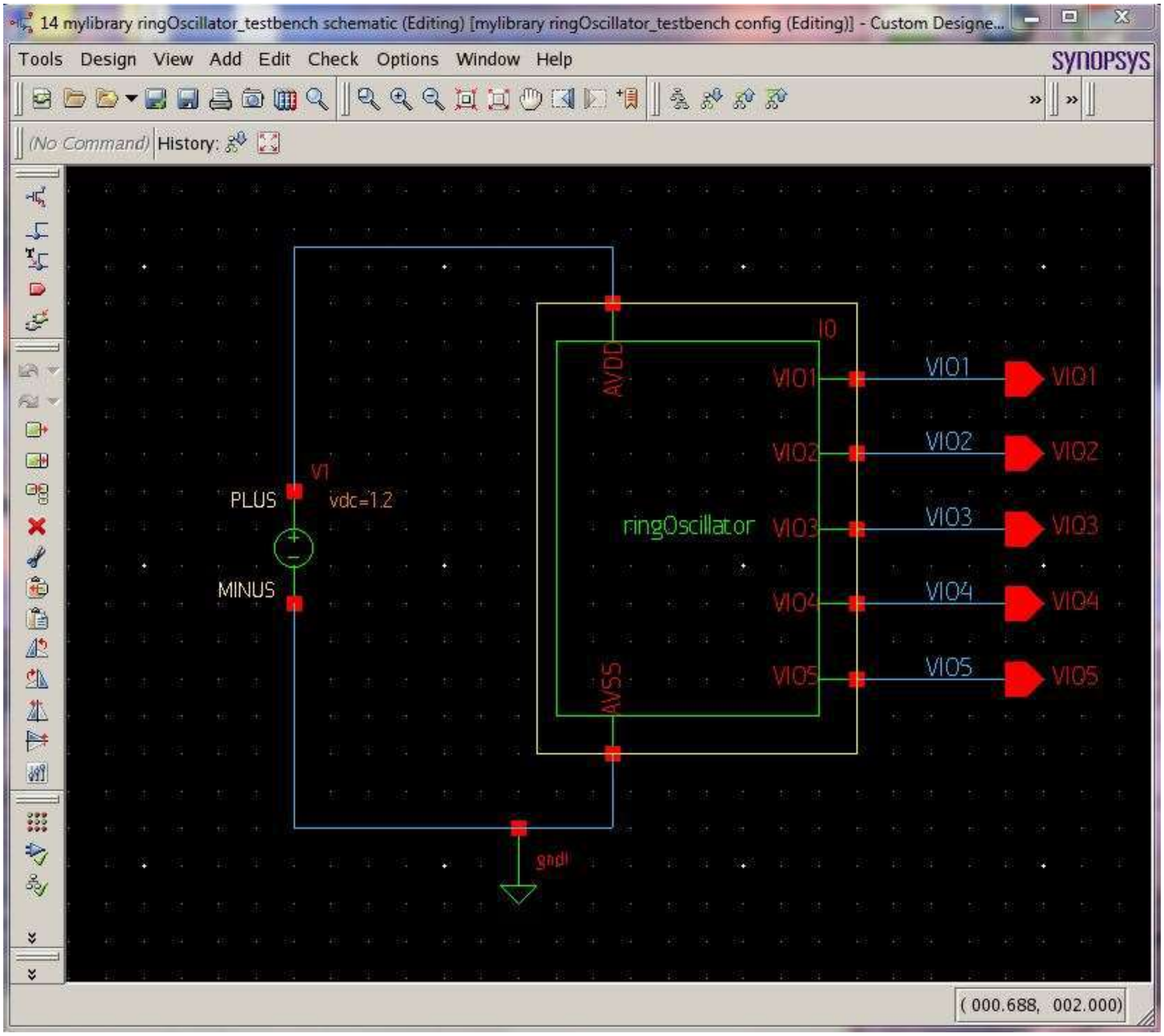


Figure 102: Ring Oscillator Testbench Circuit

From the ring oscillator testbench window, you can simulate the circuit by using SAE as noted in part 3 of the tutorial. To open SAE, go to **Tools → SAE** from the schematics window and setup the simulation for a transient analysis and plot the voltages for the VIO1, VIO2, VIO3, VIO4, and VIO5 voltages. For the transient analysis setup in SAE, use 1ps for step time and 1ns for stop time.

Side Notes for Using Convergence Aids to Initialize Voltages:
Also note that it may be helpful to give a wire in the circuit an initial voltage before running simulation. This particular setup applies to circuits such as a five stage ring oscillator circuit shown in figure 103. In addition to the setup noted in part 3 for SAE, before running the simulation, go to **Setup → Convergence Aids** in the SAE window.

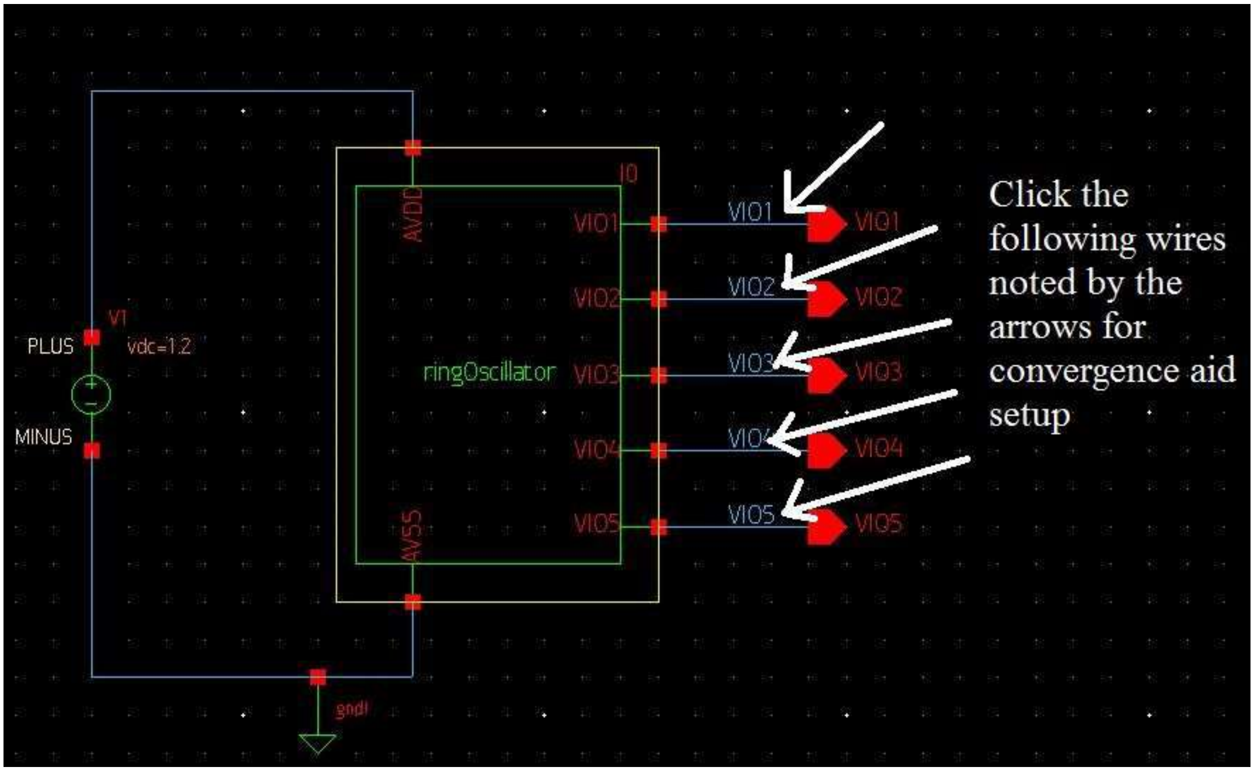


Figure 103: Where to Click on Schematic for Node Setup in Ring Oscillator

Setup the options as noted in figure 104 below. You may need to setup multiple initial voltages to drive the inverters since one initial voltage may not be enough to drive the entire ring oscillator. It is suggested that you setup at least two initial voltages using alternating voltages of 0 and 1.2 for consecutive inverter nodes in the ring oscillator circuit. See figure 105 for multi-node initialization and see figure 103 on where to click in the schematic for node setups. Click **OK**, when done and run the simulation as noted in part 3 of the tutorial.

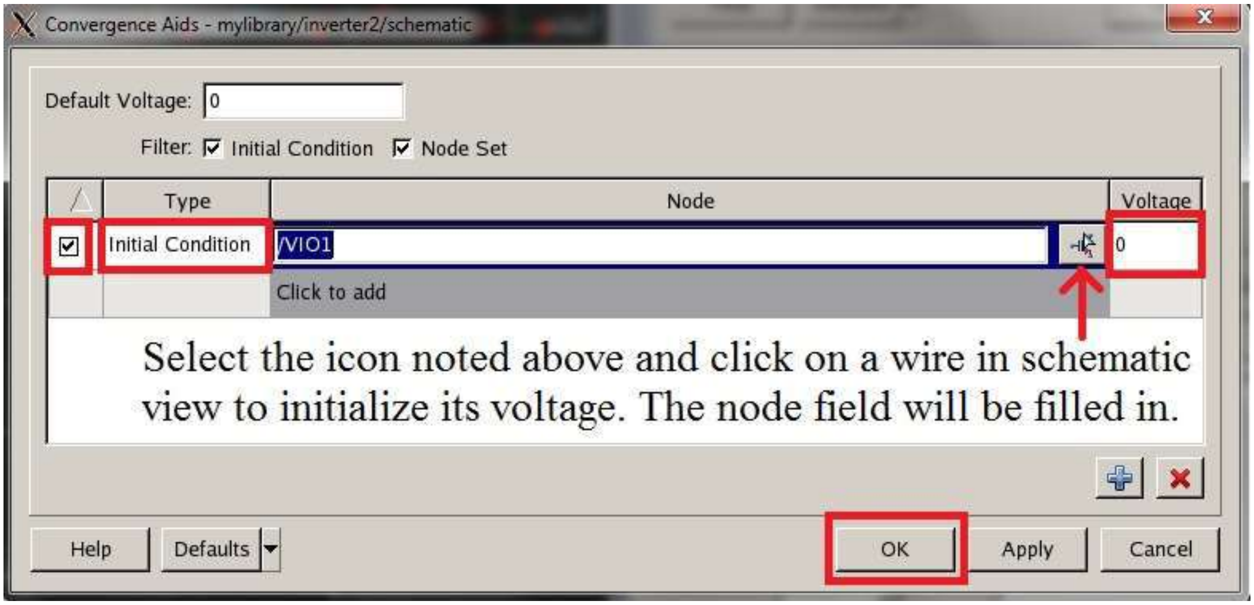


Figure 104: Setting up Convergence Aids

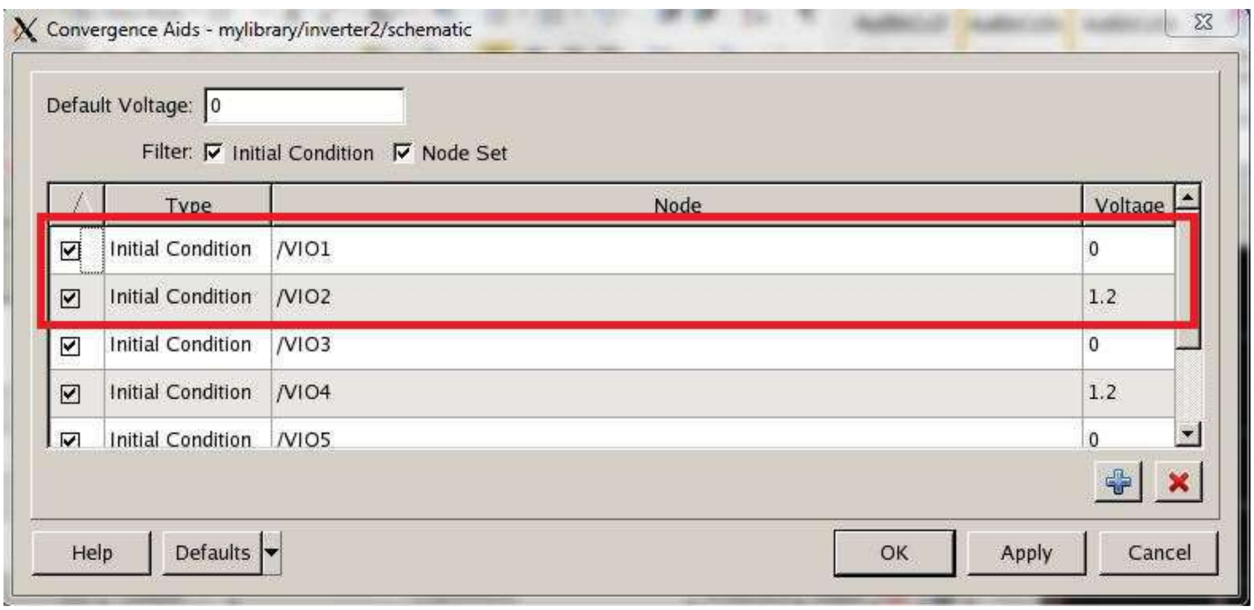


Figure 105: Setup for Multiple Initial Voltages

You have now finished transient simulation of the ring oscillator circuit with applied parasitics.

Additional Notes

Using the M2 Layer in Layout
For designs that require require an extra metal layers in layout, designers can use a metal layer higher up (like M2) to make connections if the lower metal layers (like M1) are too constricting to allow any other connections.

In the ring oscillator layout, M1 (blue layer) is replaced with M2 (pink layer). See figure 106 for reference. Also note that M1 layers can run under M2 layers without physically connecting unless there is a VIA1 layer in between them.

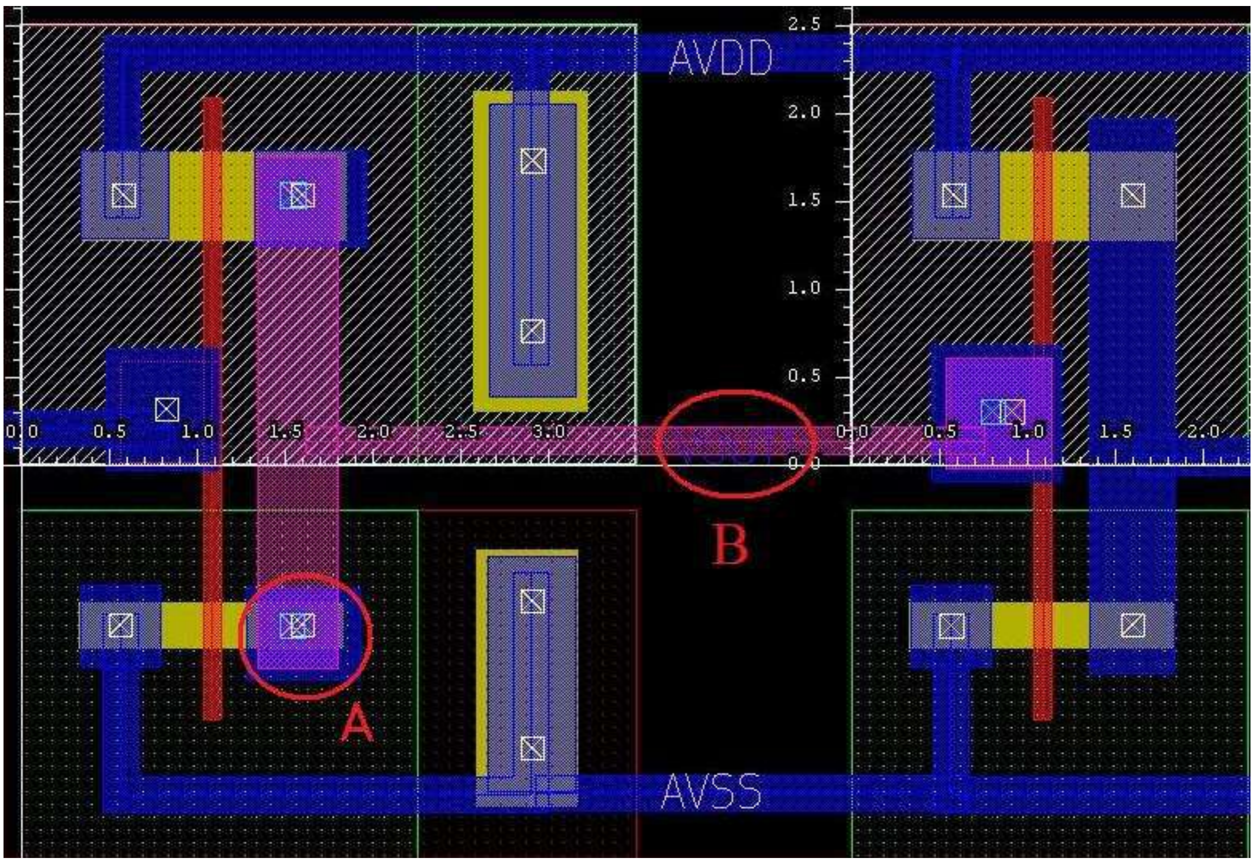


Figure 106: Using M2 in Layout

In order to connect different layers to each other, all metal layers and contacts/VIAs must exist in between them. For example, to connect a metal 2 layer (M2) to diffusion (DIFF), a contact layer (CO), metal 1 layer (M1), and a VIA1 layer must exist between them. See figure 107 for reference. Also see figure 106 in circle A for a layout drawing example.

To use higher metal layers not shown, use their corresponding VIA layers to interconnect between the desired layers. For example, to connect M2 to M3, a VIA2 layer/contact must be drawn in between them.

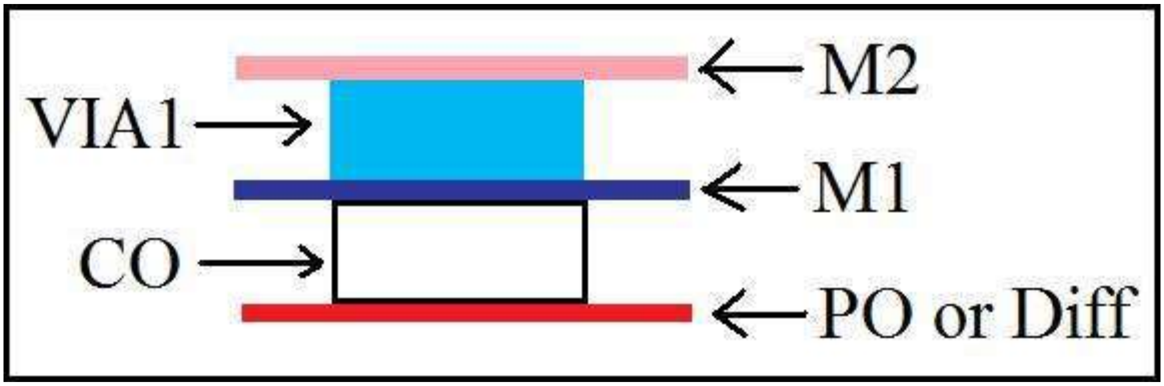


Figure 107: Layers Needed to Connect M2 to Poly

When dealing with multiple layers, be sure that enough metal/poly/diff surrounds the contact or VIA connection on each side as per the design rules for using higher layers. Also note that VIA connections have specific dimensions that need to be observed when drawn. M2 and VIA1 layers can be found under the LPP panel as shown in figure 108. Also note that every layer has its own pin for labeling. For example, use the M2PIN layer (violet color) to label the M2 layer. See figure 106 in circle B for a layout example.

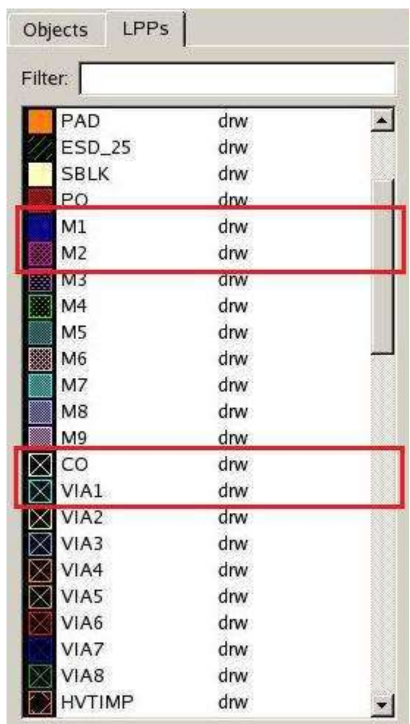


Figure 108: Selecting VIA1 and M2 Layers

Measuring Device Capacitances after Operating Point Simulations

Sometimes it is helpful to have estimates for transistor capacitances to facilitate design hand calculations (e.g. estimated fan out of an inverter cell). After running an operating point simulation, the resulting device capacitances (for that static DC test condition) can be measured using the Results Analyzer tool in SAE. To get to the Results Analyzer tool after running an operating point simulation, click on Results → Analyzer from the top of the SAE window or click on  in the bottom right of the SAE window. Afterwards, the Results Analyzer window will launch. To measure a particular transistor's capacitance parameters, click on **op** in the **Analysis** menu on the left. Then in the **Expression** menu on the right, select **operating point** and then **Instance** in the **Selection** menu. In the **Parameter Name** drop down menu, select the device capacitance parameter of interest (for example, **cgtot** for the transistor's total gate capacitance). Check the **Immediate Evaluation** box to force the value to be evaluated immediately and reported in the field at the bottom of the screen. The necessary fields that must be selected in the Results Analyzer window are shown in Figure 109.

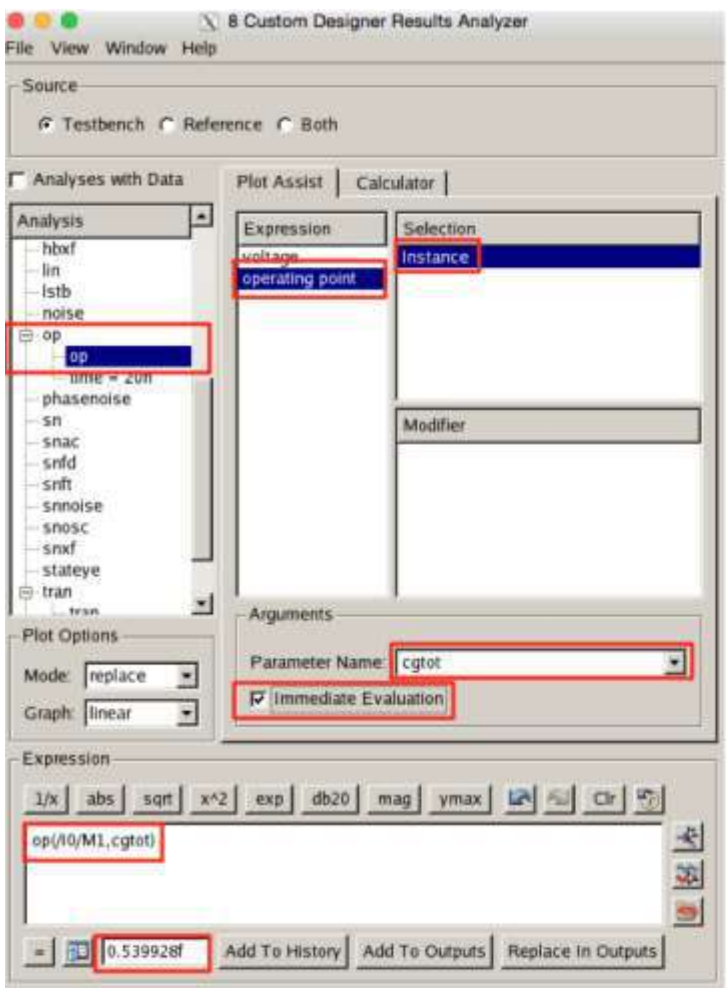


Figure 109: Measuring a Transistor's Capacitance with the Results Analyzer

Troubleshooting

Reference library, tech file, or runset will not load:
Check to see the library path is correct after you select a library in the file browser. Sometimes there is a glitch in the fields. If there is a glitch try typing in the file path manually.

Schematic or layout has glitches:
Inside the schematic or layout window scroll away from the object and then return to the object. It should be refreshed. When you select the option to visualize or hide a layer in Full Custom Designer it is common for the change the change may not be readily apparent. Scroll away and back to refresh.

Window does not close when close window icon is clicked:
This is an issue with the x-server. Inside the window you want go to close go to **File>Quit**.

Library and cell will open but you are unable to edit cell:
Your cell has a lock on it. Open the library and check the “SCH” and “CEL” folders in the terminal or file transfer window. Delete all files with a “.lock” file extension.